Quality Management and Improvement

Kevin Otto

Schedule and Learning Outcomes

Week	Recorded Lecture Topic
1	Introduction.
2	Inspection. Yield.
3	ANOVA. Measurement Systems Analysis.
4	Statistics. Confidence Intervals.
5	Hypothesis Testing. Sample Sizes.
6	**Quality Management Processes.**
7	Regression.
8	Design of Experiments.
9	System Reliability. Availability.
	BREAK – Non Teaching Week
10	Parametric Distribution Analysis
11	Statistical Process Control. Supplier Quality.
12	Review

After completion of the course the student should be able to:

- Explain different statistical methods
- Parametric and non-parametric hypothesis tests
- Design and plan for collecting experimental data
- Concepts of reliability engineering and the relationship between TTF, R, H
- **How to improve its processes, products and services**
- **Apply TQM, 6S, and benchmarking**

Overview

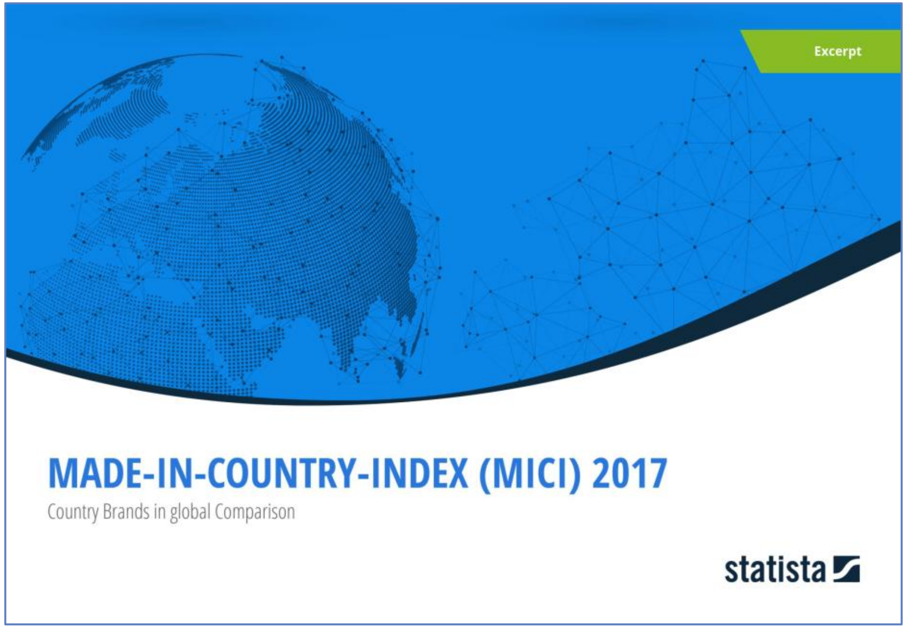
PDCA

IS09000

Six Sigma

Quality

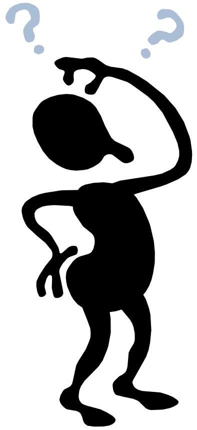
Rank	Country	Index
1	Germany	100
2	Switzerland	98
3	European Union	92
4	United Kingdom	91
5	Sweden	90
6	Canada	85
7	Italy	84
8	Japan	81
8	France	81
8	USA	81
11	Finland	77
11	Norway	77
13	Netherlands	76
14	Australia	75
15	New Zealand	73
15	Denmark	73
/…/		
20	Spain	64
/…/		
30	Argentina	42
/…/		
42	India	36
/…/		
49	China	28

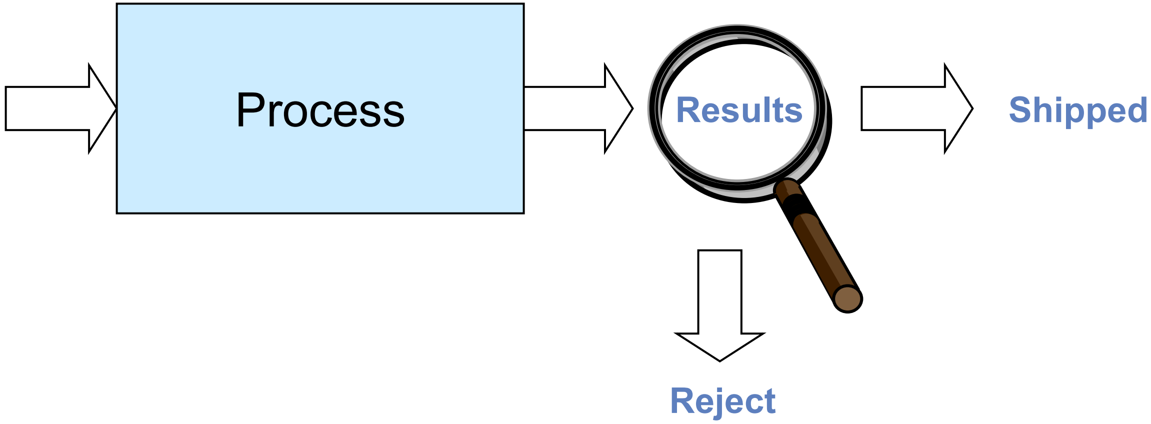


How do you raise a company to a higher level of quality?

Continuous Improvement

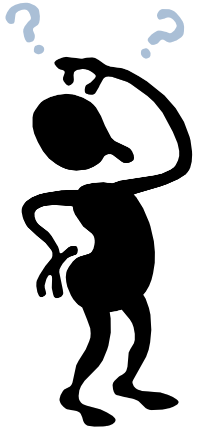
Suppose your company didn't always make it right.
You had a turnback. What do you do?

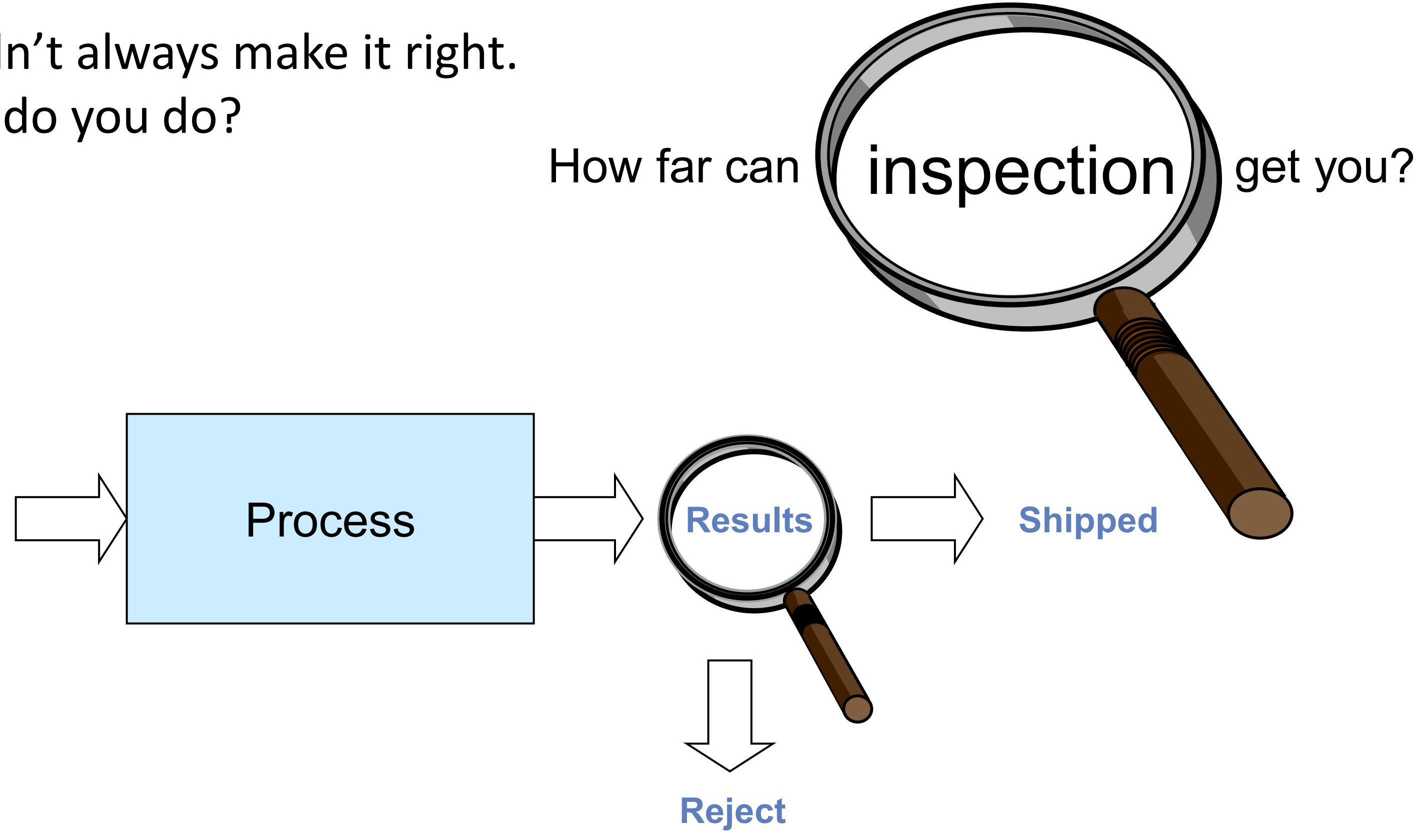




Continuous Improvement

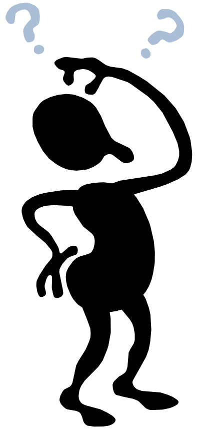
Suppose your company didn't always make it right.
You had a turnback. What do you do?
Rework it.

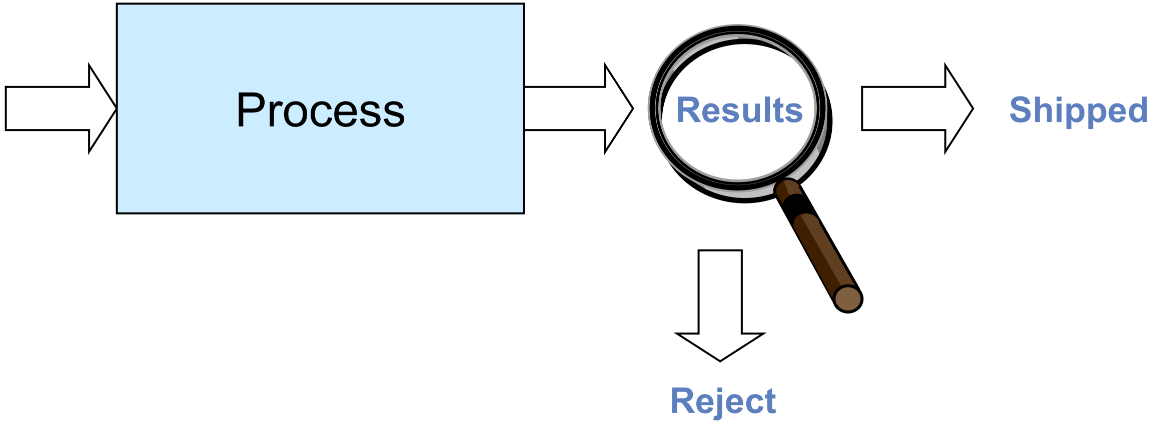




Continuous Improvement

Suppose your company didn't always make it right.
You had a turnback. What do you do?
How about ask **Why?**
Something was different in the process when it was done wrong.



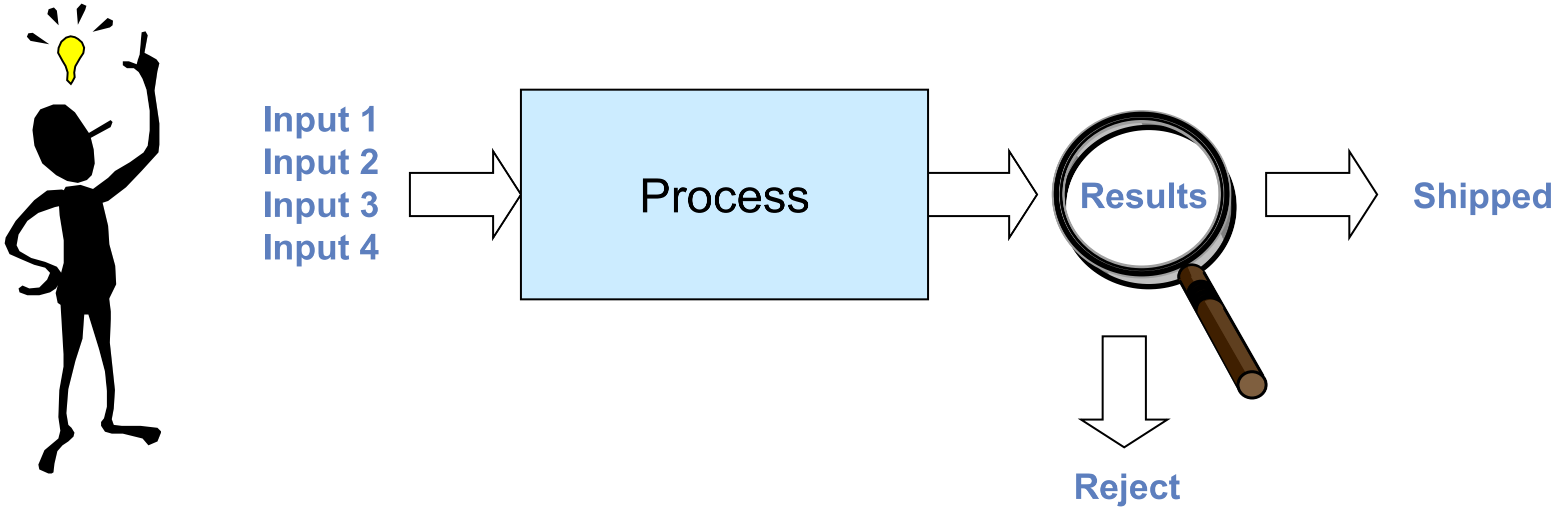


Continuous Improvement

Something was different in the process when it was done wrong right.

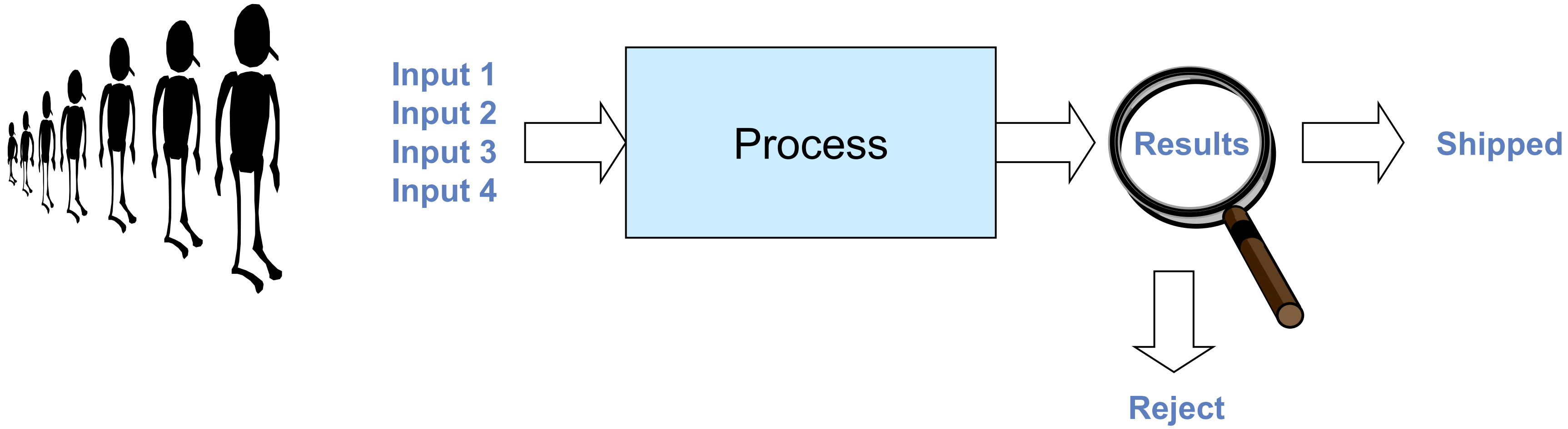
Why?

Rather than guessing and making quick incorrect decisions, be methodical.
Use the ***scientific method*****.** Find the cause. Fix the process so the cause doesn't happen.



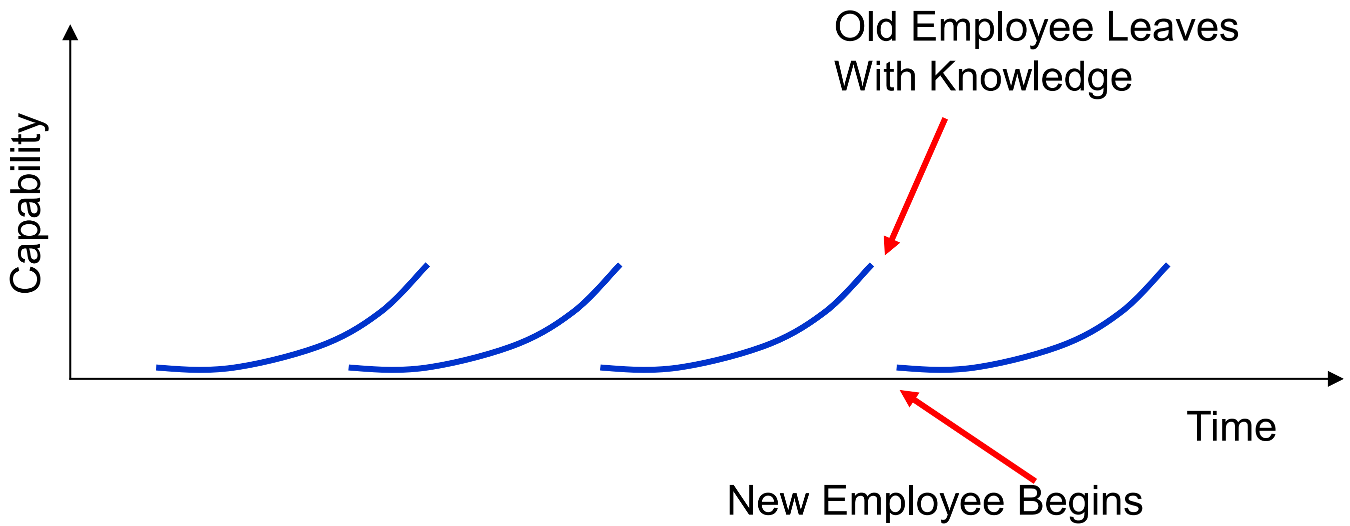
Continuous Improvement

If you figured it out and fixed it, will it stay fixed?



Building Organizational Memory

Many organizations exhibit the ***Eyelash Learning Curve***



No ***organizational memory*** to allow people to start where their predecessors left off

Nothing in place to capture the new or improved methods that produce results

***Fourth Generation Management*, Brian Joiner**

Rapid Learning Organization

Organization continues to advance their knowledge by preserving the lessons each learns

How?

How Do We Create Rapid Learning?

Two key ingredients:

1. Having best known methods documented. ***Standard Operating Procedures***.
2. Training people on those methods.

Who to train?

- New employees
- Managers
- Experience employees

Pros and Cons of Standard Methods

Disadvantages of Standard Operating Procedures:

- Stifle creativity and lead to stagnation
- Interfere with customer focus
- Add bureaucracy and red tape
- Make work inflexible and boring
- Only describe the minimal acceptable output
- Have legal implications
- Waste time

Advantages of Standard Methods

Advantages of Standard Operating Procedures

- Everyone has bought into them
- Customer progress is more visible and can be tracked over time
- Capture and share lessons learned
- System itself does not become a source of variation

Finding a Balance

The difficulty is that the arguments for and against standardization are both true

- To achieve a balance, develop standards judiciously – where it matters the most – and treat them as living, evolving guidelines that can and must be constantly improved.

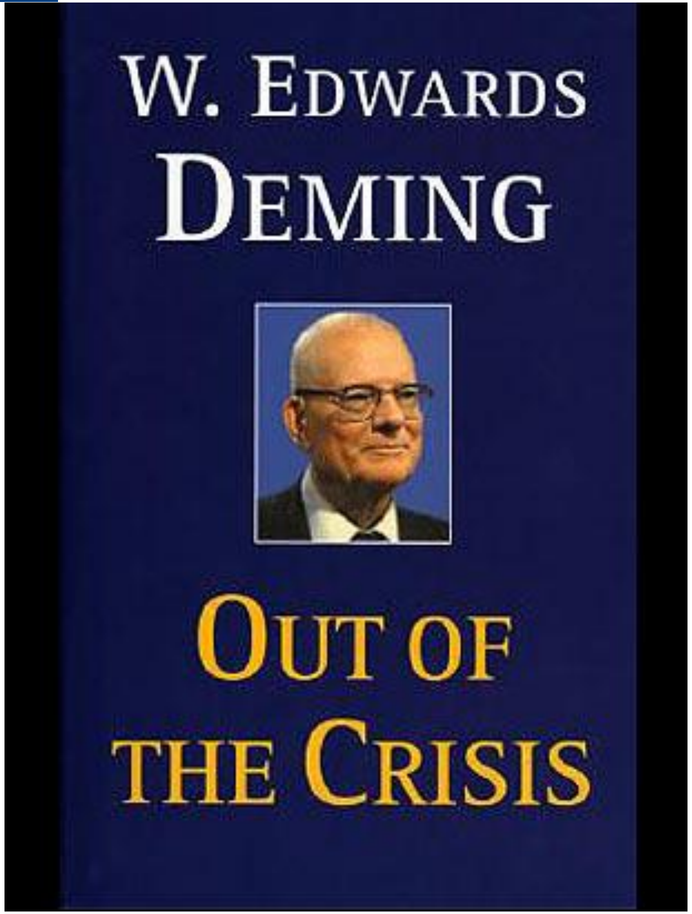
Companies run into trouble when methods are changed before understanding why those methods were there in the first place

Resist changing standard work until you know:

- Are the documented standards the best?
- What is the impact on the rest of the system?
- Are the methods actually being followed?
- Can we eliminate steps or are they there for a reason?

Use the scientific method.

Deming Quality Philosophy



Published in 1986

W. Edwards Deming

1900 – 1993

Quality philosophy:

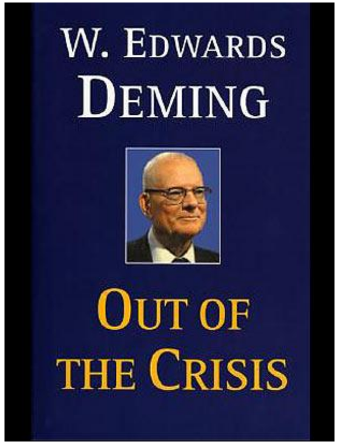
Continuous improvement.

Focus on the process. PDCA.

Organizational learning is much harder than individual learning. You need standard procedures for individuals to follow. Then improve them.

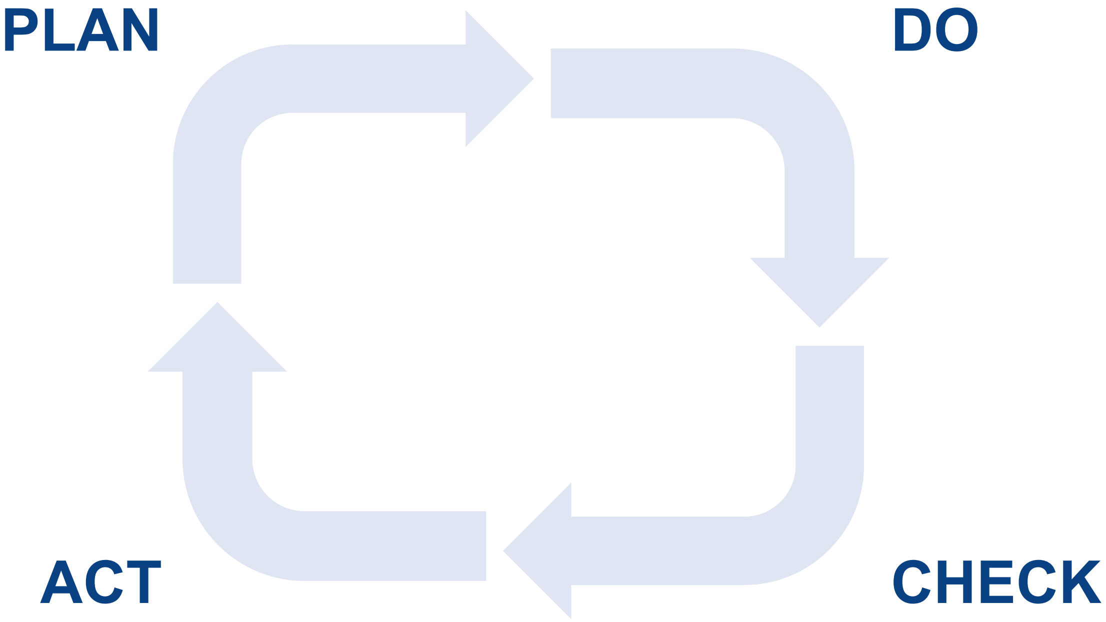
Deming's 14 Management Principles

1. Create a constant purpose toward improvement.
2. Adopt the new philosophy.
3. Stop depending on inspections.
4. Use a single supplier for any one item.
5. Improve constantly and forever.
6. Use on the job training.
7. Implement leadership.
8. Eliminate fear.
9. Break down barriers between departments.
10. Get rid of unclear slogans.
11. Eliminate management by objectives.
12. Remove barriers to pride of workmanship.
13. Implement education and self-improvement.
14. Make "transformation" everyone's job.

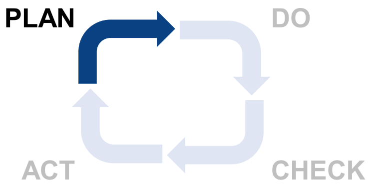


Continuous Improvement

The **PDCA continuous improvement process** was developed by Deming for organizations to understand and eliminate failures. It is the scientific method applied to organizations. It is used to this day.



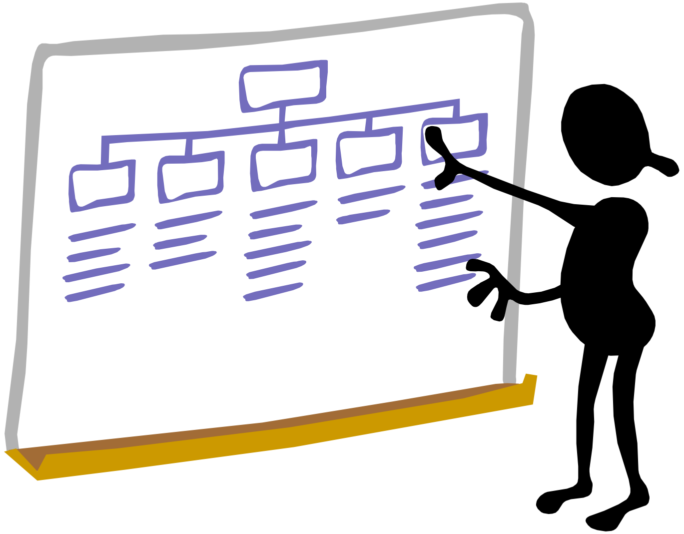
PDCA – Plan



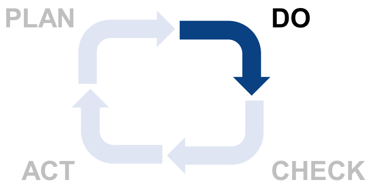
Investigate why there are defects.

Enumerate a set of possible improvements to the process.

In manufacturing, this means **do a process FMEA** and **collect data on causes.**



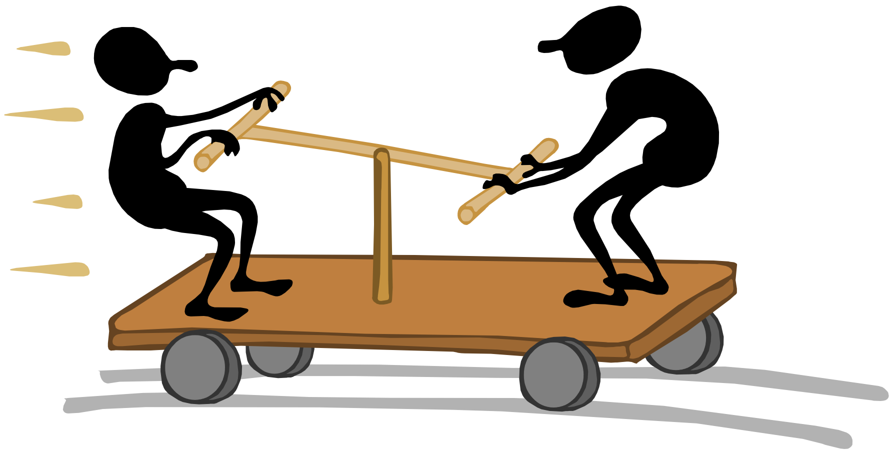
PDCA – Do



Select an ‘improvement’.

Experimentally test it.

In manufacturing, this means **form a random experiment against an improvement.**

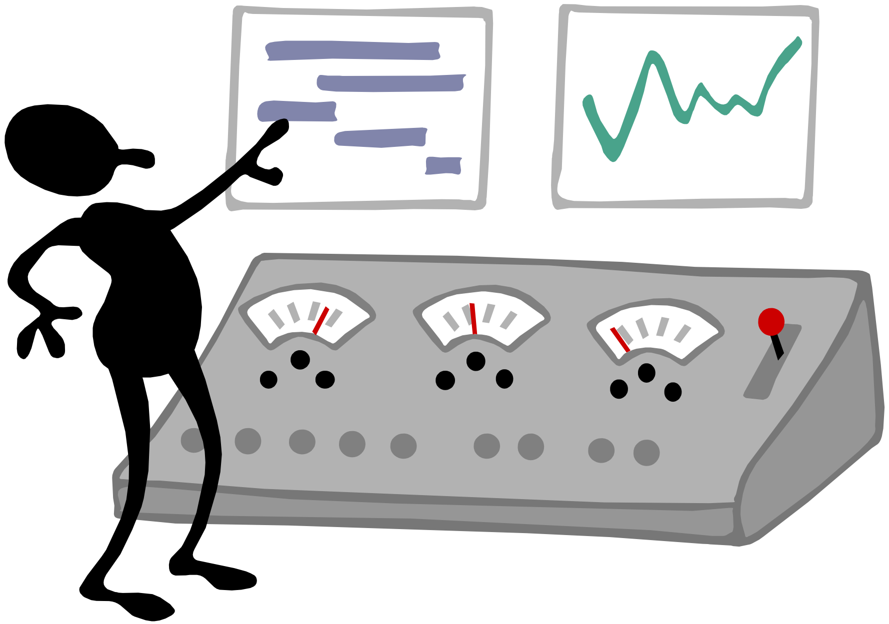


PDCA – Check

PLAN DO ACT CHECK

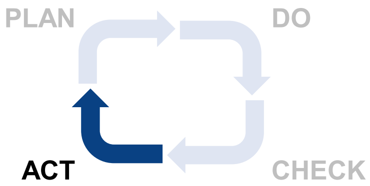
Evaluate the ‘improvement.’

Did it improve the process?



In manufacturing, this means **do a hypothesis test.**
Did it statistically make a significant improvement?

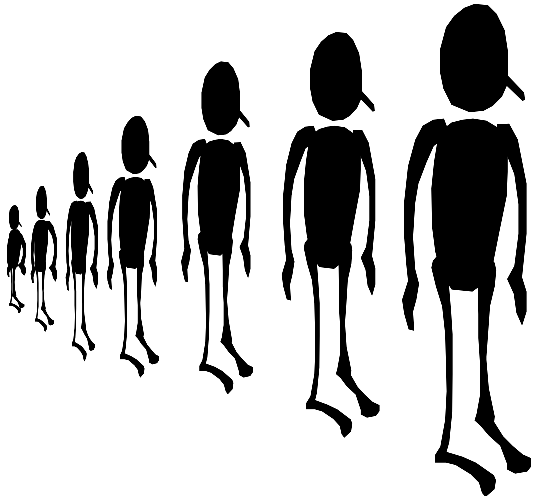
PDCA – Act



If the process was improved, now revise the old standard work procedure into a new standard work procedure.

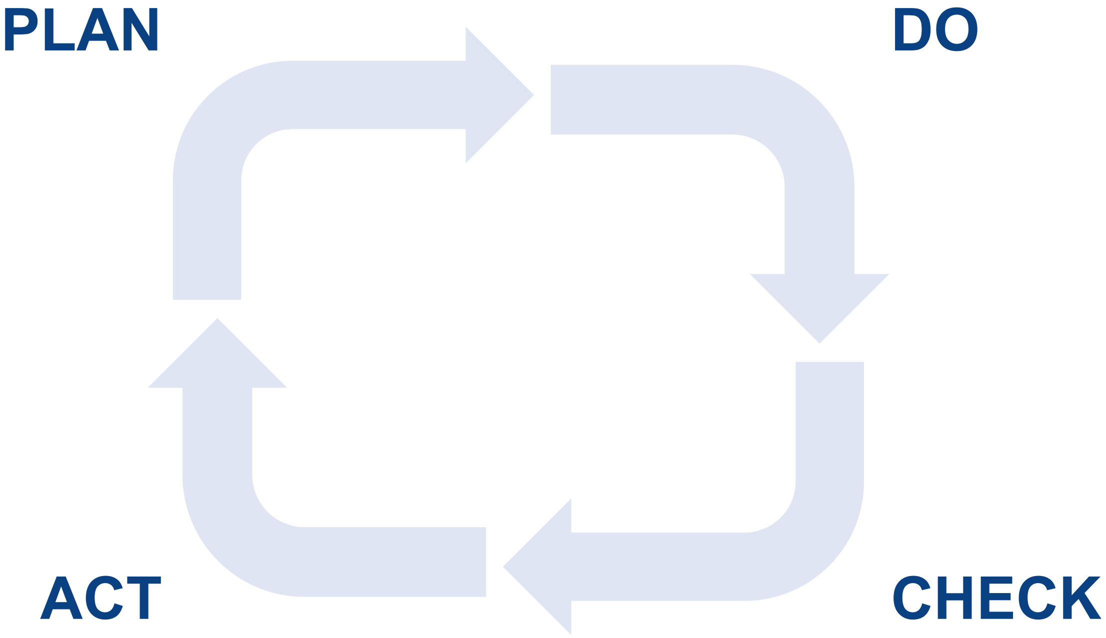
Deploy the new standard work.

In manufacturing this means **change the process controls** and standard work instructions.



PDCA Continuous Improvement

Continuously repeat the PDCA continuous improvement process.
Quality improvement never stops.



LESSONS TO LEARN:

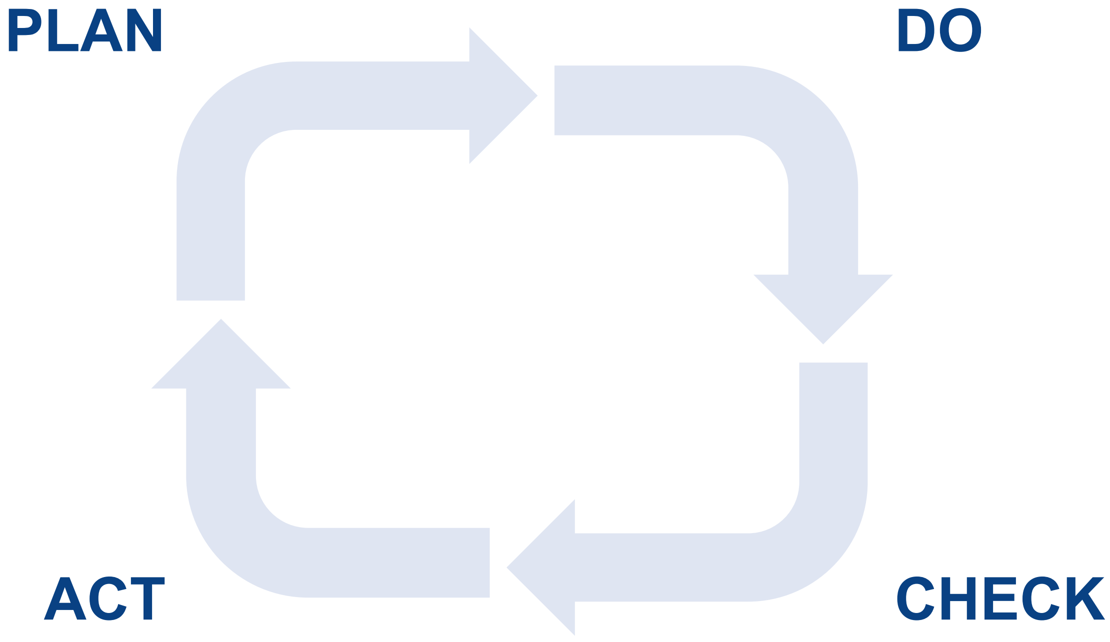
Continuously repeat the PDCA continuous improvement process.
Quality improvement never stops. Focus on the process. Understand it. Improve it.

Constantly.

PDCA.

PDCA remains an excellent quality management framework.

Use it.



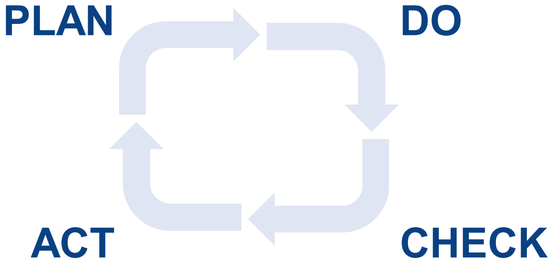
EXERCISE

Suppose you need to please your significant other. You decide to bake them some cookies. Unfortunately, the cookies came out dry and not very good tasting.

To correct this, apply PDCA.

1. Execute the PLAN phase of PDCA. What might have gone wrong?

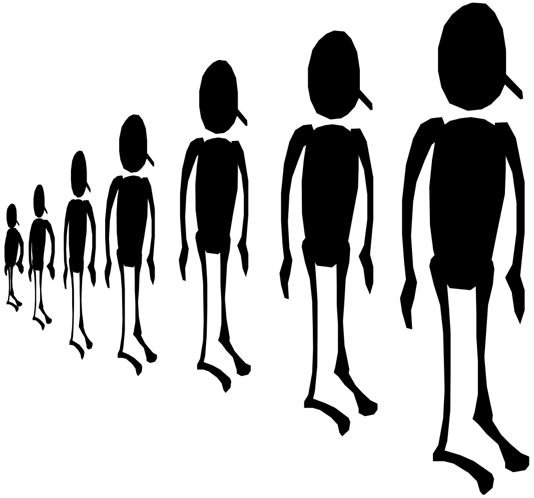
2. Lay out a task sequence to finish the DCA phases.



Organizational Quality Management

Suppose you discover a cause of a defect occurring in manufacturing, and come up with a solution to prevent it from ever occurring.

How do you deploy this into production so the workers doing the tasks adopt it?



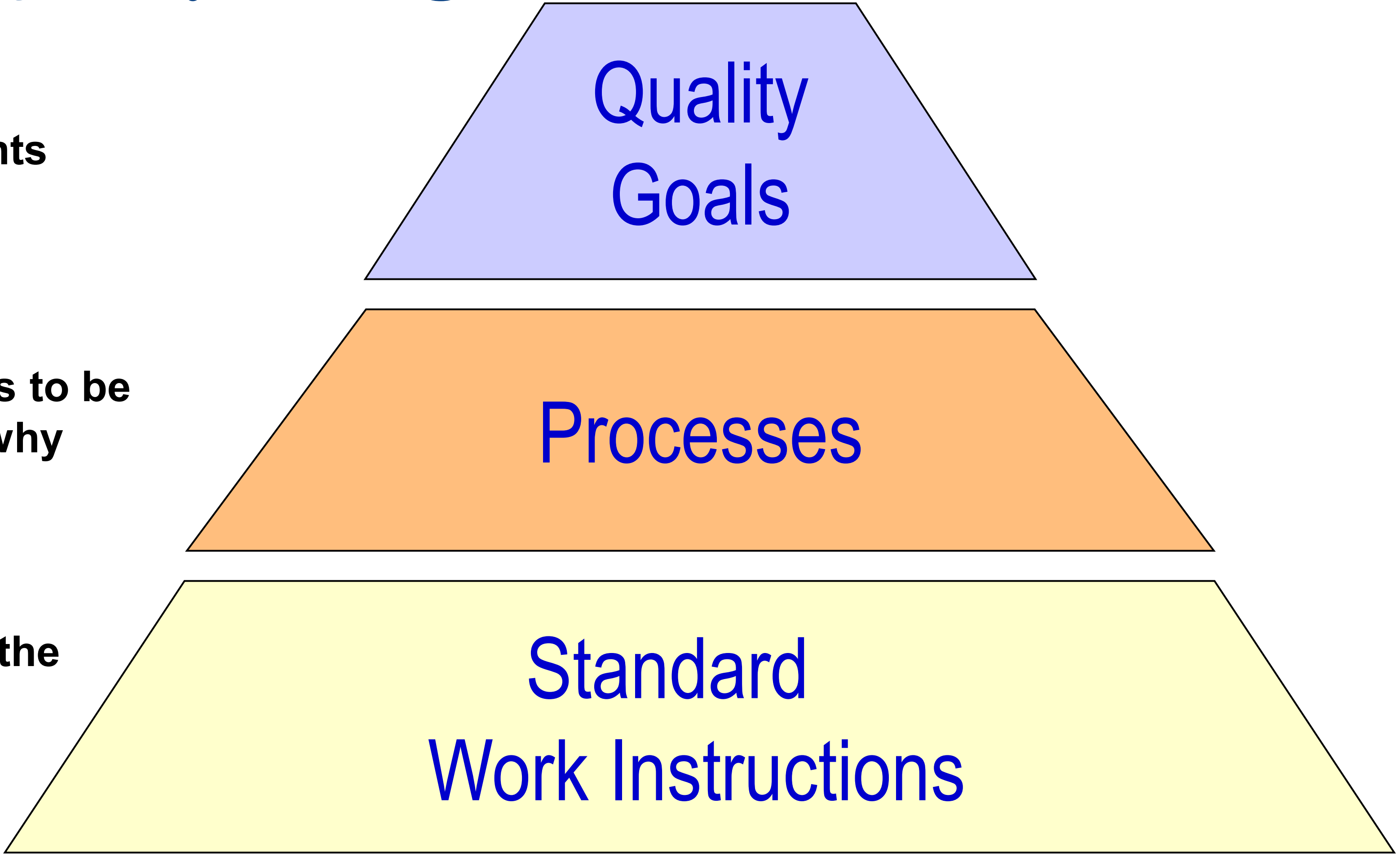
Quality management is the means by which an organization ensures quality.

Quality Management

Goals and requirements

What needs to be done and why

How to do the processes



Undocumented Quality Management

Often companies will manage quality using ad-hoc methods and procedures

- Dependent on the skills of individuals. No organizational learning
- Unclear dependencies of quality objectives on particular critical processes

The result is sporadic quality levels and random occasional serious quality issues.

Outline

PDCA

IS09000

Six Sigma

Managing Quality Internationally

Suppose you are the CEO of a firm that executes manufacturing using remote 3rd party foreign manufacturer.

How do you ensure each unit
they build meets all requirements?

Zero defects?

Delights the customer?

You don't want to run your supplier's business.

You want them to have a quality focus, to be doing PDCA and continuous improvement.

ISO9001 Quality Management Standard

To produce high quality systems, you must ensure consistency and accuracy of all work processes of your suppliers.

To achieve this consistently across the supply chain, the ISO-9000 Quality Process standards was created.

What’s in it?

What does ISO9000 certification mean?

ISO 9001

5. painos 1 (1 + 76)

SFS/ICS 03.120.10; 03.100.02; 03.100.03

Korvaa standardit SFS-EN ISO 9001:2008 ja SFS-EN ISO 9001/AC:2009

Replaces the standards SFS-EN ISO 9001:2008 and SFS-EN ISO 9001/AC:2009

Ristiriitatapauksissa pätee englanninkielinen teksti.
Suomenkielisen käännöksen päivämäärä 2015-11-05

In case of interpretation disputes the English text applies.
Date of translation into Finnish 2015-11-05

Quality management systems. Requirements (ISO 9001:2015)

Tämä standardi sisältää eurooppalaisen standardin EN ISO 9001:2015 ”Quality management systems. Requirements (ISO 9001:2015)” englanninkielisen tekstin.

This standard consists of the English text of the European Standard EN ISO 9001:2015 ”Quality management systems. Requirements (ISO 9001:2015)”.

Standardi sisältää myös englanninkielisen tekstin suomenkielisen käännöksen.

The Standard also contains a Finnish translation of the English text.

Eurooppalainen standardi EN ISO 9001:2015 on vahvistettu suomalaiseksi kansalliseksi standardiksi.

The European Standard EN ISO 9001:2015 has the status of a Finnish national standard.

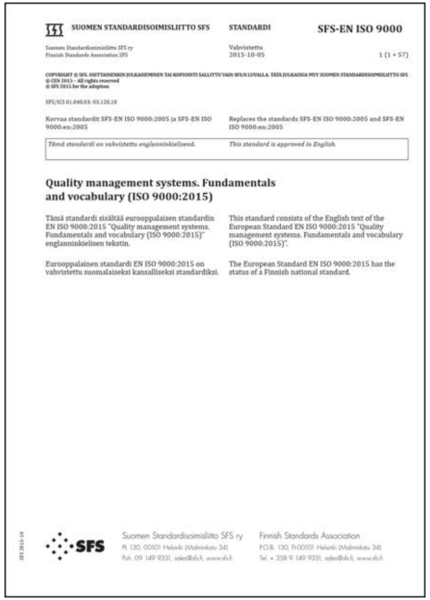
ISO 9000 Standards

ISO 9000 Quality management systems. Fundamentals and vocabulary

ISO 9001 Quality management systems. Requirements

ISO 9002 Quality management systems. Guidelines for the application of ISO 9001

ISO 9004 Managing for the sustained success of an organization – A quality management approach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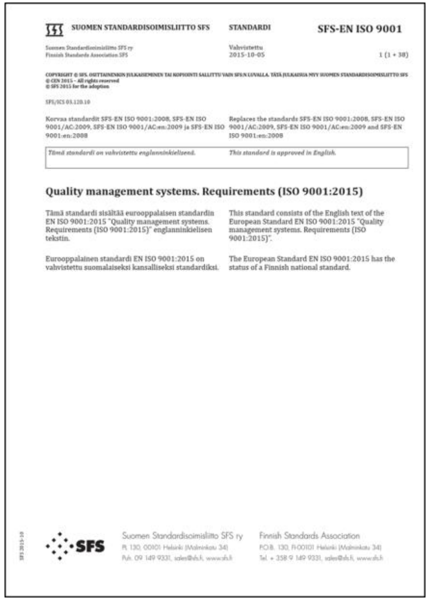
Quality management systems. Fundamentals and vocabulary (ISO 9000:2015)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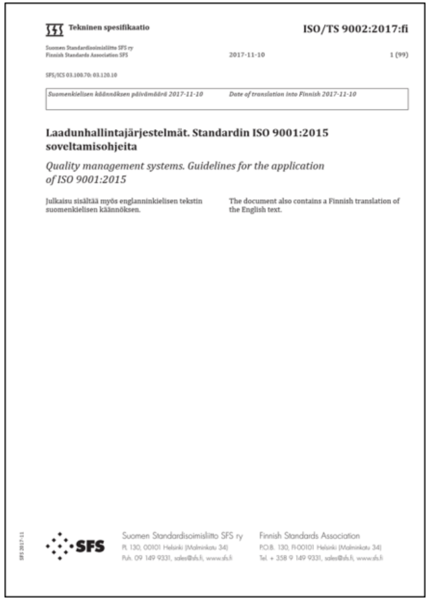
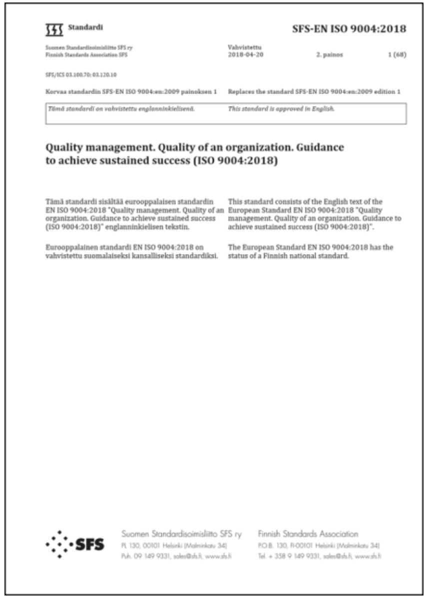
Quality management systems. Requirements (ISO 9001:2015)


Tekninen spesifikaatio ISO/TS 9002:2017:fi

Laadunhallintajärjestelmät. Standardin ISO 9001:2015 soveltamisohjeita

Quality management systems. Guidelines for the application of ISO 9001:2015


Standardi SFS-EN ISO 9004:2018

Quality management. Quality of an organization. Guidance to achieve sustained success (ISO 9004:2018)

ISO9000: Quality Principles

The ISO9000 quality management principles are:

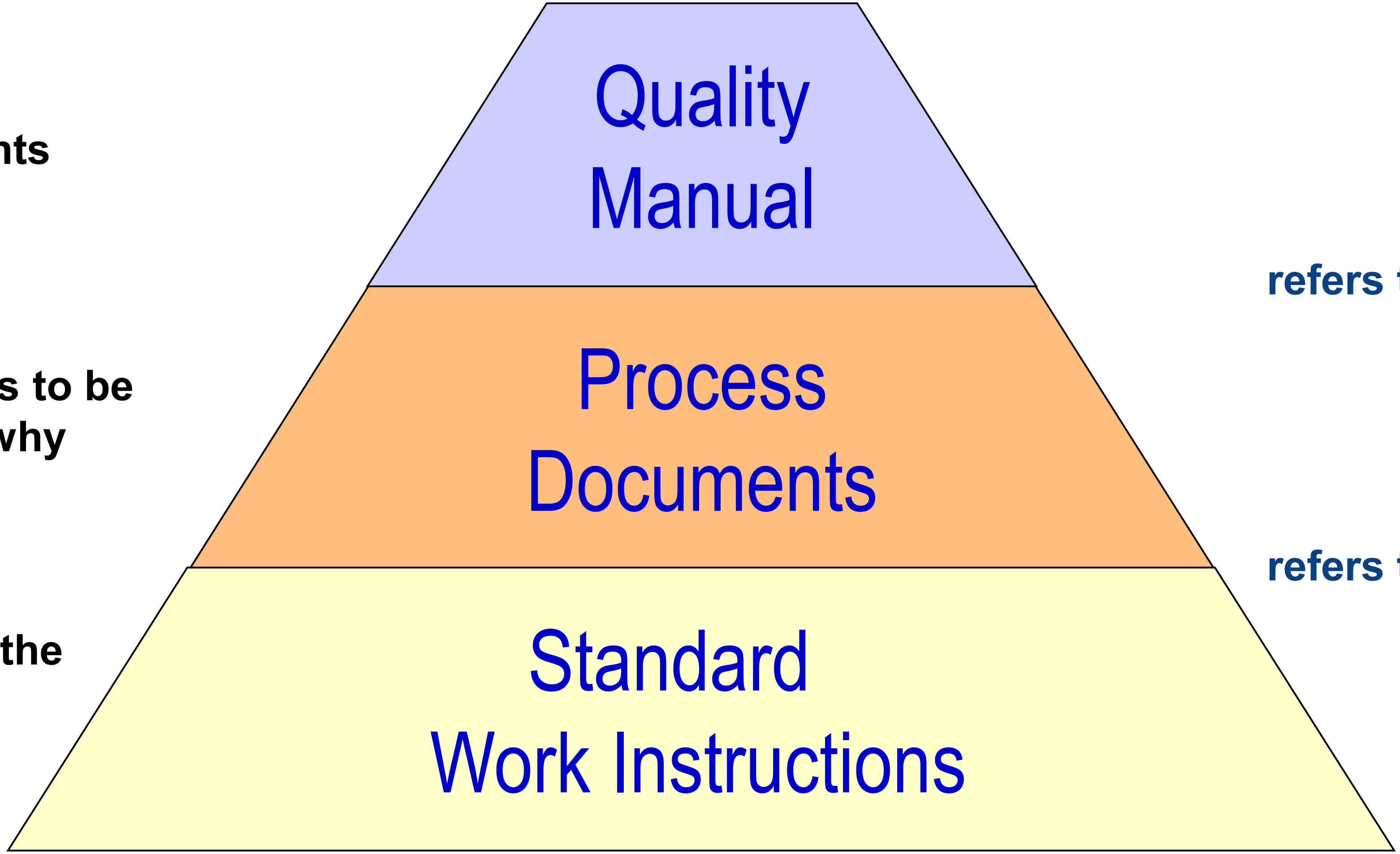
- customer focus;
- leadership;
- engagement of people;
- process approach;
- improvement;
- evidence-based decision making;
- relationship management.

ISO 9001: Documented Information

Goals and requirements

What needs to be done and why

How to do the processes



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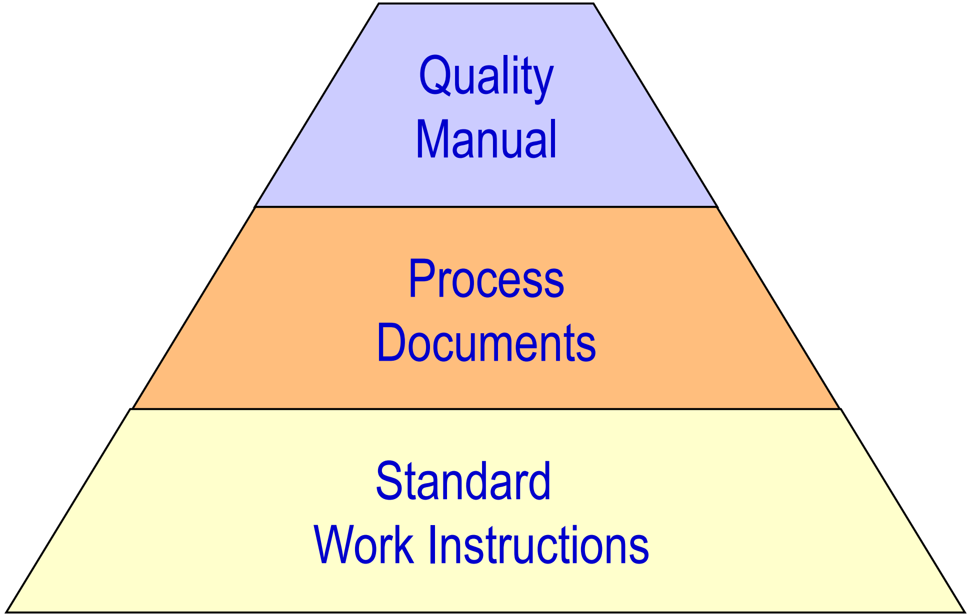
Quality Management System

The combination of these documented goals, manuals, processes and standard work is the ***quality management system*** of a company.

The set of business processes to consistent deliver customer satisfaction and high quality.



ISO9000 concerns a company's Quality Management System.

A company's quality management System can be certified as ISO9000 compliant. Or not.

Quality Manual

A ***quality manual*** is a document that explains how quality is managed at an organization.

What the quality goals are.

Who has responsibility.

What resources are used.

How quality management is organized.

What processes are used.

Quality Process Documents

Quality ***process documents*** are descriptions of how quality is managed within particular organizational functions.

- Manufacturing plants
- Suppliers
- Research and Development
- Customer Sales

The process documents explain how each of these groups measure, maintain and improve quality. Prevent mistakes, errors, defects, and how they improve the customer focus.

Standard Work Instructions

Standard Work Instructions are the detailed written instructions to do the tasks described in the process documents.

- How to perform the manufacturing process tasks
- How to detect when the process has changed for some reason
- What to do to correct the process if it has changed
- Inspection procedures
- Maintenance procedures

All the instructions needed to execute tasks to ensure the quality goals.

ISO 9000 Quality Manual Sections

1 Scope

2 Normative references

3 Terms and definitions

4 Context of the organization

5 Leadership

6 Planning

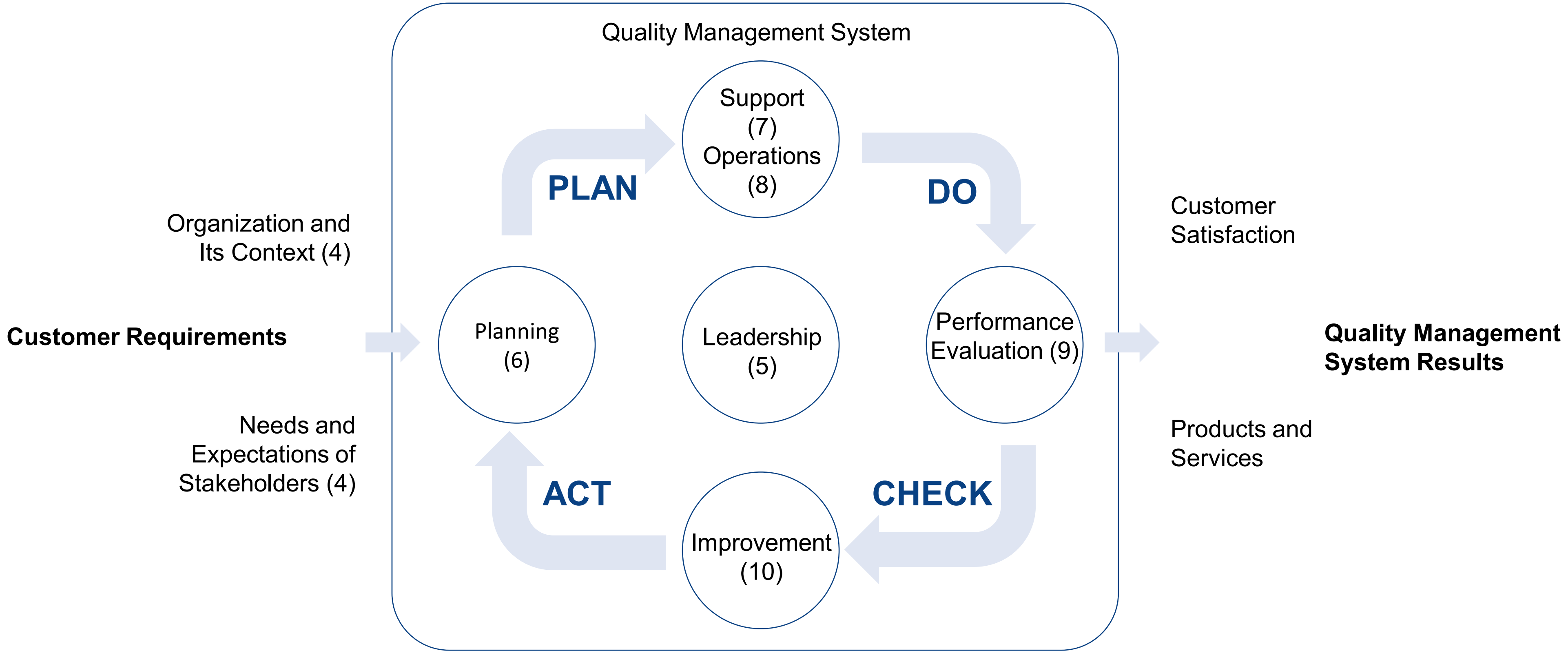
7 Support

8 Operation

9 Performance evaluation

10 Improvement

PDCA Basis



Quality Management System
Support (7) Operations (8)
PLAN
DO
Organization and Its Context (4)
Customer Satisfaction
Customer Requirements
Planning (6)
Leadership (5)
Performance Evaluation (9)
Quality Management System Results
Needs and Expectations of Stakeholders (4)
Products and Services
ACT
CHECK
Improvement (10)

4 Context of the organization

Section 4 describes the organization.

4.1 Understanding the organization and its context

4.2 Understanding the needs and expectations of interested parties

4.3 Determining the scope of the quality management system

4.4 Quality management system and its processes

4 Context of the organization

4.1 Understanding the organization and its context

The organization shall determine external and internal issues that are relevant to its purpose and its strategic direction and that affect its ability to achieve the intended result(s) of its quality management system.

The organization shall monitor and review information about these external and internal issues.

NOTE 1 Issues can include positive and negative factors or conditions for consideration.

NOTE 2 Understanding the external context can be facilitated by considering issues arising from legal, technological, competitive, market, cultural, social and economic environments, whether international, national, regional or local.

NOTE 3 Understanding the internal context can be facilitated by considering issues related to values, culture, knowledge and performance of the organization.

4.2 Understanding the needs and expectations of interested parties

Due to their effect or potential effect on the organization's ability to consistently provide products and services that meet customer and applicable statutory and regulatory requirements, the organization shall determine:

a) the interested parties that are relevant to the quality management system;

b) the requirements of these interested parties that are relevant to the quality management system.

The organization shall monitor and review information about these interested parties and their relevant requirements.

4.3 Determining the scope of the quality management system

The organization shall determine the boundaries and applicability of the quality management system to establish its scope.

When determining this scope, the organization shall consider:

a) the external and internal issues referred to in 4.1;

b) the requirements of relevant interested parties referred to in 4.2;

c) the products and services of the organization.

The organization shall apply all the requirements of this International Standard if they are applicable within the determined scope of its quality management system.

The scope of the organization's quality management system shall be available and be maintained as documented information. The scope shall state the types of products and services covered, and provide justification for any requirement of this International Standard that the organization determines is not applicable to the scope of its quality management system.

Conformity to this International Standard may only be claimed if the requirements determined as not being applicable do not affect the organization's ability or responsibility to ensure the conformity of its products and services and the enhancement of customer satisfaction.

4.4 Quality management system and its processes

4.4.1 The organization shall establish, implement, maintain and continually improve a quality management system, including the processes needed and their interactions, in accordance with the requirements of this International Standard.

The organization shall determine the processes needed for the quality management system and their application throughout the organization, and shall:

a) determine the inputs required and the outputs expected from these processes;

b) determine the sequence and interaction of these processes;

c) determine and apply the criteria and methods (including monitoring, measurements and related performance indicators) needed to ensure the effective operation and control of these processes;

d) determine the resources needed for these processes and ensure their availability;

e) assign the responsibilities and authorities for these processes;

f) address the risks and opportunities as determined in accordance with the requirements of 6.1;

g) evaluate these processes and implement any changes needed to ensure that these processes achieve their intended results;

h) improve the processes and the quality management system.

4.4.2 To the extent necessary, the organization shall:

a) maintain documented information to support the operation of its processes;

b) retain documented information to have confidence that the processes are being carried out as planned.

5 Leadership

Section 5 describes why quality is managed, particularly the **quality policy**.

5.1 Leadership and commitment

5.2 Policy

5.3 Organizational roles, responsibilities and authorities

5 Leadership

5.1 Leadership and commitment

5.1.1 General

Top management shall demonstrate leadership and commitment with respect to the quality management system by:

a) taking accountability for the effectiveness of the quality management system;

b) ensuring that the quality policy and quality objectives are established for the quality management system and are compatible with the context and strategic direction of the organization;

c) ensuring the integration of the quality management system requirements into the organization's business processes;

d) promoting the use of the process approach and risk-based thinking;

e) ensuring that the resources needed for the quality management system are available;

f) communicating the importance of effective quality management and of conforming to the quality management system requirements;

g) ensuring that the quality management system achieves its intended results;

h) engaging, directing and supporting persons to contribute to the effectiveness of the quality management system;

i) promoting improvement;

j) supporting other relevant management roles to demonstrate their leadership as it applies to their areas of responsibility.

NOTE Reference to "business" in this International Standard can be interpreted broadly to mean those activities that are core to the purposes of the organization's existence, whether the organization is public, private, for profit or not for profit.

5.1.2 Customer focus

Top management shall demonstrate leadership and commitment with respect to customer focus by ensuring that:

a) customer and applicable statutory and regulatory requirements are determined, understood and consistently met;

b) the risks and opportunities that can affect conformity of products and services and the ability to enhance customer satisfaction are determined and addressed;

c) the focus on enhancing customer satisfaction is maintained.

5.2 Policy

5.2.1 Establishing the quality policy

Top management shall establish, implement and maintain a quality policy that:

a) is appropriate to the purpose and context of the organization and supports its strategic direction;

b) provides a framework for setting quality objectives;

c) includes a commitment to satisfy applicable requirements;

d) includes a commitment to continual improvement of the quality management system.

5.2.2 Communicating the quality policy

The quality policy shall:

a) be available and be maintained as documented information;

b) be communicated, understood and applied within the organization;

c) be available to relevant interested parties, as appropriate.

5.3 Organizational roles, responsibilities and authorities

Top management shall ensure that the responsibilities and authorities for relevant roles are assigned, communicated and understood within the organization.

Top management shall assign the responsibility and authority for:

a) ensuring that the quality management system conforms to the requirements of this International Standard;

b) ensuring that the processes are delivering their intended outputs;

c) reporting on the performance of the quality management system and on opportunities for improvement (see 10.1), in particular to top management;

d) ensuring the promotion of customer focus throughout the organization;

e) ensuring that the integrity of the quality management system is maintained when changes to the quality management system are planned and implemented.

6 Planning

Section 6 describes how quality issues are identified.

6.1 Actions to address risks and opportunities

6.2 Quality objectives and planning to achieve them

6.3 Planning of changes

6 Planning

6.1 Actions to address risks and opportunities

6.1.1 When planning for the quality management system, the organization shall consider the issues referred to in 4.1 and the requirements referred to in 4.2 and determine the risks and opportunities that need to be addressed to:

a) give assurance that the quality management system can achieve its intended result(s);

b) enhance desirable effects;

c) prevent, or reduce, undesired effects;

d) achieve improvement.

6.1.2 The organization shall plan:

a) actions to address these risks and opportunities;

b) how to:

1) integrate and implement the actions into its quality management system processes (see 4.4);

2) evaluate the effectiveness of these actions.

Actions taken to address risks and opportunities shall be proportionate to the potential impact on the conformity of products and services.

NOTE 1 Options to address risks can include avoiding risk, taking risk in order to pursue an opportunity, eliminating the risk source, changing the likelihood or consequences, sharing the risk, or retaining risk by informed decision.

NOTE 2 Opportunities can lead to the adoption of new practices, launching new products, opening new markets, addressing new customers, building partnerships, using new technology and other desirable and viable possibilities to address the organization's or its customers' needs.

6.2 Quality objectives and planning to achieve them

6.2.1 The organization shall establish quality objectives at relevant functions, levels and processes needed for the quality management system.

The quality objectives shall:

a) be consistent with the quality policy;

b) be measurable;

c) take into account applicable requirements;

d) be relevant to conformity of products and services and to enhancement of customer satisfaction;

e) be monitored;

f) be communicated;

g) be updated as appropriate.

The organization shall maintain documented information on the quality objectives.

6.2.2 When planning how to achieve its quality objectives, the organization shall determine:

a) what will be done;

b) what resources will be required;

c) who will be responsible;

d) when it will be completed;

e) how the results will be evaluated.

6.3 Planning of changes

When the organization determines the need for changes to the quality management system, the changes shall be carried out in a planned manner (see 4.4).

7 Support

Section 7 describes quality management resources in the organization.

7.1 Resources

7.2 Competence

7.3 Awareness

7.4 Communication

7.5 Documented information

7 Support

7.1 Resources

7.1.1 General

The organization shall determine and provide the resources needed for the establishment, implementation, maintenance and continual improvement of the quality management system.

The organization shall consider:

a) the capabilities of, and constraints on, existing internal resources;

b) what needs to be obtained from external providers.

7.1.2 People

The organization shall determine and provide the persons necessary for the effective implementation of its quality management system and for the operation and control of its processes.

7.1.3 Infrastructure

The organization shall determine, provide and maintain the infrastructure necessary for the operation of its processes and to achieve conformity of products and services.

NOTE Infrastructure can include:

a) buildings and associated utilities;

b) equipment, including hardware and software;

c) transportation resources;

d) information and communication technology.

7.1.4 Environment for the operation of processes

The organization shall determine, provide and maintain the environment necessary for the operation of its processes and to achieve conformity of products and services.

NOTE A suitable environment can be a combination of human and physical factors, such as:

a) social (e.g. non-discriminatory, calm, non-confrontational);

b) psychological (e.g. stress-reducing, burnout prevention, emotionally protective);

c) physical (e.g. temperature, heat, humidity, light, airflow, hygiene, noise).

These factors can differ substantially depending on the products and services provided.

8 Operation

Section 8 describes how quality is managed.

8.1 Operational planning and control

8.2 Requirements for products and services

8.3 Design and development of products and services

8.4 Control of externally provided processes, products and services

8.5 Production and service provision

8.6 Release of products and services

8.7 Control of nonconforming outputs

8 Operation

8.1 Operational planning and control

The organization shall plan, implement and control the processes (see 4.4) needed to meet the requirements for the provision of products and services, and to implement the actions determined in Clause 6, by:

a) determining the requirements for the products and services;

b) establishing criteria for:

1) the processes;

2) the acceptance of products and services;

c) determining the resources needed to achieve conformity to the product and service requirements;

d) implementing control of the processes in accordance with the criteria;

e) determining, maintaining and retaining documented information to the extent necessary:

1) to have confidence that the processes have been carried out as planned;

2) to demonstrate the conformity of products and services to their requirements.

The output of this planning shall be suitable for the organization's operations.

The organization shall control planned changes and review the consequences of unintended changes, taking action to mitigate any adverse effects, as necessary.

The organization shall ensure that outsourced processes are controlled (see 8.4).

8.2 Requirements for products and services

8.2.1 Customer communication

Communication with customers shall include:

a) providing information relating to products and services;

b) handling enquiries, contracts or orders, including changes;

c) obtaining customer feedback relating to products and services, including customer complaints;

d) handling or controlling customer property;

e) establishing specific requirements for contingency actions, when relevant.

8.2.2 Determining the requirements for products and services

When determining the requirements for the products and services to be offered to customers, the organization shall ensure that:

a) the requirements for the products and services are defined, including:

1) any applicable statutory and regulatory requirements;

2) those considered necessary by the organization;

b) the organization can meet the claims for the products and services it offers.

8.2.3 Review of the requirements for products and services

8.2.3.1 The organization shall ensure that it has the ability to meet the requirements for products and services to be offered to customers. The organization shall conduct a review before committing to supply products and services to a customer, to include:

9 Performance evaluation

Section 9 describes how quality is monitored.

9.1 Monitoring, measurement, analysis and evaluation

9.2 Internal audit

9.3 Management review

9 Performance evaluation

9.1 Monitoring, measurement, analysis and evaluation

9.1.1 General

The organization shall determine:

a) what needs to be monitored and measured;

b) the methods for monitoring, measurement, analysis and evaluation needed to ensure valid results;

c) when the monitoring and measuring shall be performed;

d) when the results from monitoring and measurement shall be analysed and evaluated.

The organization shall evaluate the performance and the effectiveness of the quality management system.

The organization shall retain appropriate documented information as evidence of the results.

9.1.2 Customer satisfaction

The organization shall monitor customers' perceptions of the degree to which their needs and expectations have been fulfilled. The organization shall determine the methods for obtaining, monitoring and reviewing this information.

NOTE Examples of monitoring customer perceptions can include customer surveys, customer feedback on delivered products and services, meetings with customers, market-share analysis, compliments, warranty claims and dealer reports.

9.1.3 Analysis and evaluation

The organization shall analyse and evaluate appropriate data and information arising from monitoring and measurement.

The results of analysis shall be used to evaluate:

a) conformity of products and services;

b) the degree of customer satisfaction;

c) the performance and effectiveness of the quality management system;

d) if planning has been implemented effectively;

e) the effectiveness of actions taken to address risks and opportunities;

f) the performance of external providers;

g) the need for improvements to the quality management system.

NOTE Methods to analyse data can include statistical techniques.

9.2 Internal audit

9.2.1 The organization shall conduct internal audits at planned intervals to provide information on whether the quality management system:

a) conforms to:

10 Improvement

Section 10 describes how quality

is continuously improved.

10.1 General

10.2 Nonconformity and corrective action

10.3 Continual improvement

10 Improvement

10.1 General

The organization shall determine and select opportunities for improvement and implement any necessary actions to meet customer requirements and enhance customer satisfaction.

These shall include:

a) improving products and services to meet requirements as well as to address future needs and expectations;

b) correcting, preventing or reducing undesired effects;

c) improving the performance and effectiveness of the quality management system.

NOTE Examples of improvement can include correction, corrective action, continual improvement, breakthrough change, innovation and re-organization.

10.2 Nonconformity and corrective action

10.2.1 When a nonconformity occurs, including any arising from complaints, the organization shall:

a) react to the nonconformity and, as applicable:

  1) take action to control and correct it;

  2) deal with the consequences;

b) evaluate the need for action to eliminate the cause(s) of the nonconformity, in order that it does not recur or occur elsewhere, by:

  1) reviewing and analysing the nonconformity;

  2) determining the causes of the nonconformity;

  3) determining if similar nonconformities exist, or could potentially occur;

c) implement any action needed;

d) review the effectiveness of any corrective action taken;

e) update risks and opportunities determined during planning, if necessary;

f) make changes to the quality management system, if necessary.

Corrective actions shall be appropriate to the effects of the nonconformities encountered.

10.2.2 The organization shall retain documented information as evidence of:

a) the nature of the nonconformities and any subsequent actions taken;

b) the results of any corrective action.

10.3 Continual improvement

The organization shall continually improve the suitability, adequacy and effectiveness of the quality management system.

The organization shall consider the results of analysis and evaluation, and the outputs from management review, to determine if there are needs or opportunities that shall be addressed as part of continual improvement.

Example

ABB Automation

Quality Manual

Table of Contents:

Quality Manual

4.0 Quality Management System

POLICY:

PURPOSE:

To outline the Quality Management System used at ABB Inc. Allentown site.

RESPONSIBILITIES:

DESCRIPTION:

4.1 General Requirements:

Process Approach:

The ABB Inc. Allentown site takes a process approach to quality management. The status and adequacy of the Quality Assurance Program and Quality Manual is reviewed by the General Manager and his staff. The Quality Assurance Program is reviewed regularly by several means, some of which include:

Management Review - Review of Quality Policy and objectives (quarterly meetings) by the management staff and the Quality Assurance Supervisor. The management staff is ultimately responsible for:

a) identifying the processes needed for the quality management system and their application throughout the ABB Inc. Allentown site,

b) determining the sequence and interaction of processes,

c) determining criteria and methods needed to ensure that both the operation and control of processes are effective,

d) ensuring the availability of resources and information necessary to support the operation and monitoring of processes,

e) monitoring, measuring and analyzing processes and

f) implementing actions necessary to achieve planned results and continual improvement of processes.

Quality Manual

Quality Council - Review of audit results, inspections, warranty information and a review of both internal and external reported conditions perceived to be adverse to quality. Corrective action assignments and follow up are reviewed at each meeting. (meeting every 2 weeks).

Non-conformance Material Report Review (NCMR) - Review of rejected material with reports distributed to appropriate management or supervision for review and corrective action.

External Audits - Review of audit findings and corrective action identified from customer or supplier audits (as required).

Internal Audits - Review of audit findings and corrective action identified during internal Quality System audits.

Product Integrity Committee - Review of product performance reports (as required). Records of meeting minutes, documentation or any resulting Product Advisory Letters (PAL) are maintained in files in the QA department.

Training - Quality Assurance training for Supervisors is accomplished by seminars and, or on-the-job training to cover those aspects of Quality Assurance and Quality Control programs applicable to fulfilling their responsibilities. Quality Assurance and Quality Control training for persons performing quality functions is primarily accomplished by on-the-job training. A log of QA/QC personnel is maintained to provide evidence of personnel qualifications.

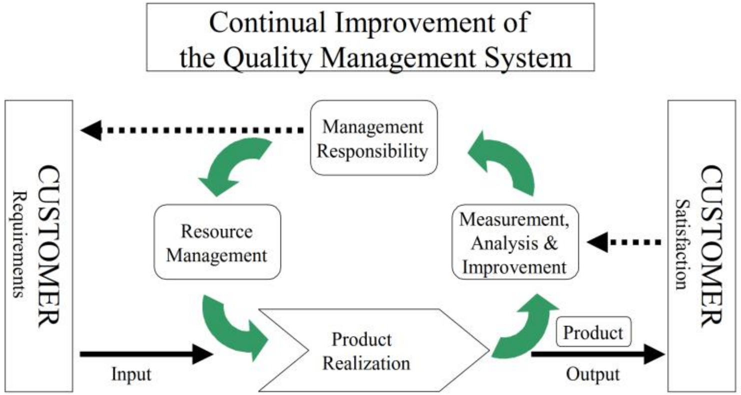
The ABB Inc. Allentown site may choose to outsource processes that affect product conformity with requirements. If these types of processes are outsourced, the ABB Inc. Allentown site will oversee and maintain control over such processes.

It is the direction of the management to establish, implement and continually maintain an effective quality management system that complies with the ISO 9001:2000 international standard.

The QMS is implemented in practice by a process management approach.
This pictorial representation and the table below depicts how the organization intends to translate requirements of the ISO 9001:2000 into its business processes.

ISO 9001:2000
Model of a Process-based Quality Management System

Quality Manual



ISO 9001:2000	ABB Business Processes (Interactions of Activities)	
Customer Requirements	Sell, Project/Order Execution	Identify customer requirements; set plans to capture customer needs and ABB opportunities. Project management process
Management Responsibility	Commitment to develop business processes	Development of the QMS, deploy quality policy and objectives that are commensurate with corporate objectives and customer needs.
Product Realization	Project execution, product delivery, supply and demand chain management	Project management, Product management, Plant design, software production, hardware assembly, tests, service and support Execution and delivery of ordered product Supply management, purchasing control and logistic support

Quality Manual

Measurement, analysis and improvement	Develop performance indicators, nonconformance control, corrective and preventive actions, Internal audit and reviews	Identify control areas in the business chain and develop process performance indicators (set quality objectives), ensure conformity of products during processes; Controls, corrections on nonconformities and prevention of potential problems; conduct product, process, internal audits and QMS reviews.
Continual Improvement of the QMS	Periodic review of the QMS and processes	Identify areas for improvements in all of the above
Customer Satisfaction	Sales process, project execution, service and support, CCRP, SupportLine, survey process	Alignment with ABB strategy of customer focus in customer relationship management: Deliver products that meet or exceed customer's satisfaction. Seek customer feedback in all processes where communication with customer takes place. Resolve customer issues and complaints.

4.2 Documentation:

4.2.1 General:

The quality system documentation includes the following:

Quality Manual:
Describes the scope of the Quality Management System at the ABB Allentown site.

Quality Policy:
Describes the organization's Quality Policy and provides a framework for establishing and reviewing quality objectives.

Quality Assurance Procedures (QAPs):
A compilation of procedures and instructions for all aspects of quality assurance programs, 10CFR 50 Appendix B, ISO 9001, quality control, test and inspection.

Engineering Procedures (ESPs):
Procedures for development of new products, design control, etc., including documentation for product qualification for Class 1E applications, mechanical and electrical standards.

Quality Manual

It is the responsibility of the area manager or supervisor to maintain the records in his/her area and review his/her files once each year for disposition of records.

Quality Assurance data and records for the products are submitted to the customer in accordance with their purchase order requirements. For Class 1E products a copy of this information is placed in the QA order file. If a customer desires to be notified prior to the destruction of these records, a note to this effect can be attached to the sales order file.

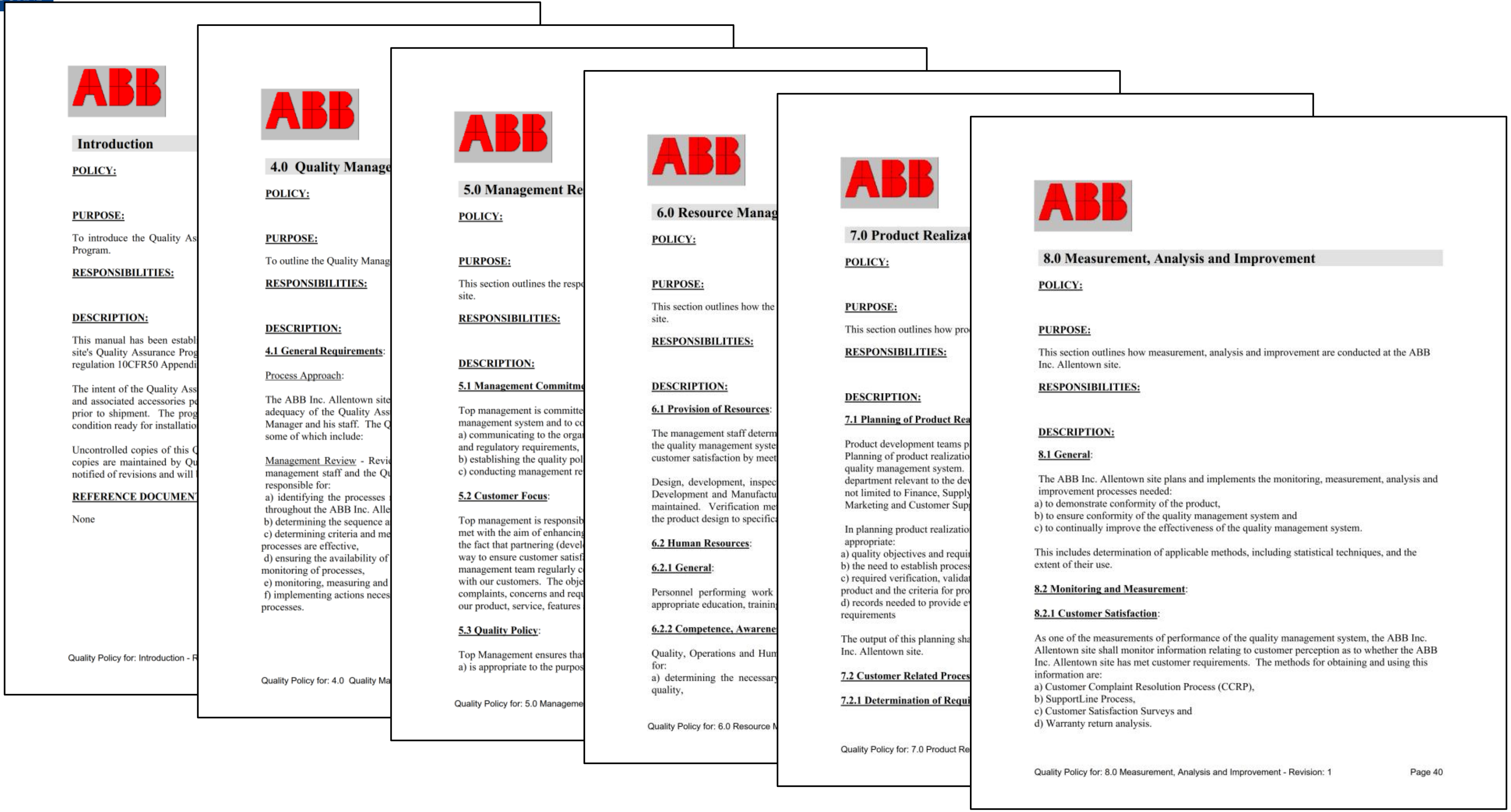
Class 1E records have a 40 year retention as identified in the Quality Assurance record index file.

Quality Assurance records other than Class 1E records do not require long term storage and protection. These records are retained for internal use such as warranty records, etc. The product itself carries permanent inspection and test verification in the form of Quality Assurance stamps. Other records such as sales orders, certificates, and/or required test data are delivered to the customer with the shipment of material as requested.

REFERENCE DOCUMENTATION:

QAP 2.1 -- QUALITY ASSURANCE PROGRAM
QAP 2.2 -- QUALITY PROGRAM REQUIREMENTS
QAP 2.3 -- QA TRAINING
QAP 2.4 -- ANNUAL REVIEW OF QUALITY ASSURANCE PROCEDURES
QAP 6.1 -- PREPARATION AND CONTROL OF QUALITY ASSURANCE PROCEDURES
QAP 6.2 -- ENGINEERING RELEASE PROCESSING
QAP 6.3 -- QUALITY ASSURANCE MANUAL - CONTROLLED COPIES
QAP 6.4 -- CONTROL OF INDUSTRY STANDARDS REFERENCE DOCUMENTS
QAP 17.1 -- RETENTION AND STORAGE OF QUALITY RECORDS
ESP-201 -- ENGINEERING CHANGE DATABASE
ESP-202 -- ENGINEERING DRAWING RELEASE
ESP-203 -- ENGINEERING DRAWING RELEASE

Referenced Process Documents



ABB

8.0 Measurement, Analysis and Improvement

POLICY:

PURPOSE:

This section outlines how measurement, analysis and improvement are conducted at the ABB Inc. Allentown site.

RESPONSIBILITIES:

DESCRIPTION:

8.1 General:

The ABB Inc. Allentown site plans and implements the monitoring, measurement, analysis and improvement processes needed:
a) to demonstrate conformity of the product,
b) to ensure conformity of the quality management system and
c) to continually improve the effectiveness of the quality management system.

This includes determination of applicable methods, including statistical techniques, and the extent of their use.

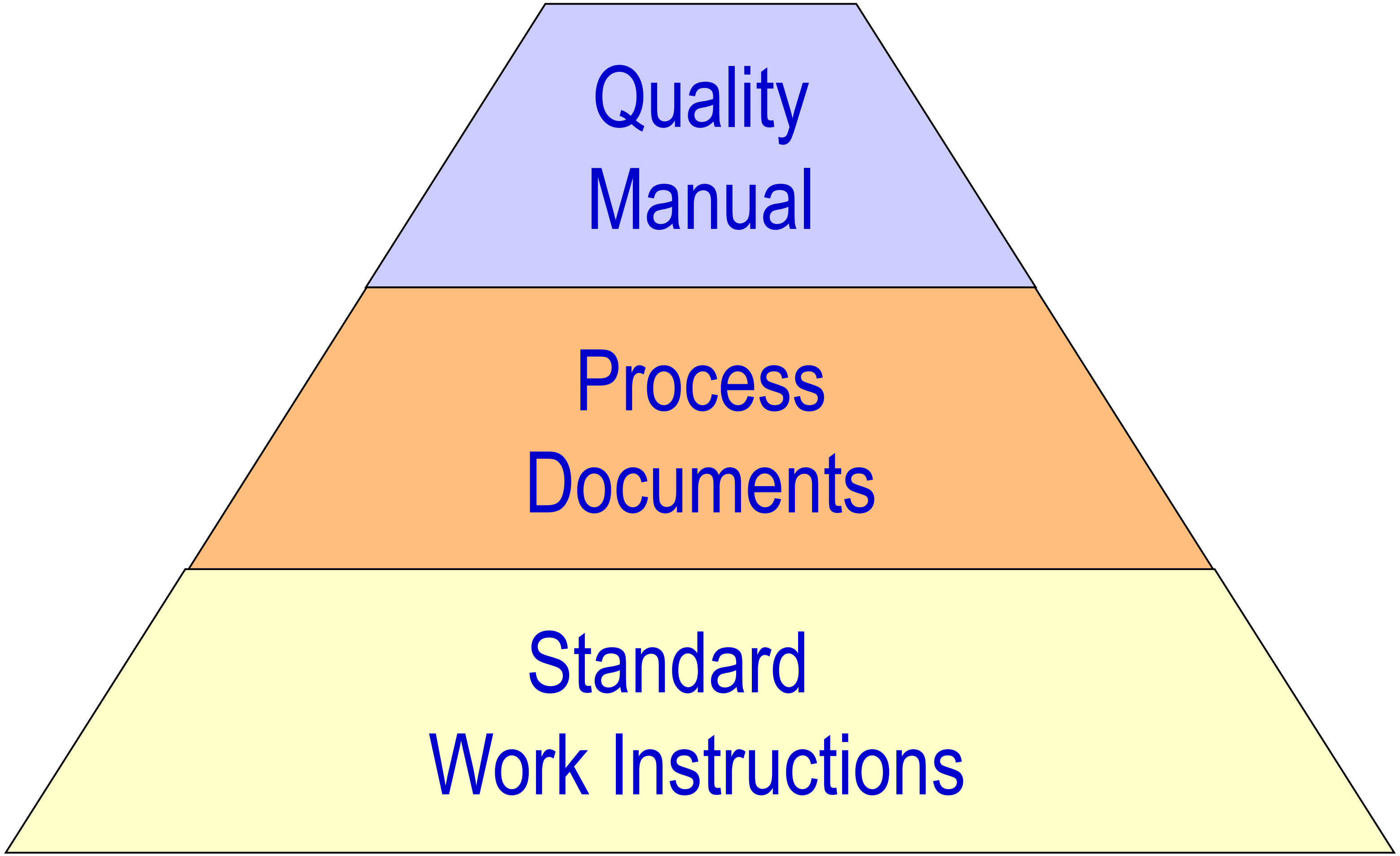
8.2 Monitoring and Measurement:

8.2.1 Customer Satisfaction:

As one of the measurements of performance of the quality management system, the ABB Inc. Allentown site shall monitor information relating to customer perception as to whether the ABB Inc. Allentown site has met customer requirements. The methods for obtaining and using this information are:
a) Customer Complaint Resolution Process (CCRP),
b) SupportLine Process,
c) Customer Satisfaction Surveys and
d) Warranty return analysis.

Quality Policy for: 8.0 Measurement, Analysis and Improvement - Revision: 1 Page 40

ISO 9001: Documented Information



The Quality Policy

The **quality policy** is the written expression of the quality directive from the executive management. The “why”.

The policy is written within the quality manual (Section 5).

It is important as the overall communication to customers, employees, and other stakeholders as a legal statement on quality goals.

The Quality Policy

5.2.1 Establishing the quality policy

Top management shall establish, implement and maintain a quality policy that:

a) is appropriate to the purpose of the organization and supports its strategic direction;

b) provides a framework for setting quality objectives;

c) includes a commitment to satisfy applicable requirements;

d) includes a commitment to continual improvement of the quality management system.

5.2.2 Communicating the quality policy

The quality policy shall:

a) be available and be maintained as documented information;

b) be communicated, understood and applied within the organization;

c) be available to relevant interested parties, as appropriate.

Quality Policy

Allentown's Quality Vision is to be incredibly customer focused and thereby improve customer satisfaction, growth and profit for both our customers and our company.

The Substation Automation and Protection Group is committed to outrageous levels of customer service and quality. All employees strive to produce defect free products and are dedicated to continuous improvement of processes. Our Quality Objectives are determined by all employees and are approved, reviewed and maintained by the management team. Quality objectives are located in the Quality Plan.

ABB Automation

Konecranes Quality Policy

Our goal in making products is perform zero-defect work conformant with customer requirements.

We commit to supply products and services precisely as specified and on time.

We commit to clearly define each task, make sure information is communicated and understood accurately and perform each job properly the first time – every time.

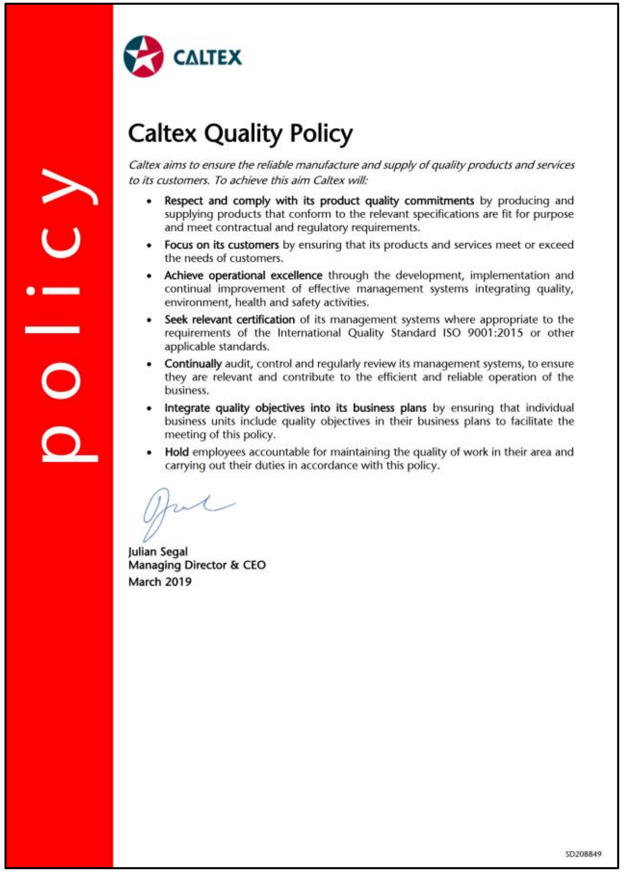
In all our products and operations the key principle is: Reliability first!

Orrcon Steel Quality Policy

At Orrcon Steel we recognise that our success is derived from a flexible approach to market opportunities and our long-term success depends on our ability to meet or exceed the needs and expectations of our customers. We also understand that the quality of our products; services and delivery performance are critical. “We'll see it through” is more than a tag line next to our brand. It is a statement that says a lot about our business and the people who work for Orrcon Steel. It is a part of our culture of being honest, ethical and reliable and it reflects our relationships both internally and externally.

Caltex

https://www.caltex.com.au/-/media/documents/caltex/corporate-governance/2019-corporate-governance-docs/caltex-quality-policy.ashx?la=en



policy

CALTEX

Caltex Quality Policy

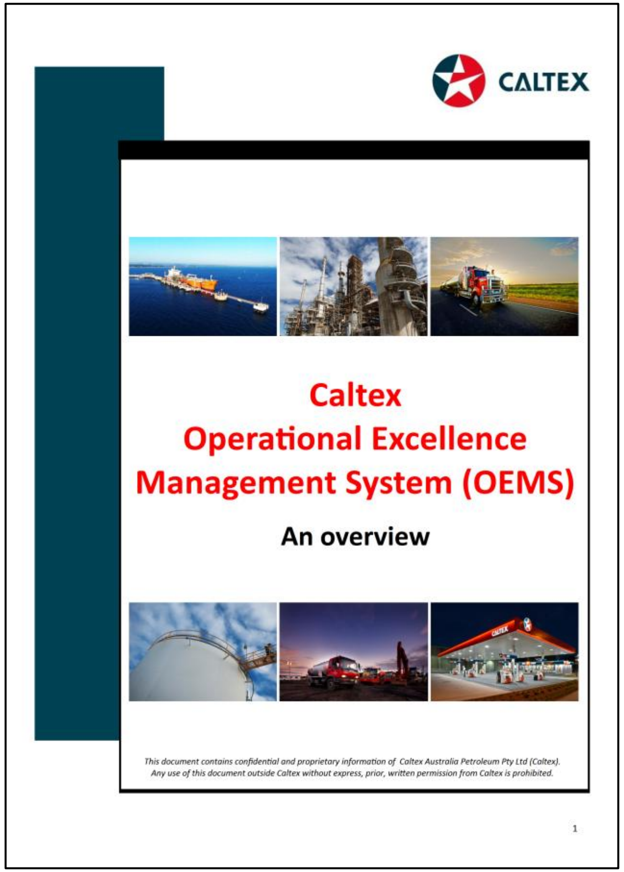
Caltex aims to ensure the reliable manufacture and supply of quality products and services to its customers. To achieve this aim Caltex will:

- **Respect and comply with its product quality commitments** by producing and supplying products that conform to the relevant specifications are fit for purpose and meet contractual and regulatory requirements.
- **Focus on its customers** by ensuring that its products and services meet or exceed the needs of customers.
- **Achieve operational excellence** through the development, implementation and continual improvement of effective management systems integrating quality, environment, health and safety activities.
- **Seek relevant certification** of its management systems where appropriate to the requirements of the International Quality Standard ISO 9001:2015 or other applicable standards.
- **Continually** audit, control and regularly review its management systems, to ensure they are relevant and contribute to the efficient and reliable operation of the business.
- **Integrate quality objectives into its business plans** by ensuring that individual business units include quality objectives in their business plans to facilitate the meeting of this policy.
- **Hold** employees accountable for maintaining the quality of work in their area and carrying out their duties in accordance with this policy.

Julian Segal
Managing Director & CEO
March 2019

SD208849

https://www.caltex.com.au/-/media/documents/caltex/environment-documents/operational-excellence/oems-booklet-sep16-final.ashx?la=en



CALTEX

Caltex
Operational Excellence
Management System (OEMS)

An overview

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1

Fonterra

https://fonterra.com/content/dam/fonterra-public-website/phase-2/new-Zealand/pdfs-docs-infographics/pdfs-and-documents/fsq-requirements-suppliers-warehousing-and-distribution.pdf

https://www.fonterra.com/nz/en/our-co-operative/governance-and-management/code-of-conduct-and-policies.html

Quality Policy Adequacy

Is it ISO9000 certified?

Does it indicate what company provides?

Does it indicate a focus on the customer?

Does it indicate a quality goal?

Does it indicate a focus on continuous improvement?

In Processes? In Staff?

Does it indicate commitment?

ISO Certification

ISO certifies inspectors and firms who then inspect and certify companies as in compliance with the standard

International Accreditation Foundation

Who says the certifying company itself is certified?

The International Accreditation Forum is a body of certifying entities who agreed to standards.

- A single worldwide program of conformity assessment which reduces risk for business and its customers by assuring them that accredited certificates may be relied upon
- ISO9000 Auditing Group
- ISO14000 Auditing Group

ISO 9000 Certification

Certification costs

Preinformation	10 hours
Quality manual check	20 hours
Planning meeting	10 hours
System audit	80 hours
TOTAL	120 hours

120 hours * 100 $/hour = $12 000

Follow-up 1…2 times / year = $3000

ISO 9000 Certification

bsi.

Certificate of Registration

QUALITY MANAGEMENT SYSTEM - ISO 9001:2015

This is to certify that:

BlueScope
Australian Steel Products Manufacturing
a division of
BlueScope Steel Limited and
BlueScope Steel (AIS) Pty Ltd
Five Islands Road
Port Kembla NSW 2505

Holds Certificate Number: **FS 594448**

and operates a Quality Management System which complies with the requirements of ISO 9001:2015 for the following scope:

The design and manufacture of plain carbon steel and low alloy steel products for the use in the construction, manufacturing, packaging and resource industries. Steel products supplied include: continuously cast slabs, hot rolled plate (including normalized, abrasive cleaned and primed), hot rolled, pickled hot rolled, cold rolled, cold rolled annealed, metal coated and organic coated flat steel strip in coil, sheet and slit form. The supply of manufactured large structural steel products principally for use in the construction industry. The calibration, repairs and service of measuring equipment in the following fields: temperature, voltage, resistance, length, mass, pressure and process field equipment (valves, transmitters and gas analysers).

For and on behalf of BSI:

Chris Cheung, Head of Compliance & Risk - Asia Pacific

Original Registration Date: 1991-09-23
Latest Revision Date: 2021-06-02

Effective Date: 2021-05-13
Expiry Date: 2024-09-22

Page: 1 of 5

...making excellence a habit.™

ISO Certification

ISO9000 Inspectors analyze a business for conformance to the statements made in the quality policy and manuals.

All processes and staff must conform with the documentation.

Then the company will be certified.

The inspectors certify the company does what they said in their quality policy and quality manual.

Inspectors do not inspect quality levels.

EXERCISE

Suppose you are CEO of ABB Electric Motor Drives Division.

(Hypothetical) You purchase an electric motor manufacturer in a country in Southeast Asia.

They have their standard work procedures. They are not what you have described in your ISO9000 documents.

Should you revise their procedures and train them to conform to your standard work procedures?

EXERCISE

Suppose you and your friend developed a quadcopter in a design course, and you want to start a business selling them.

You open a webstore.

You are now getting 30 orders a month. It appears to be growing to 100/month in 12 months. You have a business!

Right now you make them in your living room.
You will open a new space in Fisherman's Bend next month.

Should you get ISO9000 certified?

EXERCISE

Suppose you are CEO of Konecranes.

(Hypothetical) Your product development is doing fine, following the standard work and developing new cranes. Business in one category isn't falling, but it also isn't growing.

You want new highly advanced technology in your cranes.

You decide to set up a small group off-campus to develop a new lift technology never before used in cranes. You put your smartest engineer in charge, with a proven record.

Should this new group follow your ISO9000 procedures?

ISO 9000 Summary

A quality manual describes your quality system. It tells your customers how you manage total quality across the business.

The ISO 9001 standard helps develop a practical quality management system.

The standard ephasizes processes, customer satisfaction and risk-based thinking and decision making.

IS09000 allows for judgement of how quality is managed at the firm. Start with the quality manual. Does not allow for judgement of product quality.

Customers expect to see a ISO9000 certified quality system. A price to enter as a supplier.

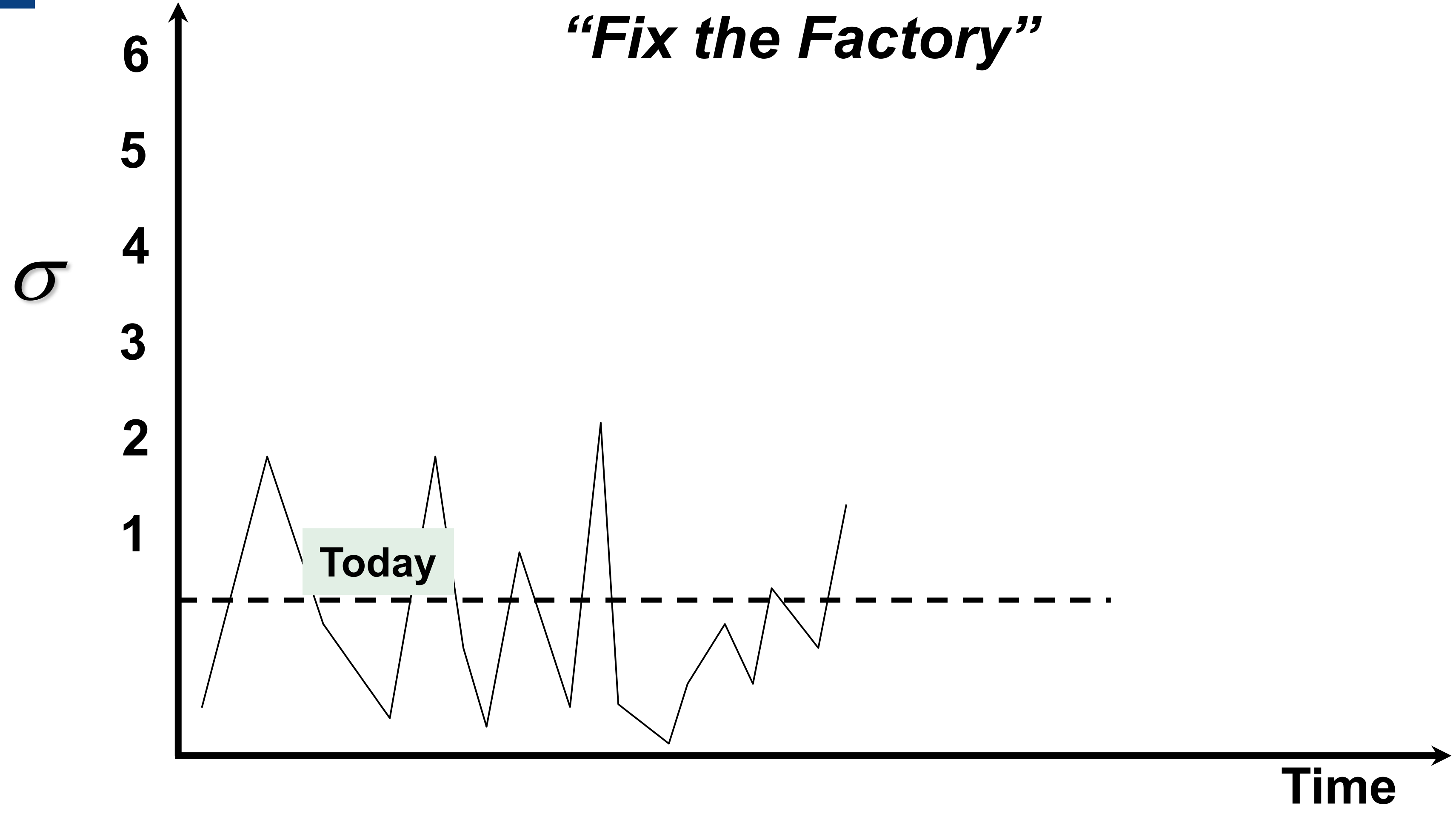
Outline

PDCA

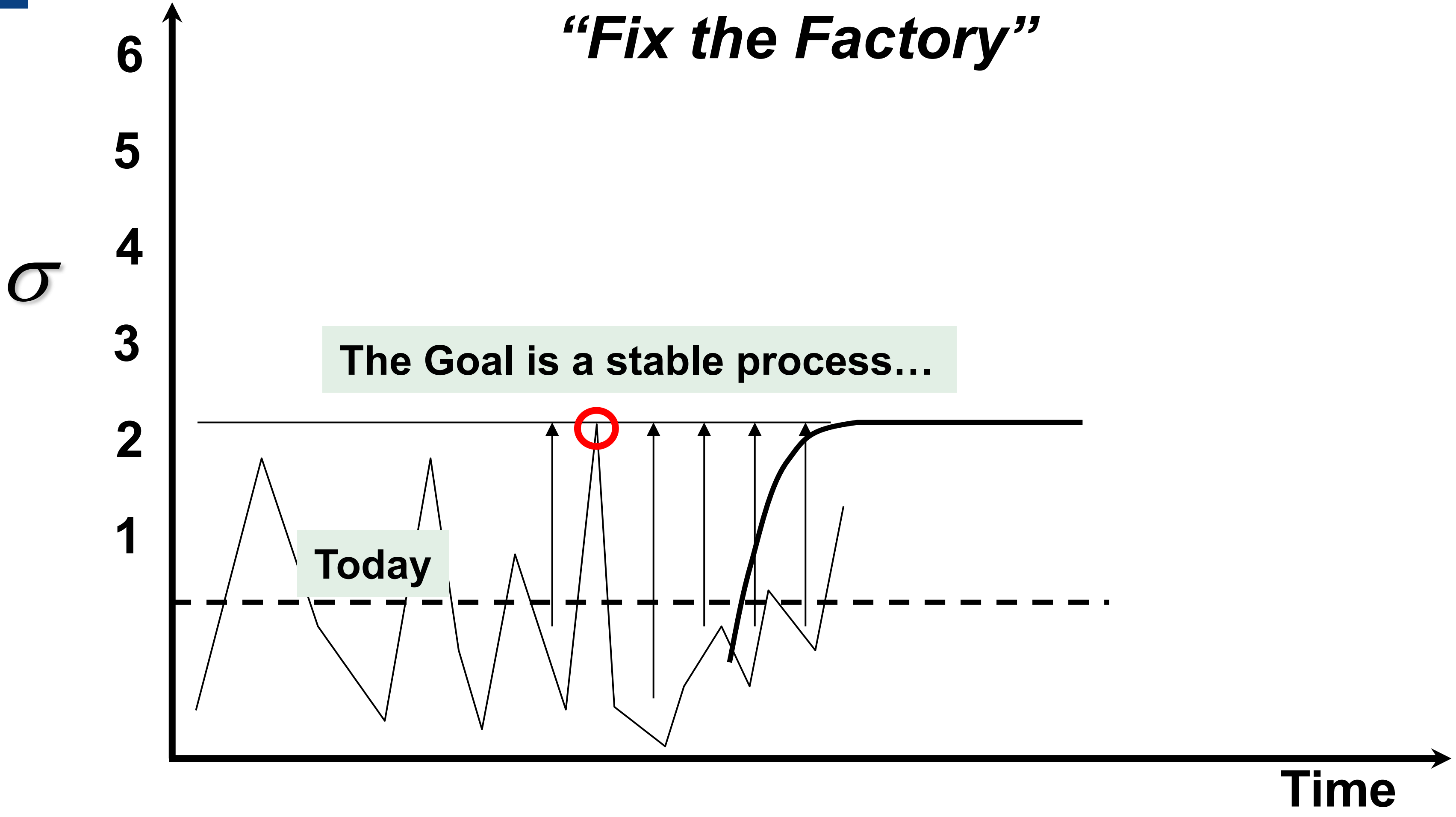
IS09000

Six Sigma

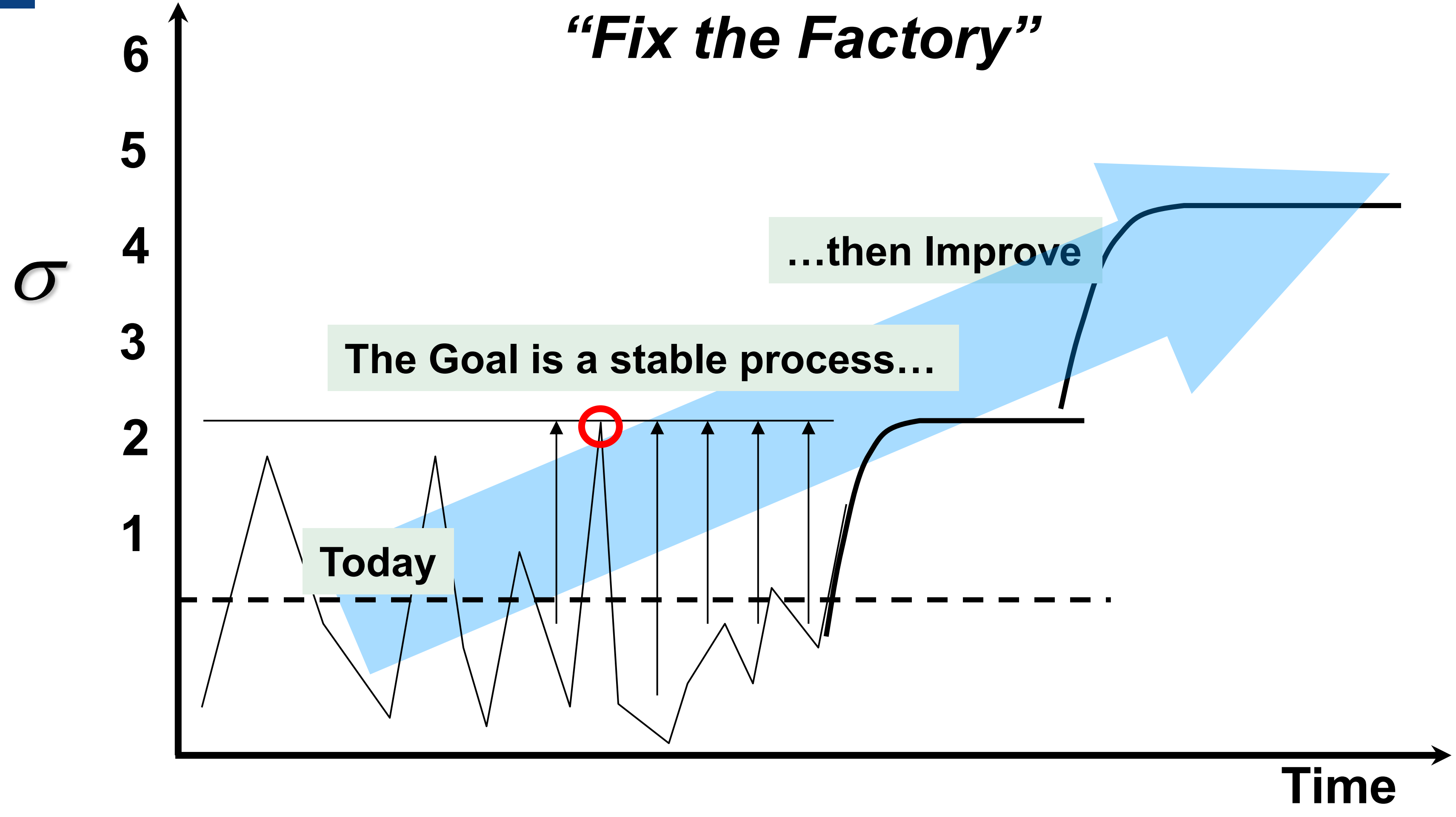
Quality Improvement



Quality Improvement

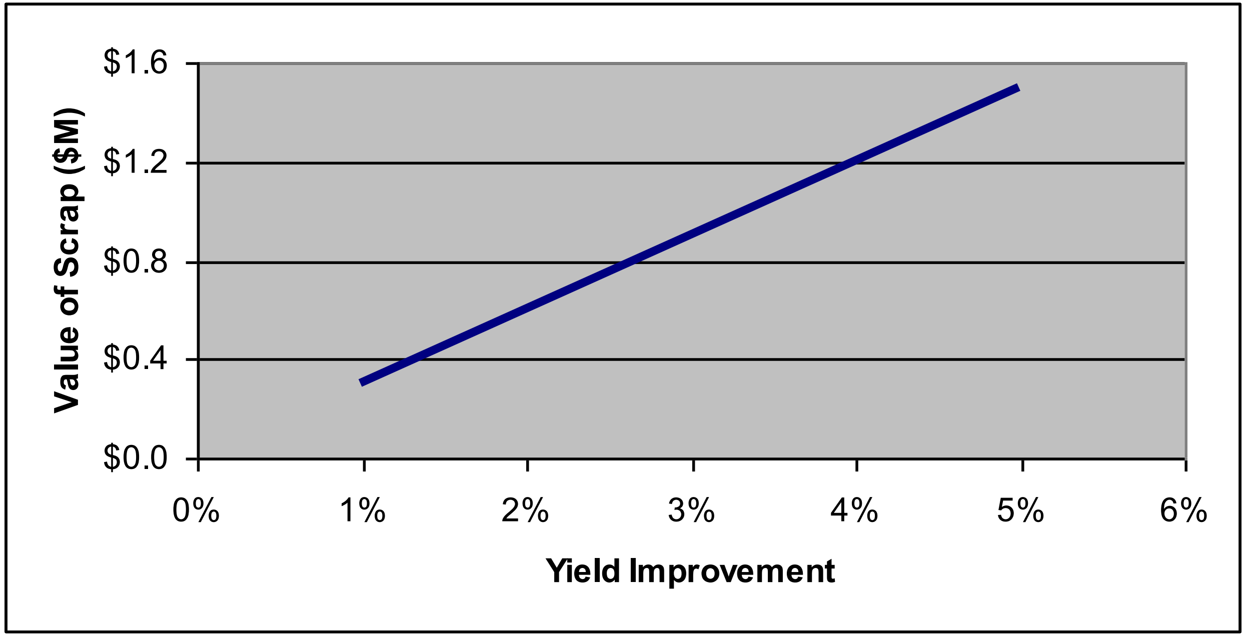


Quality Improvement



Bottom Line Value

Increasing quality reduces scrap, thereby saving money.



… but what is the value to the top line?

How?

We discussed PDCA. We discussed ISO9000.

Suppose your quality level is not adequate. How do you go about making improvements?

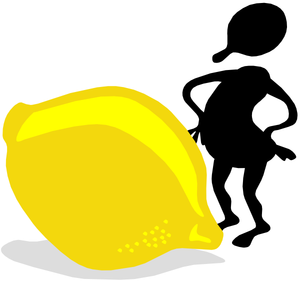
What makes a high quality manufacturer different?

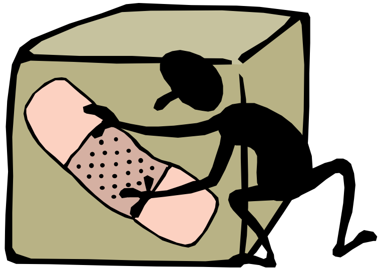
Scrap Rate Example

A Manufacturing Firm AUS Corporation makes widgets.

The Scrap Rate is a concern; it is eroding profits.

The VP of Operations tells everyone to *focus on quality*…





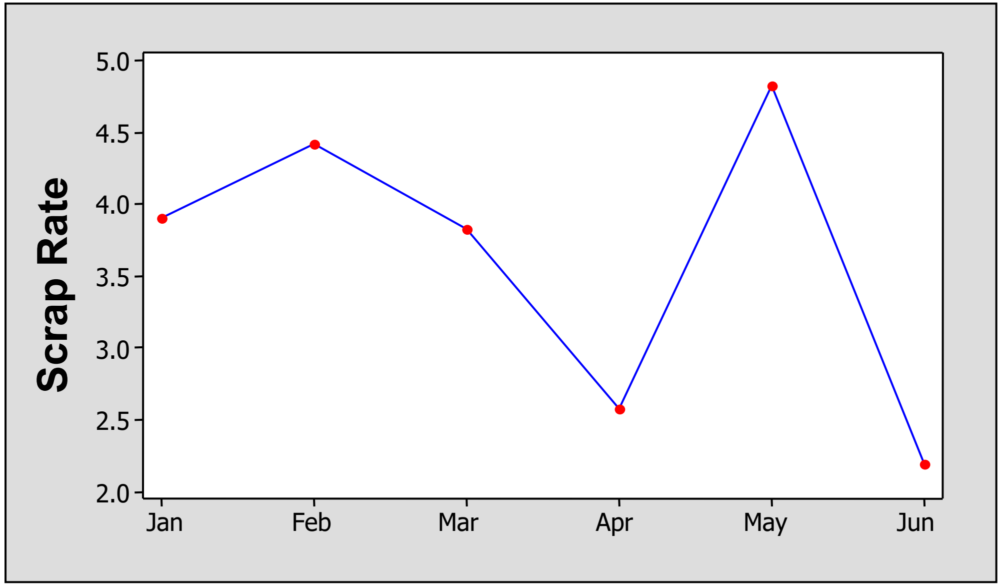
Derived from *Understanding Variation: The Key To Managing Chaos*, D. Wheeler, SPC Press. 1993.

June

Scrap Rates are at an all time low of 2.2% of sales!

The VP of Operations throws a party for all of the engineers.

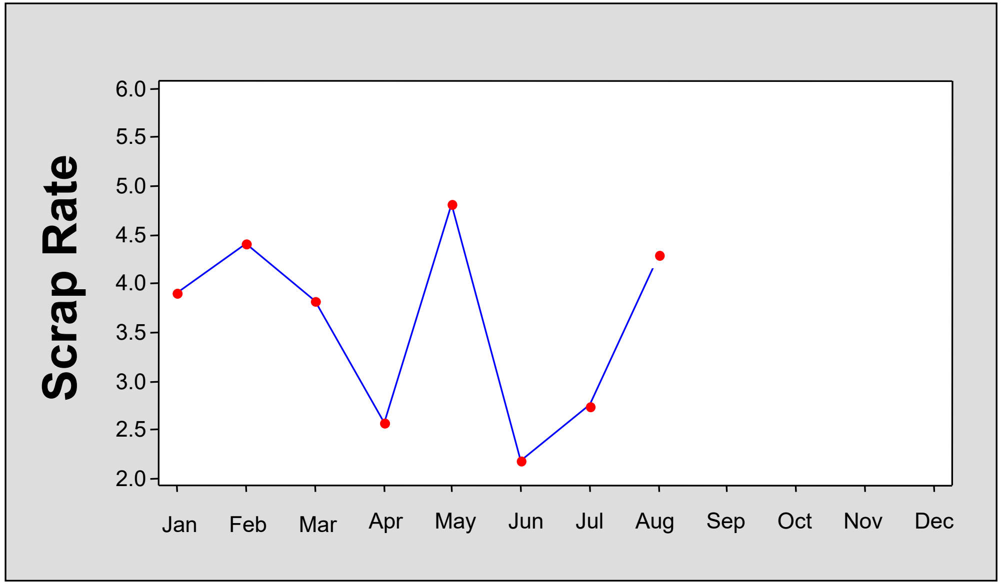
"Everyone should be proud of what we have all achieved"



August

Into to the third quarter, scrap rates went up…

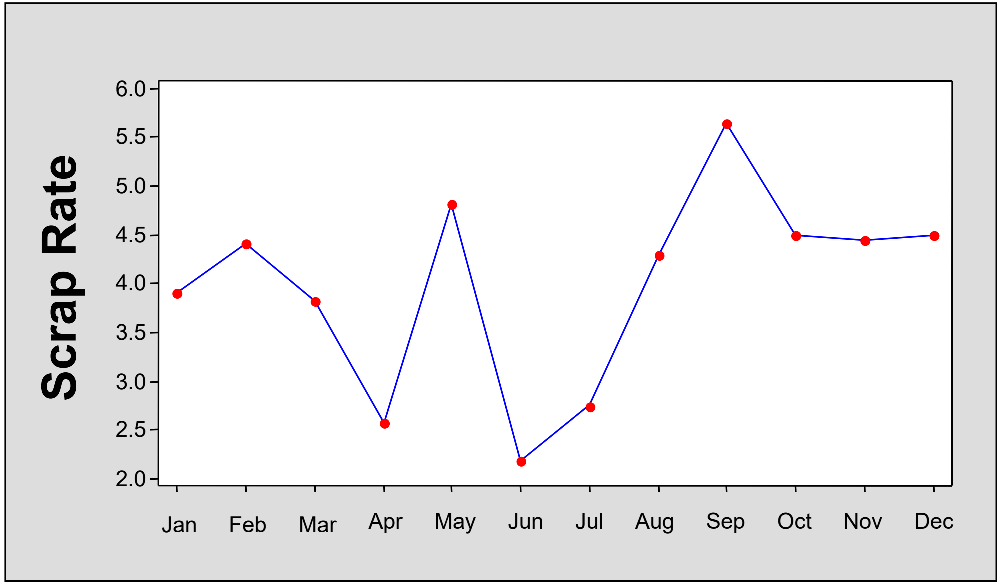
“Oh no. Giving them so much praise was the wrong thing to do”



December

Instead of maintaining the low scrap rate of the first half, it got worse and stayed that way.

The VP Operations decides
"These people tough management!"



March

Into the next year, the scrap rate continues to rise to a high over 6%

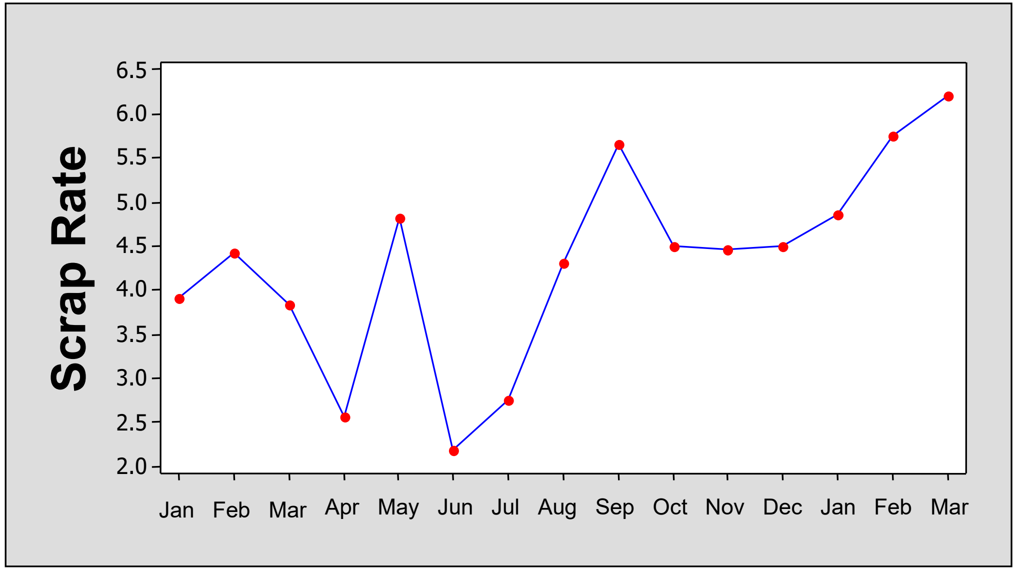
The Operations VP decides to take control

- A “special meeting” is called to solve this problem already

After a sound lecture on how important quality is to the company, the VP walks out.

The manufacturing engineers don’t really know what to do.

So they do nothing.

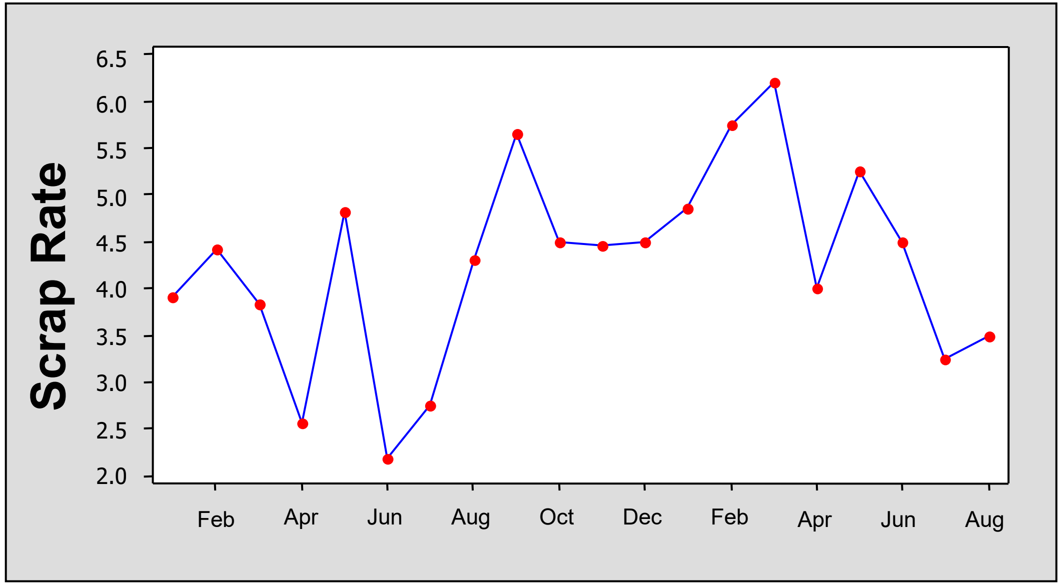


August

Out of the first half of the next year, the scrap rate gets better.

The VP Operations sees this since the special meeting. “*Its getting better!*”

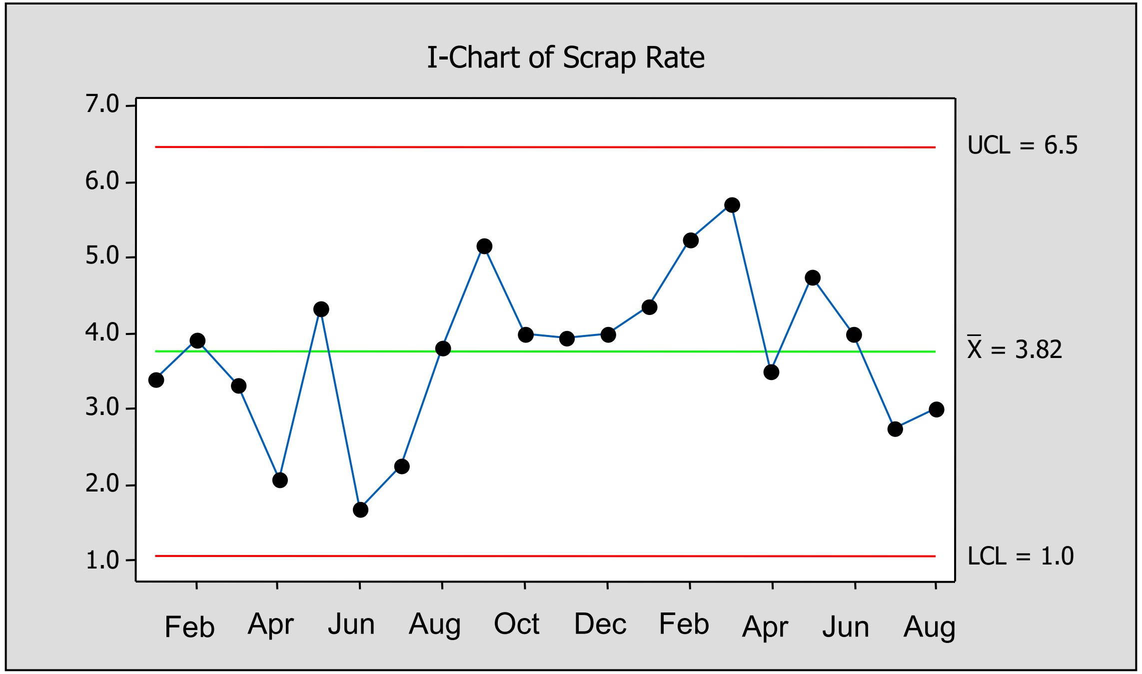
Yet all the time, nothing had changed.



A Control Chart Shows

The scrap rate was stable and predictable.

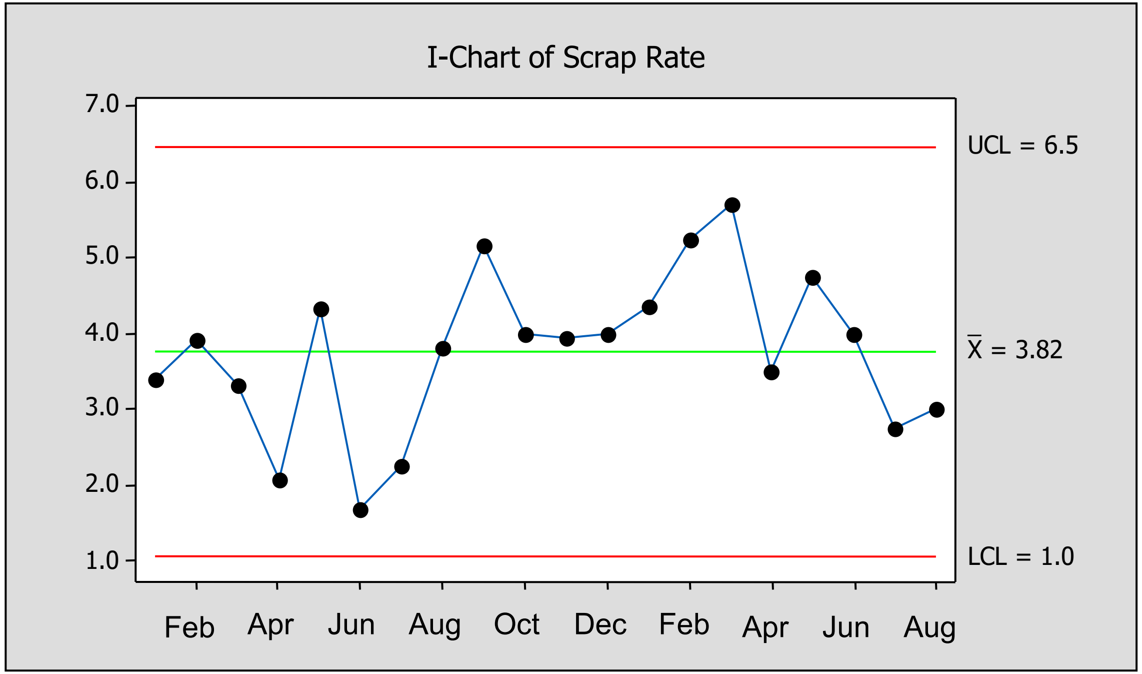
Executive decisions were made from observing high and low points as signals when really it was all normal variation.



Stable and Predictable

Yet, the scrap rate is still too high.

So then, VP Operations, what do you do?



Six Sigma

Data Driven Analysis

We use measured data to analyze, improve, and control

Process Control Maturity Level

1. We use our experience, not data, to control our process
2. We measure things, and look at the number
3. We plot the data in charts and graphs
4. We group the data to form statistics
5. We use sample plans to form periodic statistics
6. We use sample data to form statistical inferences

What level is your process?

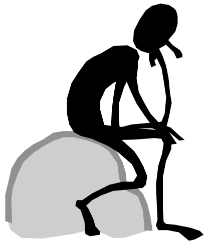
Six Sigma is a way of thinking…

"Statistical Engineering"

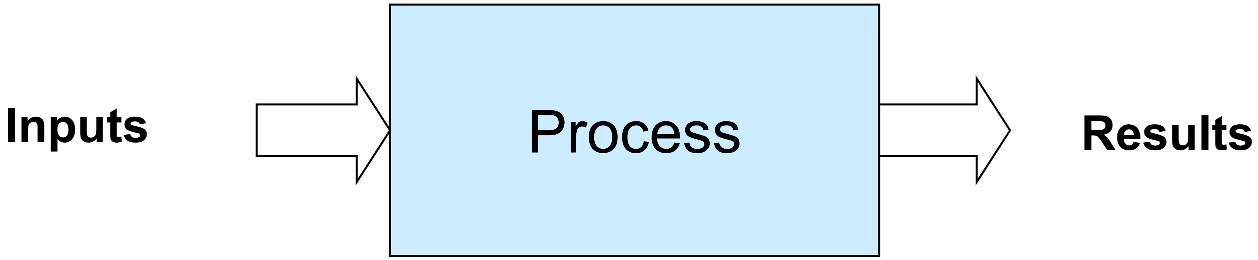
- Sample size
- Confidence
- Power
- Resolution
- Blocking
- Repeatability and Reproducibility

An evolution from thinking based on a single "fact", measurement, statement or opinion to thinking based on statistics from data.

Six Sigma is a methodology…

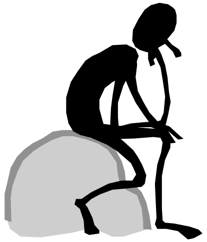


Suppose you are doing something that generates results

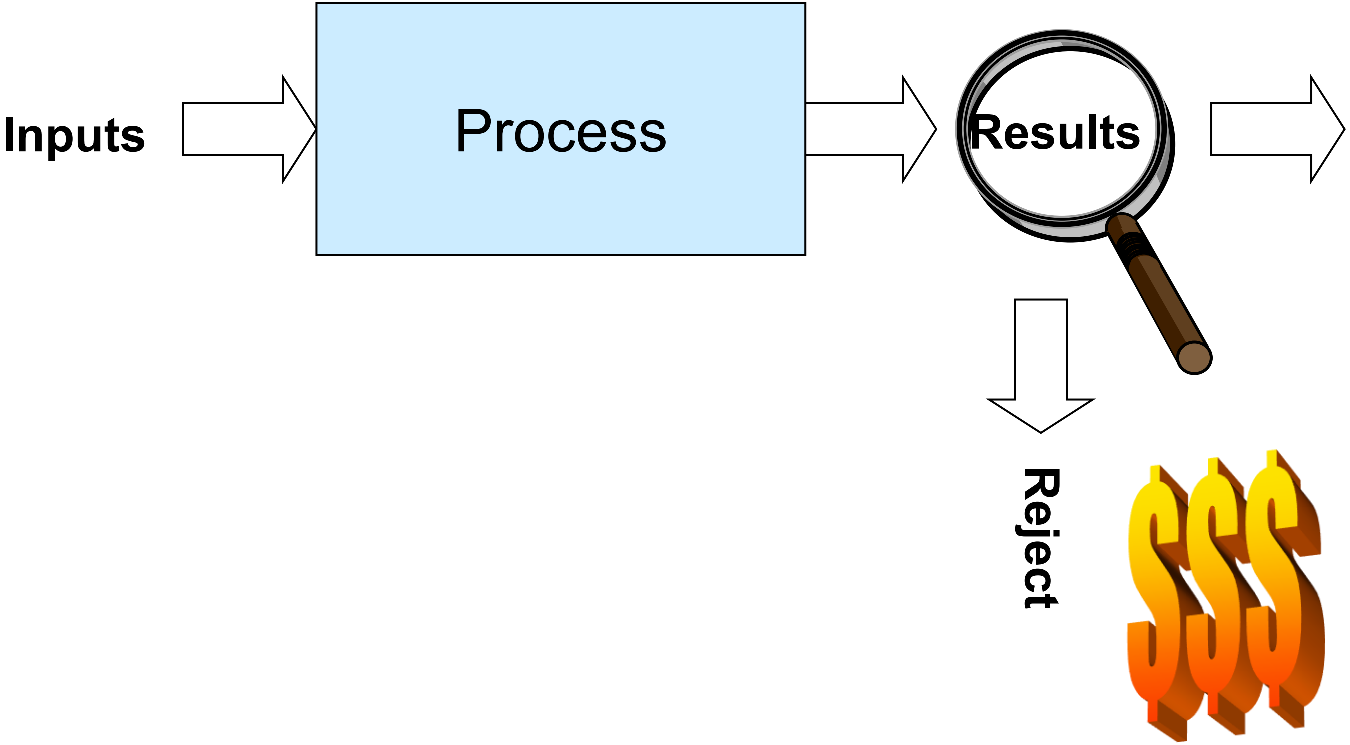


How do you ensure the process makes good results?

Six Sigma is a methodology…

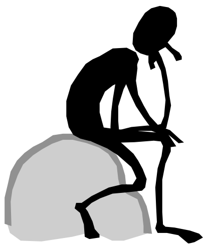


The traditional way: Inspection and scrap



Six Sigma is a methodology…

Zero Defects



The six sigma way:

KPIV
KPIV
KPIV
KPIV

Process

KPOV

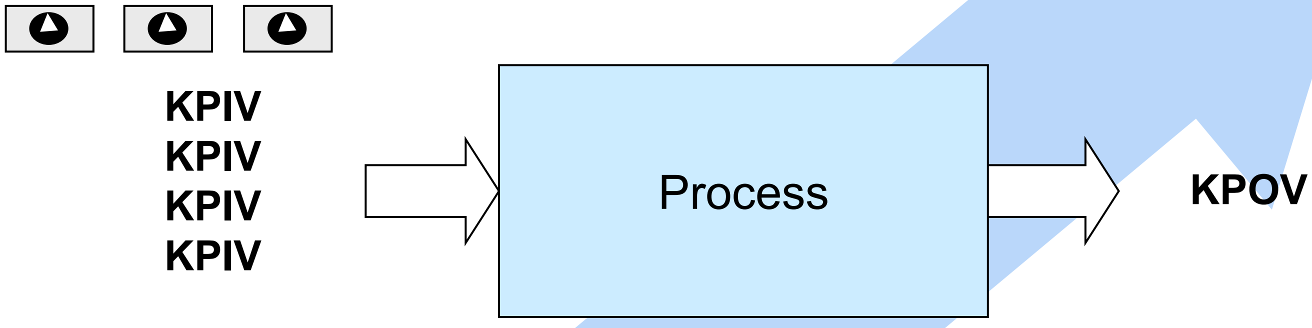
MANAGE THESE…

…AND THIS WILL BE IN SPEC

You won't need to inspect…

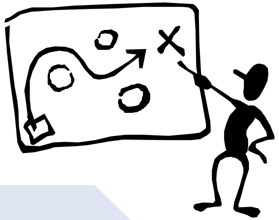
Six Sigma is a methodology…

The six sigma way:



But of the thousands of inputs, which are important to monitor and control?

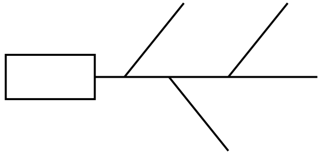
Six Sigma is a Process…



Define

What's the problem
Clarify it
- Defective data
- What defects
- What steps

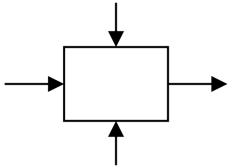
Tools:
- Process Mapping
- Fishbone diagrams
- FMEA



Measure

How to measure it
- Not defect counts
- Output variation
- Measurement system

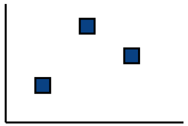
Tools:
- MSA, %GRR
- Data collection
- Graphs of Data
- Cp and Cpk



Analyze

What changes it
- Controllable
- Noise
- Signals

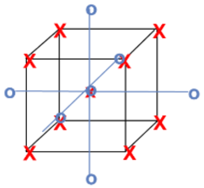
Tools:
- Hypothesis Tests
- Regression
- ANOVA



Improve

What to change
- Controllable

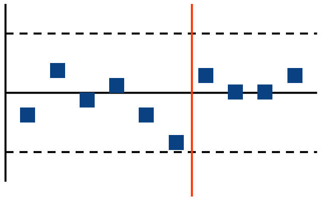
Tools:
- Hypothesis Tests
- DoE
- Robust Design



Control

How to keep it there
- Control plans

Tools:
- 5S
- Control Plans
- Empowerment
- SPC



Six Sigma is a Statistical Term

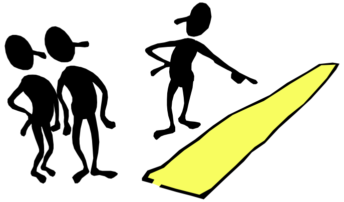
Sigma	DPMO
2	308,537
3	66,807
4	6,210
5	233
6	3.4

Defects Per Million Opportunities

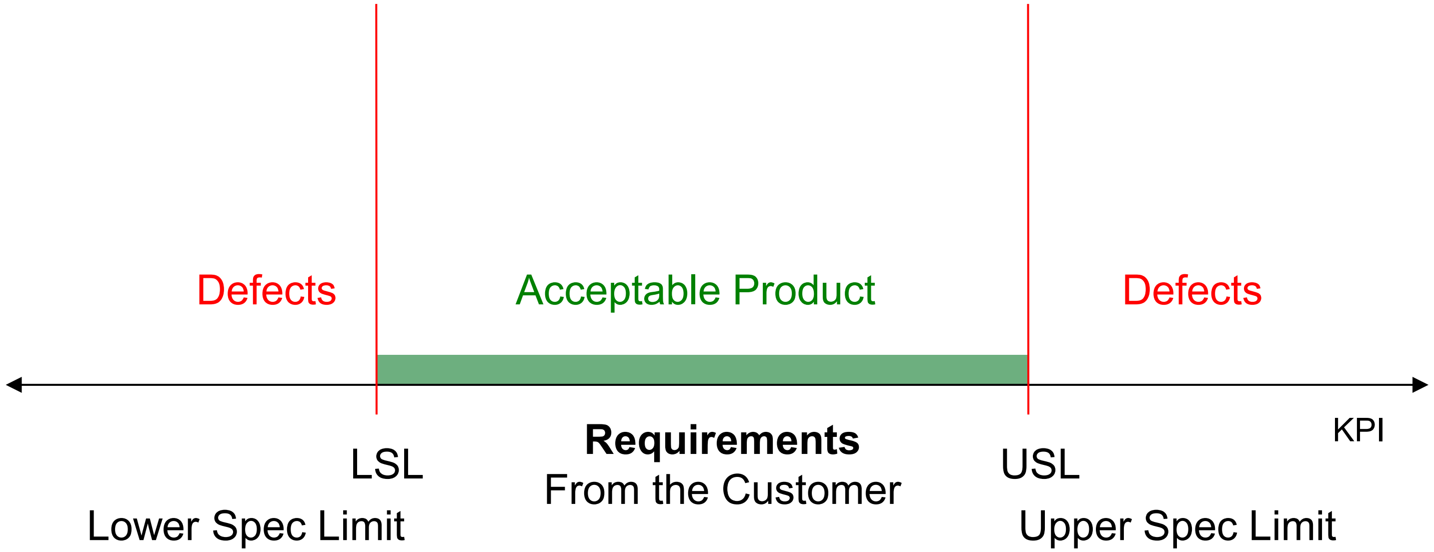
Sigma is a statistical term used to denote the performance of a process.

Six Sigma is an *aggressive* performance goal.

Requirements

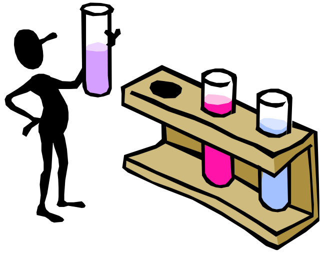


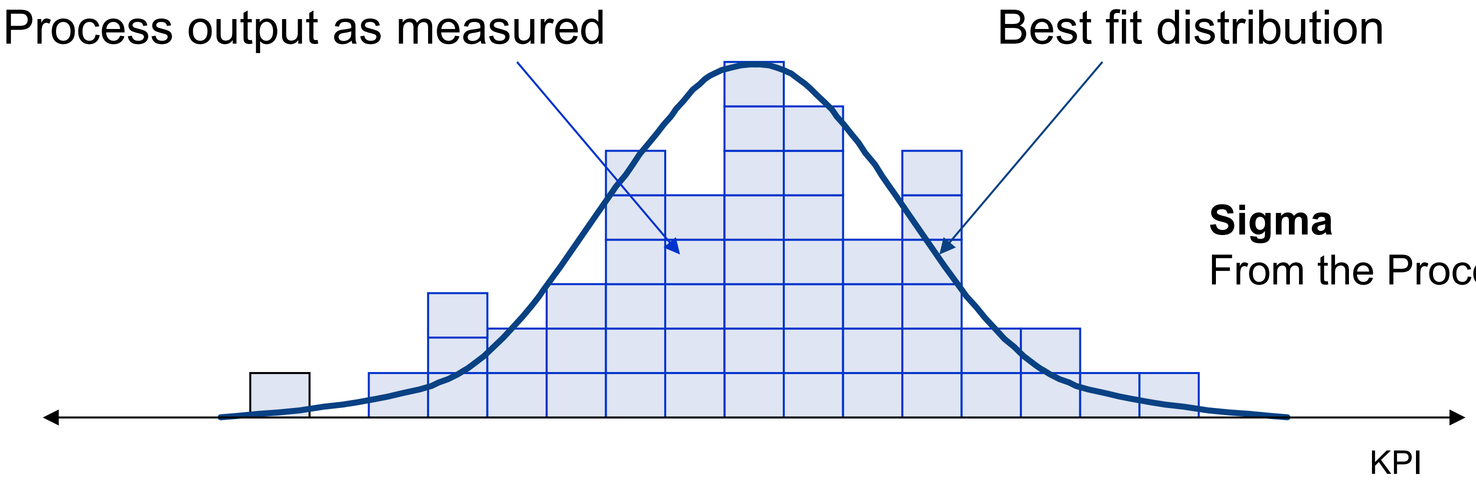
Specification Limits define when the process output is successful.



Requirements

The measurement of the process output defines a distribution…

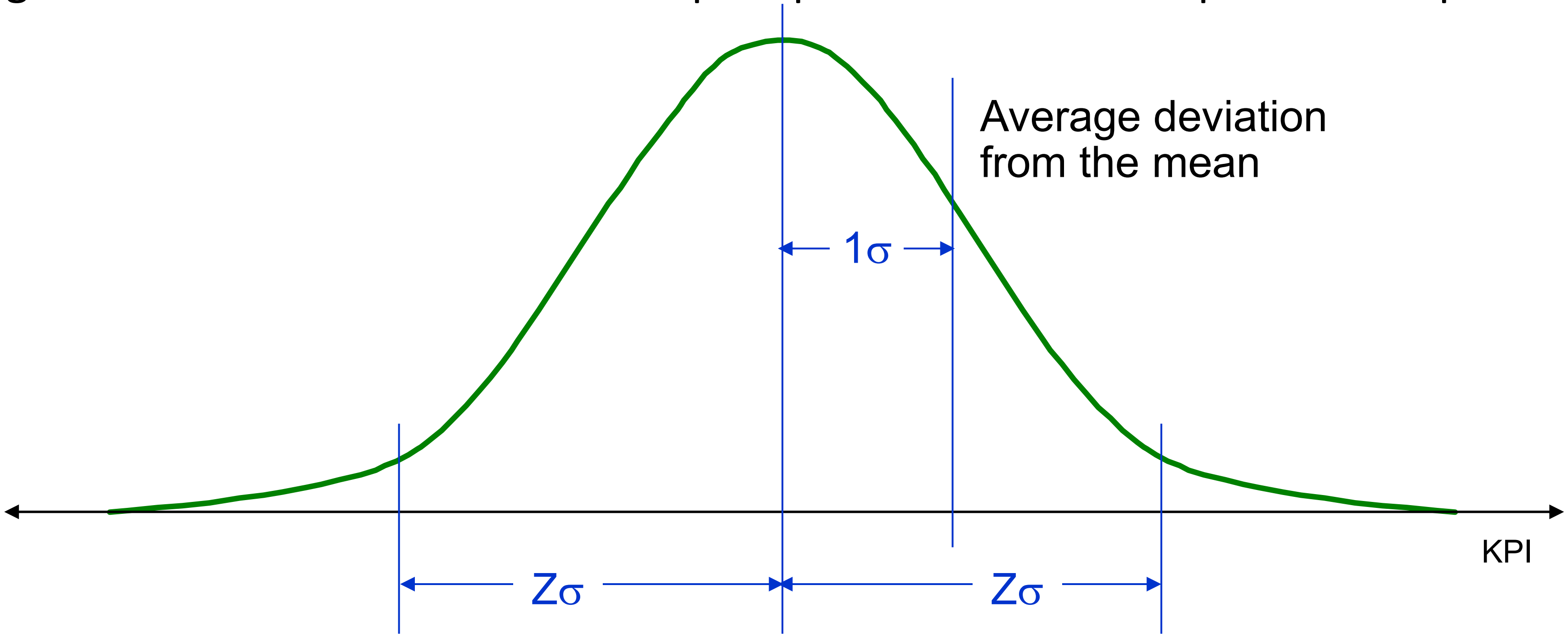




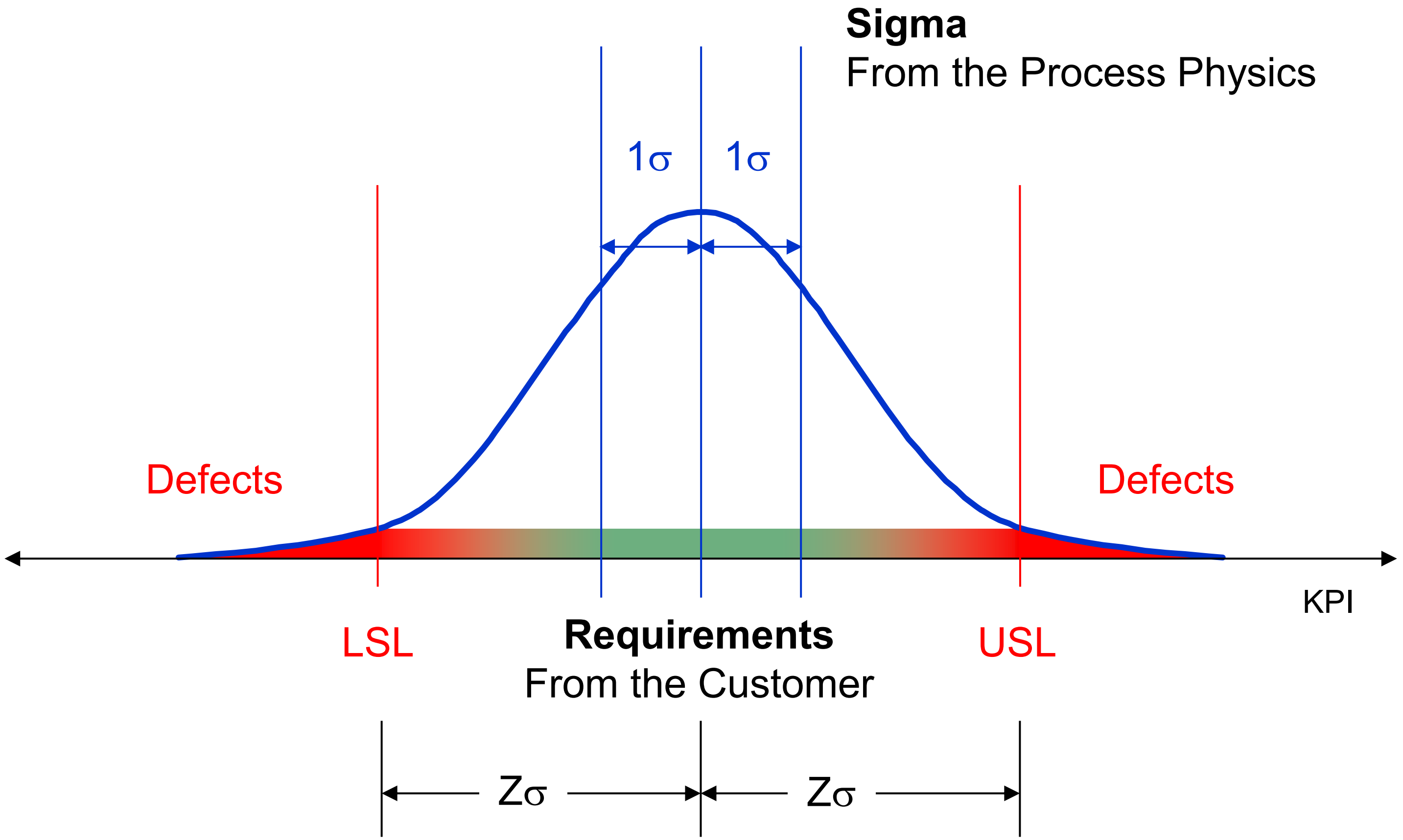
Sigma
From the Process Physics

The Z Statistic

Number of sigma's or **Z-Statistic** = number of standard deviations away from the process average. A universal metric used to compare performance across products or processes.



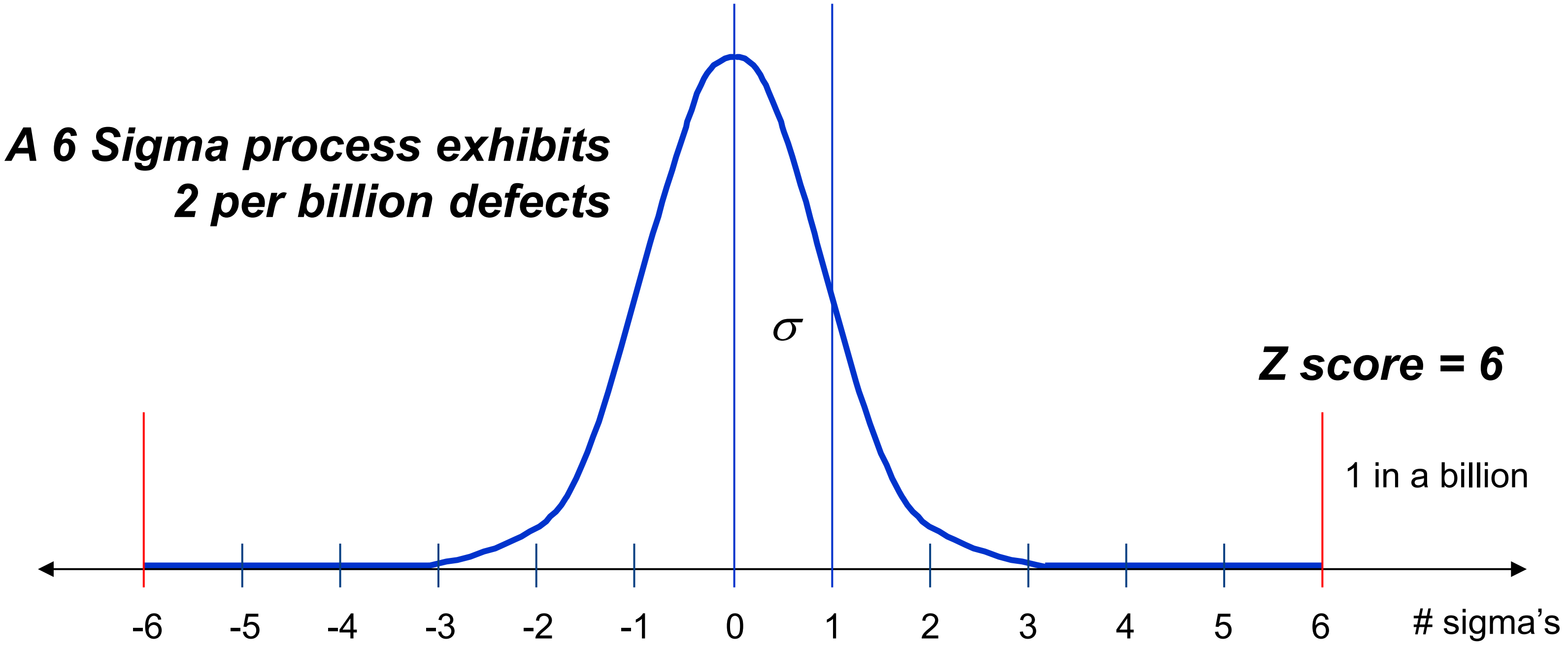
Sigma Level and the Z Statistic



How many sigma's fit in the specification limits?

A Process with 6s Specifications

The Z-statistic for a process can be calculated using measured data: sample data, or short-term data.

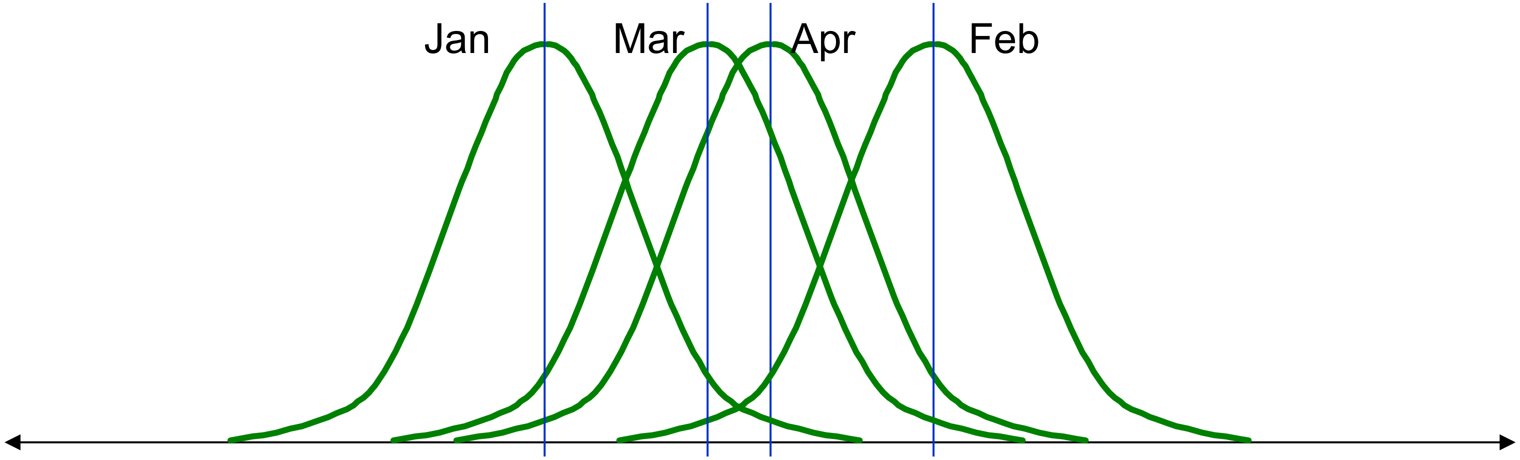


What Happens Over the Longer Term

Periodically the short-term capability is measured...

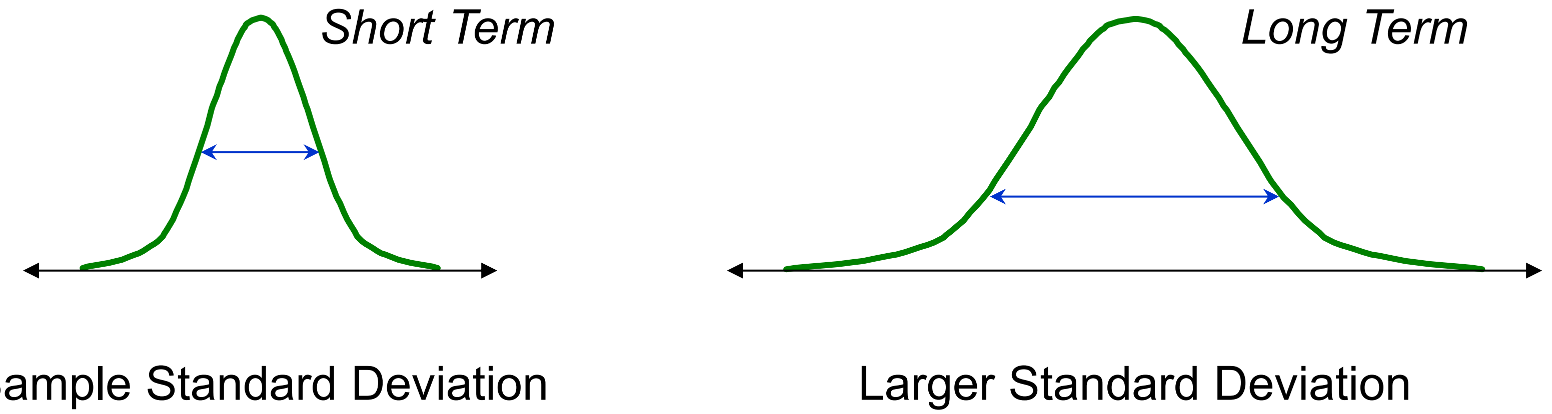
…meanwhile things shift…

…and so the long term variation is wider.



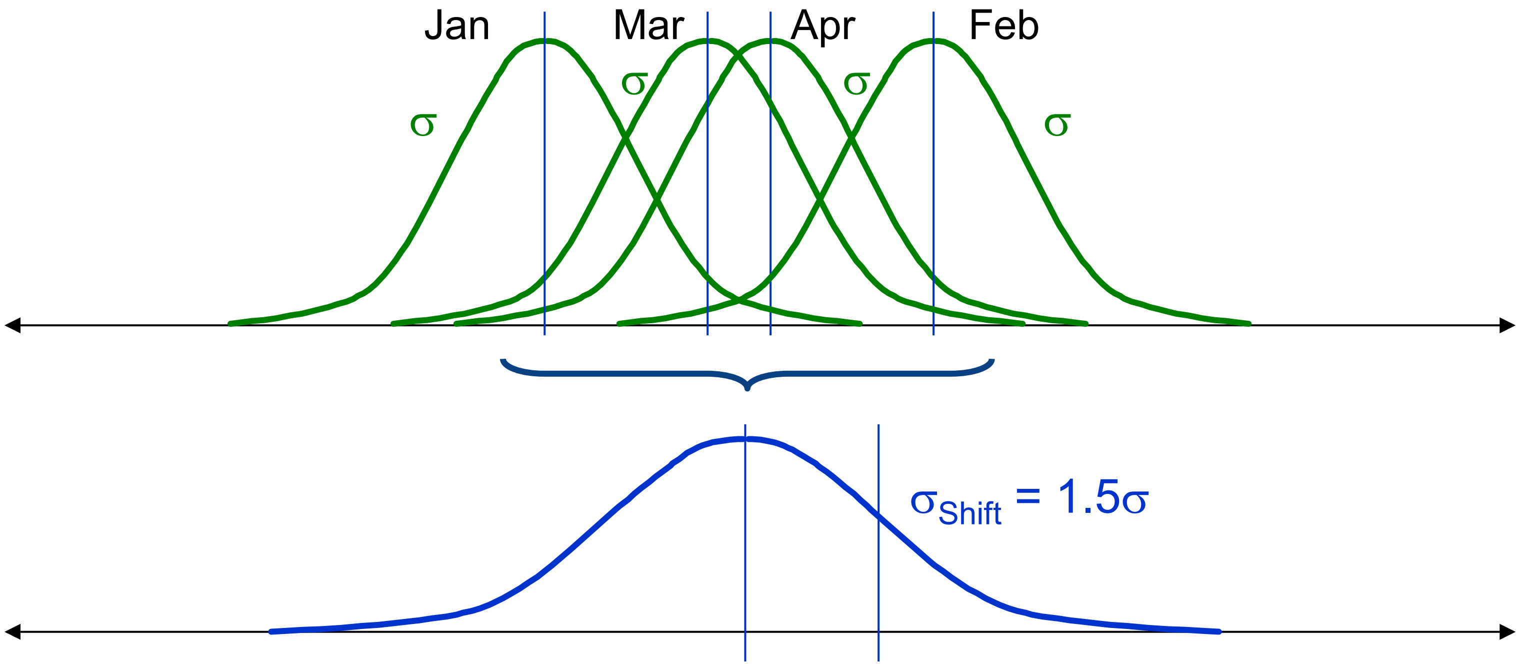
Short and Long Term Sigma

The long term estimate of sigma is larger than any short term estimates of sigma.



Determining the Difference

Generally, for most manufacturing, the short term distribution tend to shift by 1.5σ over the long term.

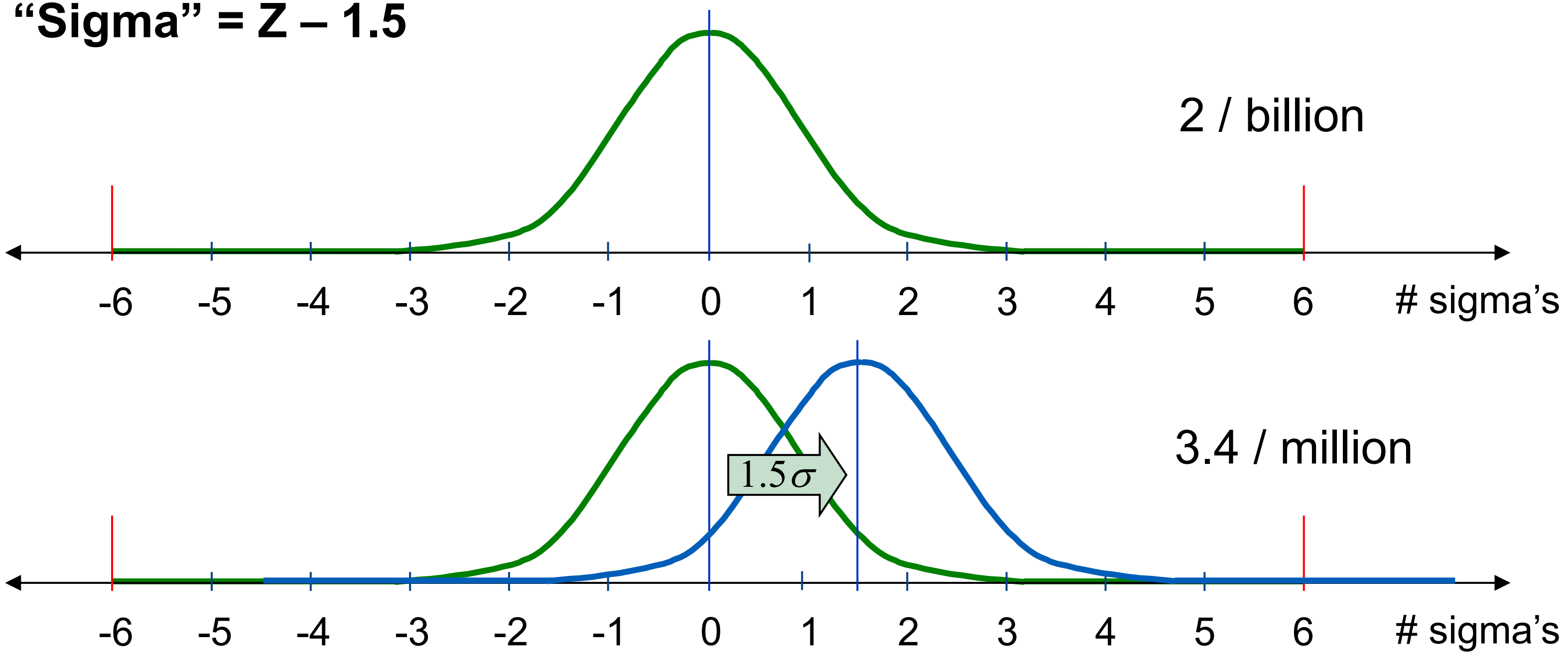


The “Sigma” Measure

A 1.5σ mean shift can be added to account for the long term shifts in the process.

It represents the amount of change a typical process will exhibit over many cycles of that process.

“Sigma” = Z – 1.5



Sigma, Cpk, Yield and DPMO

Capability Table				Short-term (2-sided)	
Sigma	Cpk	Yield	DPMO	Yield	DPMO
1	0.333	30.9%	691,462	68.3%	317,311
1.5	0.500	50.0%	500,000	86.6%	133,614
2	0.667	69.1%	308,538	95.4%	45,500
3	1.000	93.3%	66,807	99.7%	2,700
4	1.333	99.4%	6,210	99.994%	63
5	1.667	99.98%	233	99.99994%	1
6	2.000	99.9997%	3.4	99.9999998%	0.0020

"Sigma" = σ_{LT} = Z – 1.5

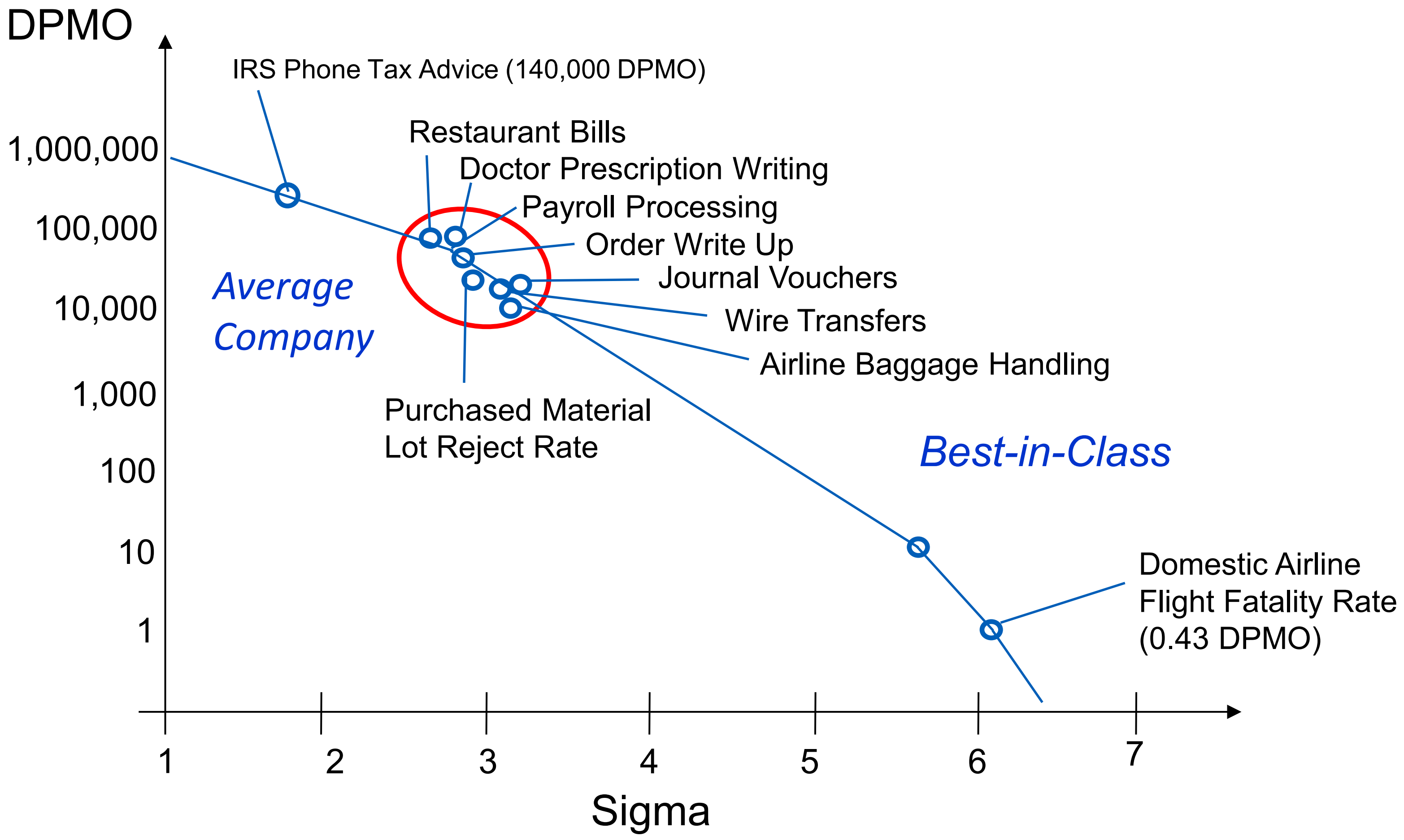
σ_{ST} = Z

Practical Interpretation

99% Good (3.8 Sigma)
Unsafe drinking water for almost 15 minutes each day
No electricity for almost seven hours each month
20,000 lost articles of mail per hour
200,000 wrong drug prescriptions each year
Two short or long landings at most major airports each day
5,000 incorrect surgical operations per week

99.9997% Good (6 Sigma)
One unsafe minute every seven months
One hour without electricity every 34 years
Seven articles lost per hour
68 wrong prescriptions per year
One short or long landing every five years
1.7 incorrect operations per week

Where Does Most Industry Stand?



Getting to Six Sigma

How far can inspection get you?

Capability Table			
Sigma	Cpk	Yield	DPMO
2	0.667	69.1%	308,538
3	1.000	93.3%	66,807
4	1.333	99.4%	6,210
5	1.667	99.98%	233
6	2.000	99.9997%	3.4

Getting to Six Sigma

Radical breakthrough process changes are needed.

Six sigma is about making real changes to the factory process and equipment.
No more work as usual.

Capability Table			
Sigma	Cpk	Yield	DPMO
2	0.667	69.1%	308,538
3	1.000	93.3%	66,807
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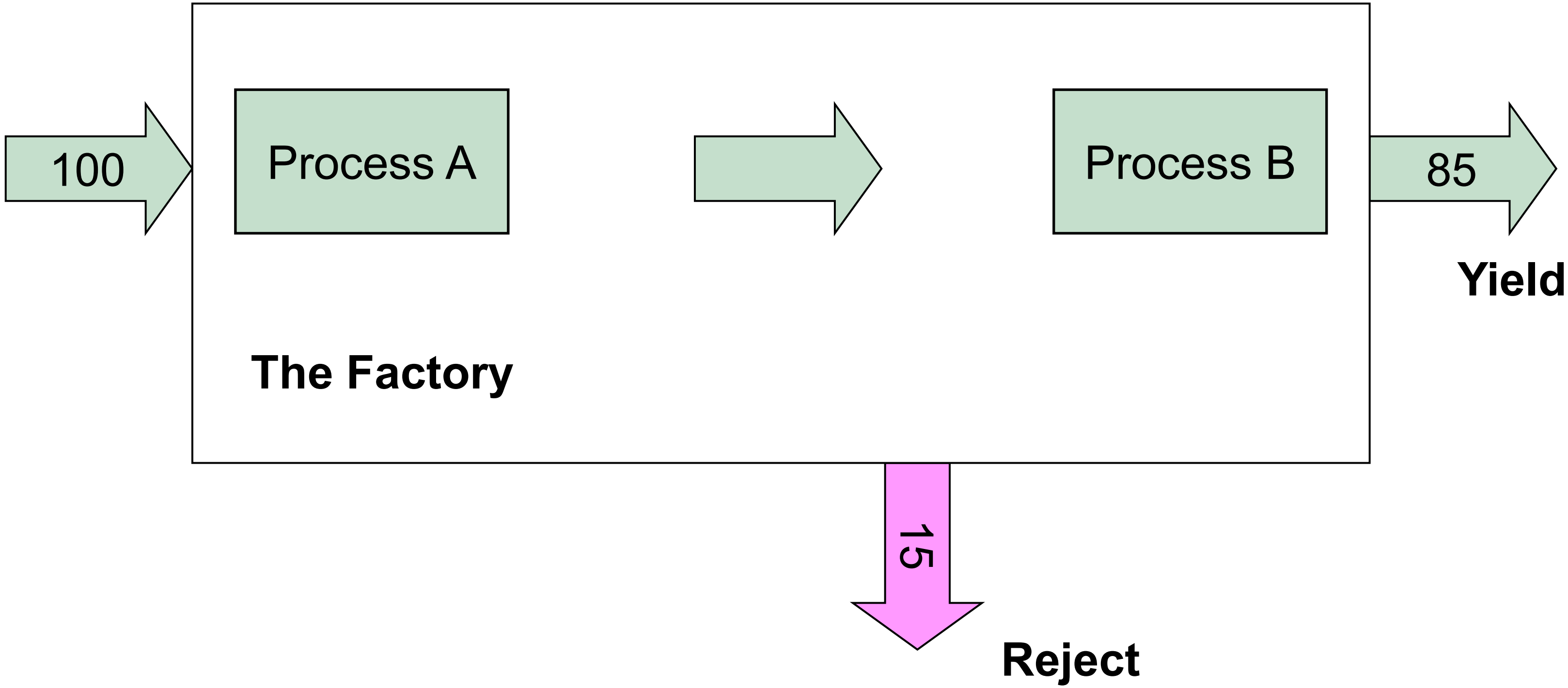
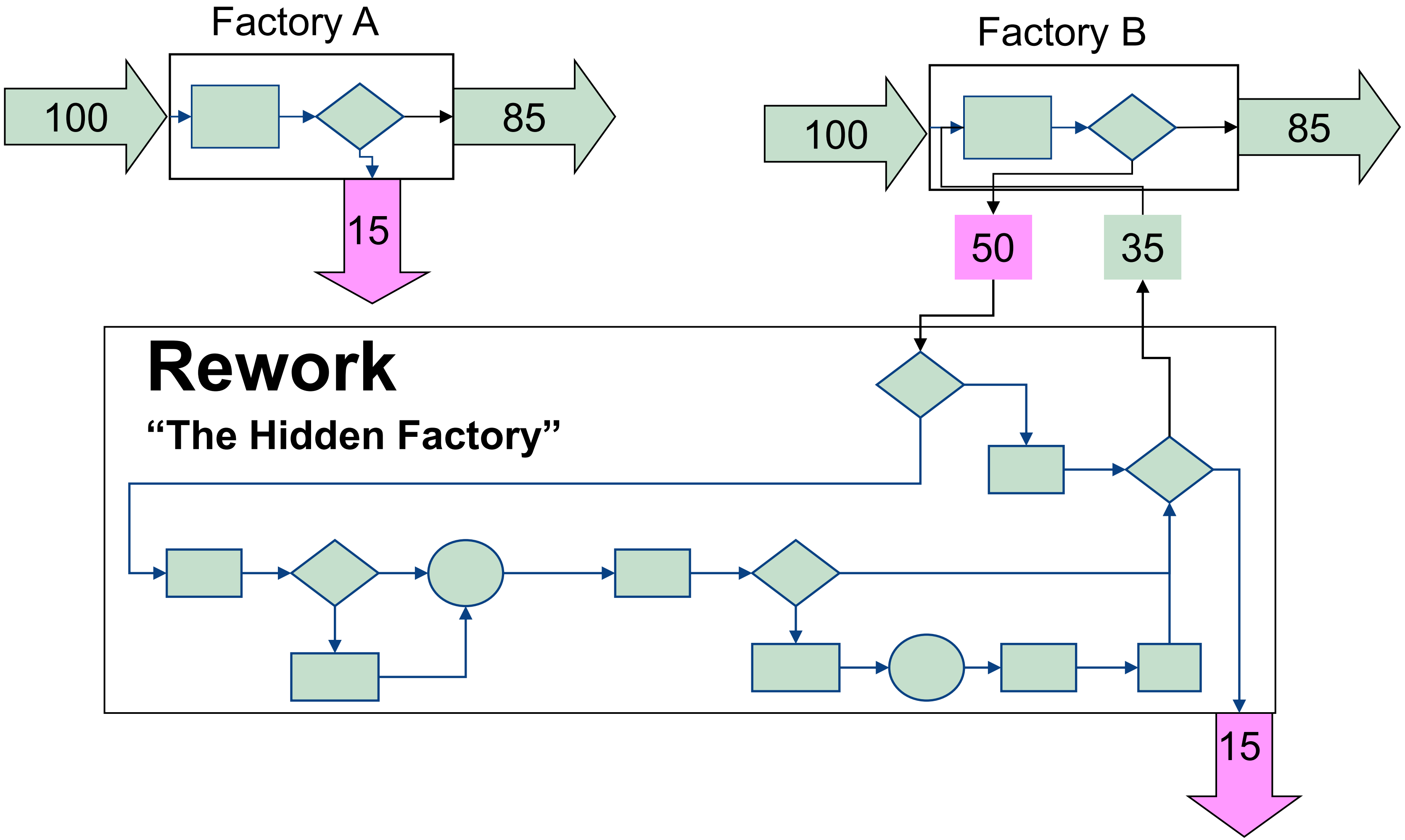
Classical Yield

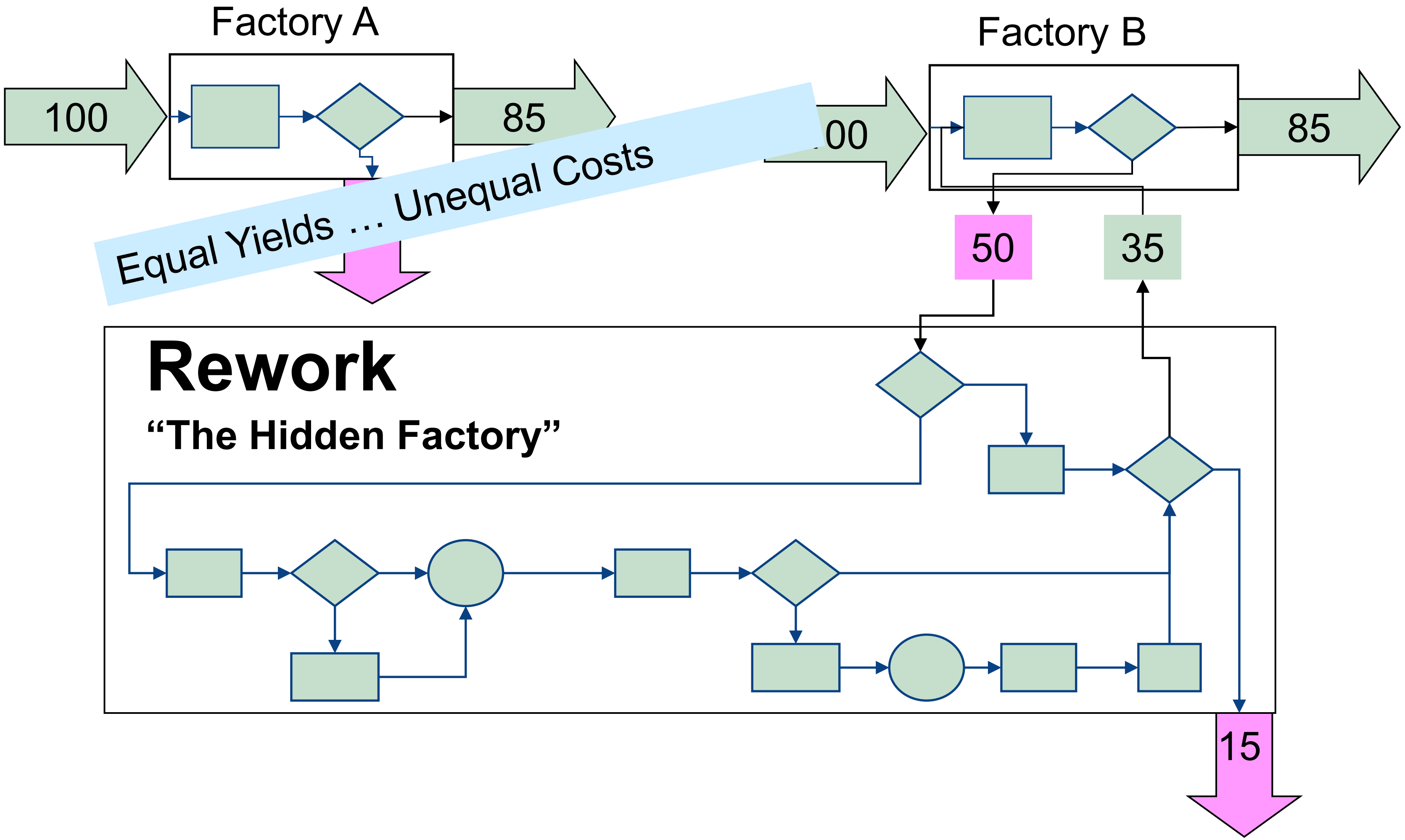


Is this a good indicator of the factory's defect rate?

Consider Two Different Factories…

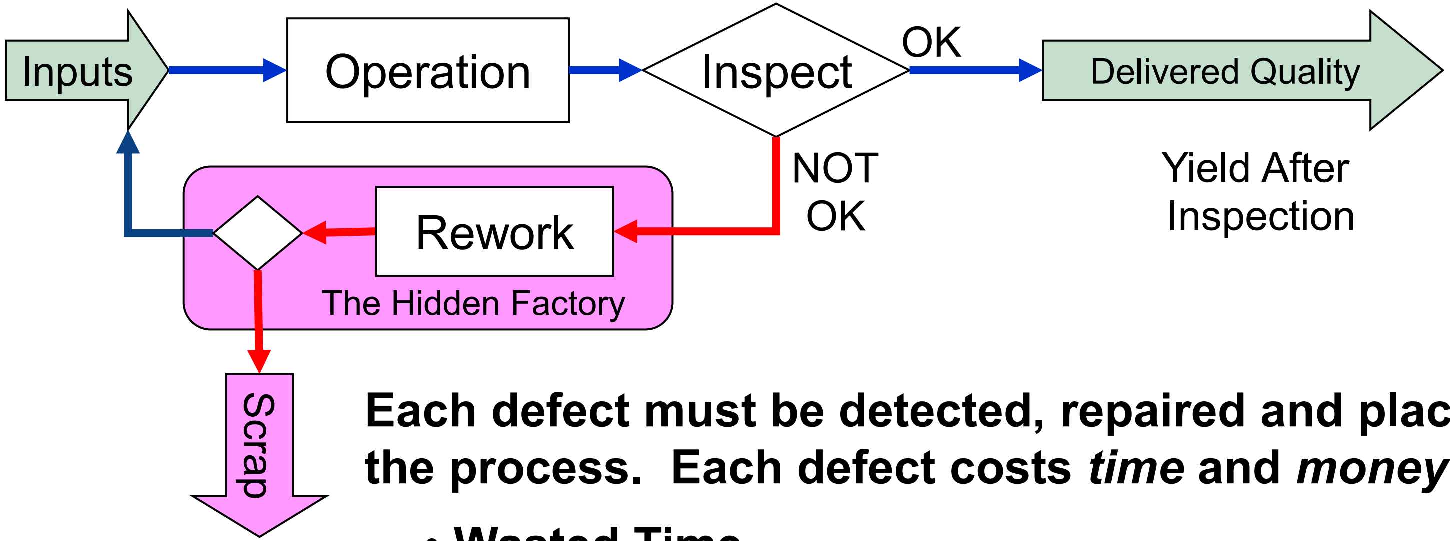


Consider Two Different Factories…



Defects and the Hidden Factory

Manufacturing Variation Causes a "Hidden Factory"



Each defect must be detected, repaired and placed back in the process. Each defect costs *time* and *money*.

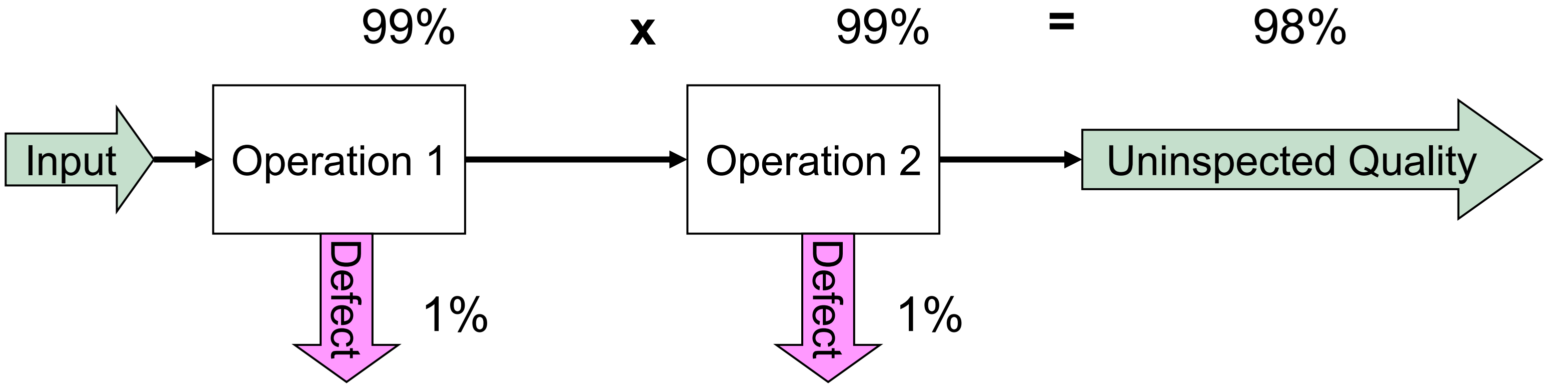
- **Wasted Time**
- **Wasted Money**
- **Wasted Resources**
- **Wasted Floor space**

Rolled Throughput Yield

Final yield is a poor measure of quality!

We need a different measure of yield

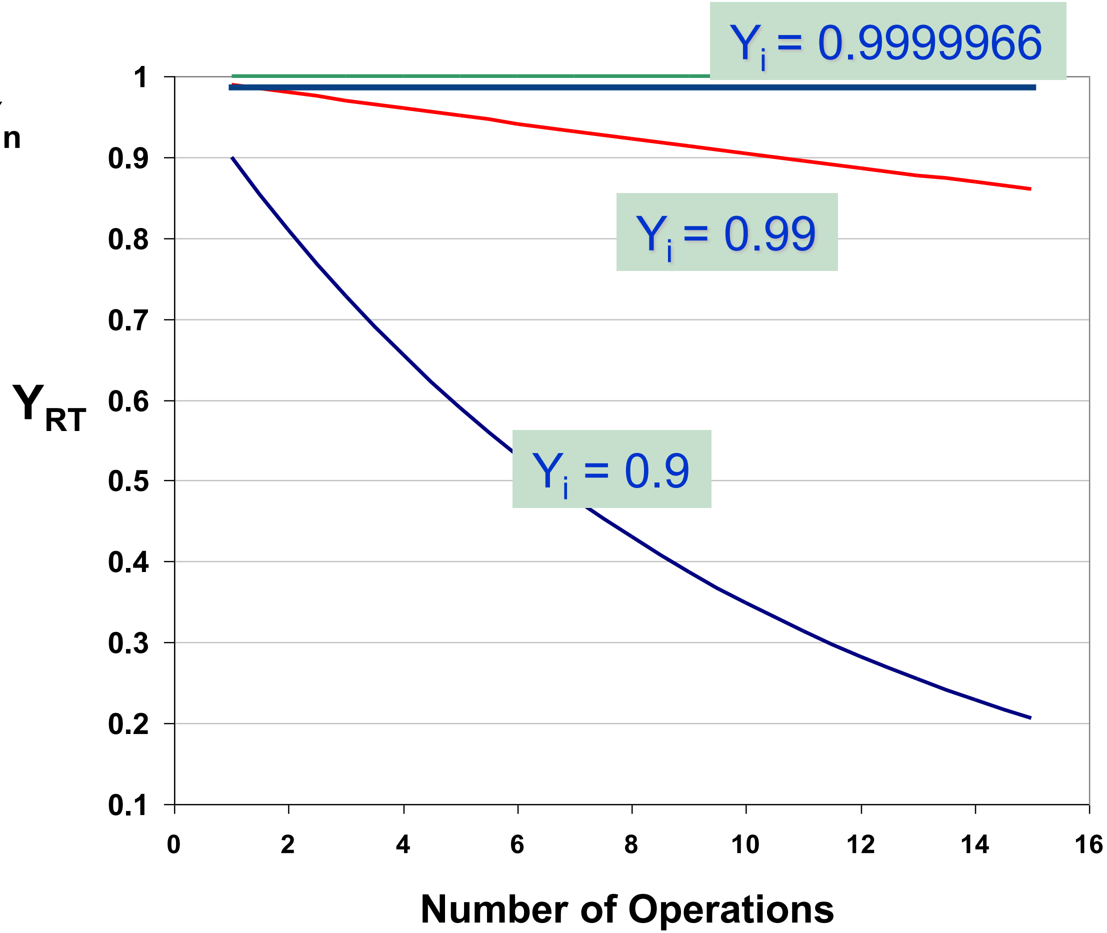
- Measure the yield of a process as if there were no inspections. ***Rolled Throughput Yield***
- What quality level would be delivered?



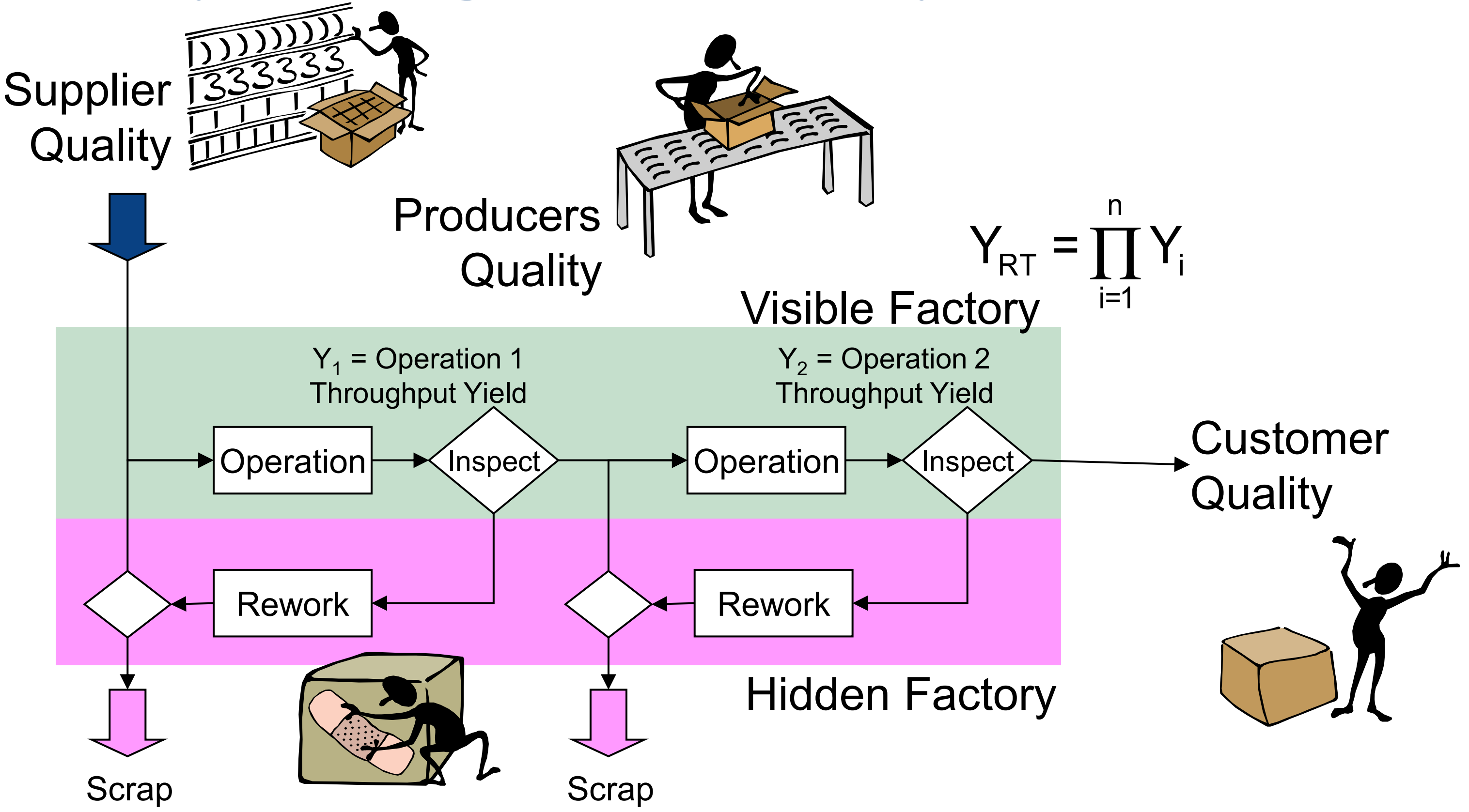
…there is a 98% probability that any given unit will pass through both operations defect free.

Rolled Throughput Yield

$$Y_{RT} = Y_1 * Y_2 * \ldots * Y_n$$



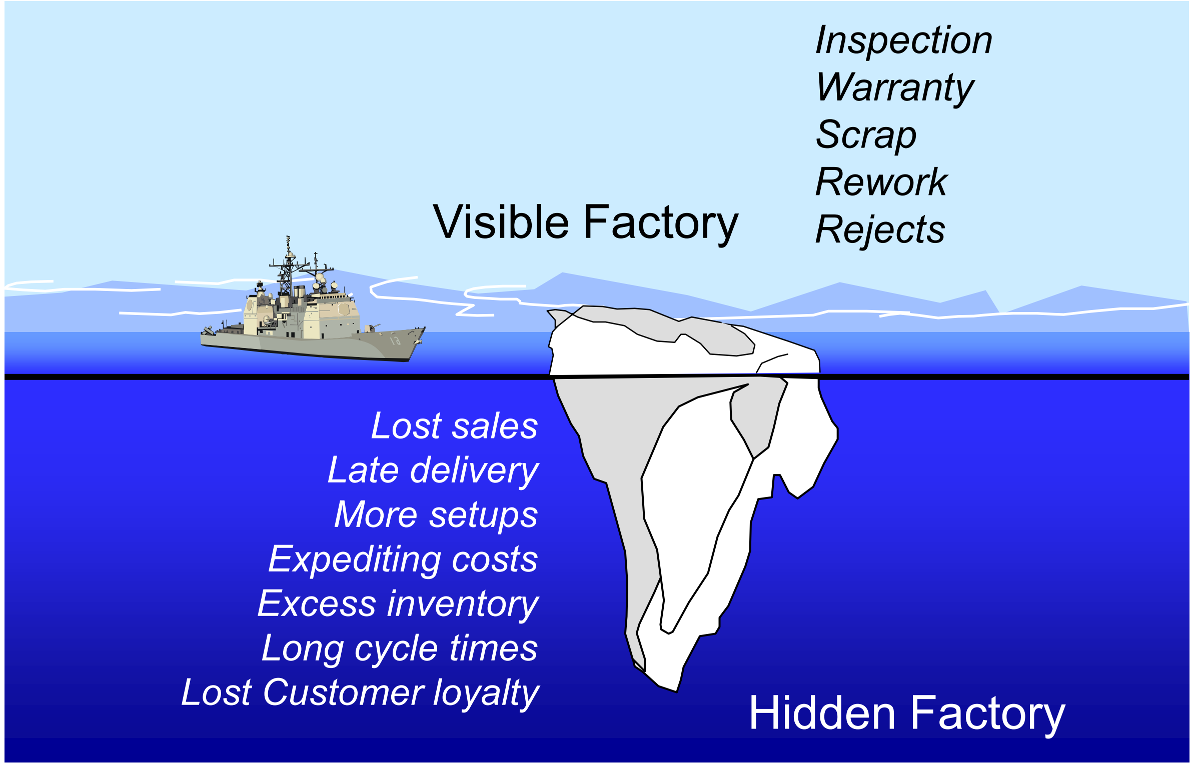
Quality through the Factory



Where do you want to spend your money?
Inspecting and reworking, or on getting it right the first time?

The Cost of Poor Quality (COPQ) "Iceberg"

Traditional Quality Costs
(tangible)



(intangible)
Additional Costs of Poor Quality

What is the Cost of Poor Quality?

In addition to the direct costs associated with finding and fixing defects, “Cost of Poor Quality” also includes:

- The hidden cost of failing to meet customer expectations the first time
- The hidden opportunity for increased efficiency
- The hidden potential for higher profits
- The hidden loss in market share
- The hidden increase in production cycle time
- The hidden labor associated with ordering replacement material
- The hidden costs associated with disposing of defects
- The hidden cost of rework

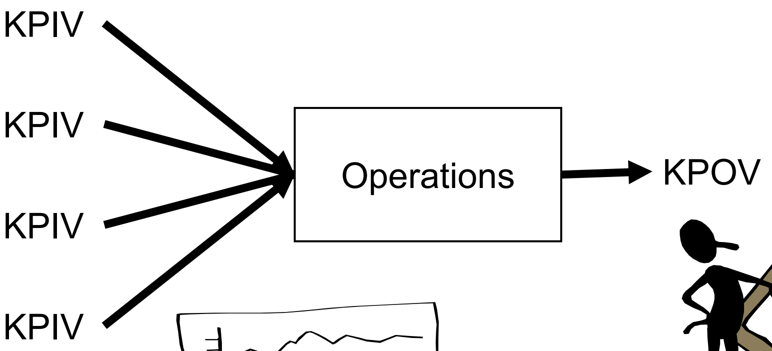
For companies starting six sigma, the cost of poor quality is often around 25% of sales.

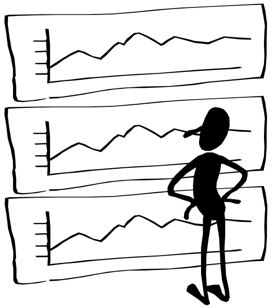
Manage Inputs, and Outputs will Follow

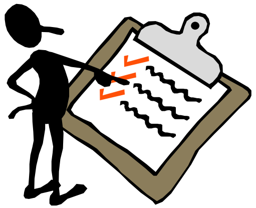
Control the Vital Few
(low variation)

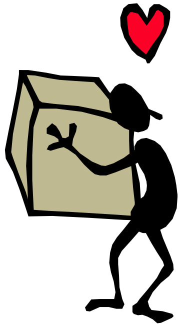
In Spec

Defect – free





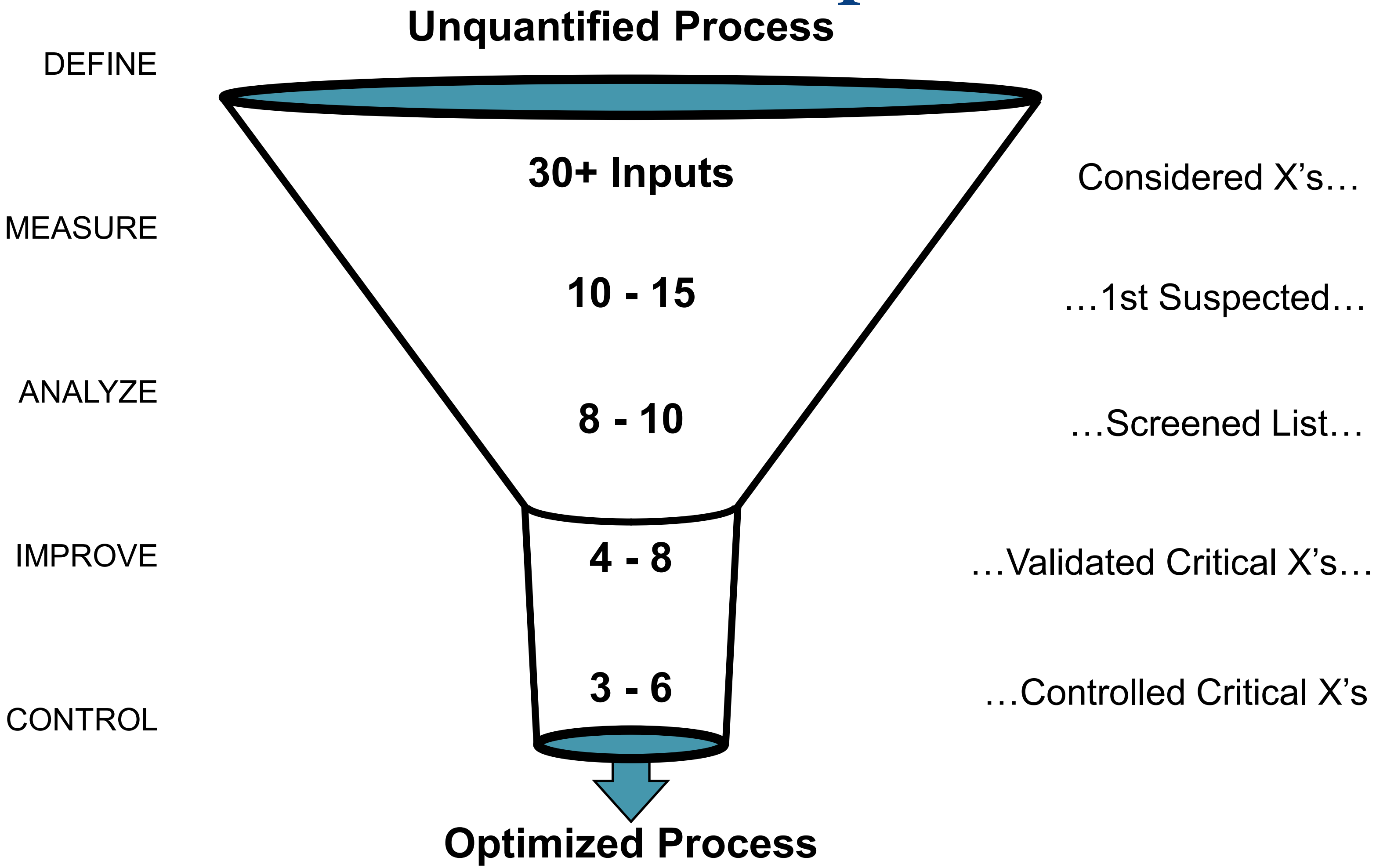




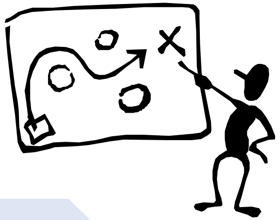
Control!
Control!
Control!

Happy Customer!

In Search of the Vital Few Inputs



Six Sigma is a Process…



Define

What's the problem
Clarify it
- Defect data
- What defects
- What productn steps

Tools:
- Process Mapping
- FMEA
- Fishbone diagrams

Measure

How to measure it
- Not defect counts
- Output variation
- Measurement system

Tools:
- MSA, %GRR
- Data collection
- Graphs of Data
- Cp and Cpk

Analyze

What changes it
- Controllable
- Noise
- Signals

Tools:
- Hypothesis Tests
- Regression
- ANOVA

Improve

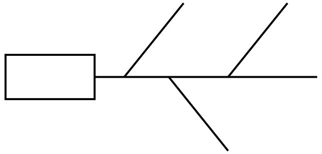
What to change
- Controllable

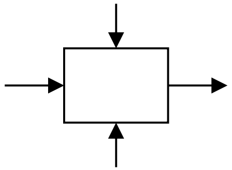
Tools:
- Hypothesis Tests
- DoE
- Robust Design

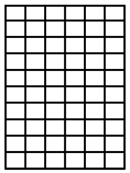
Control

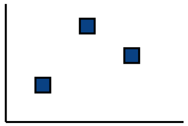
How to keep it there
- Control plans

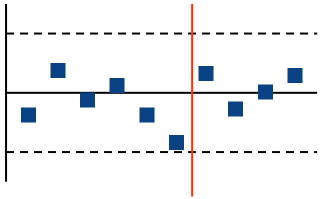
Tools:
- 5S
- Control Plans
- Empowerment
- SPC



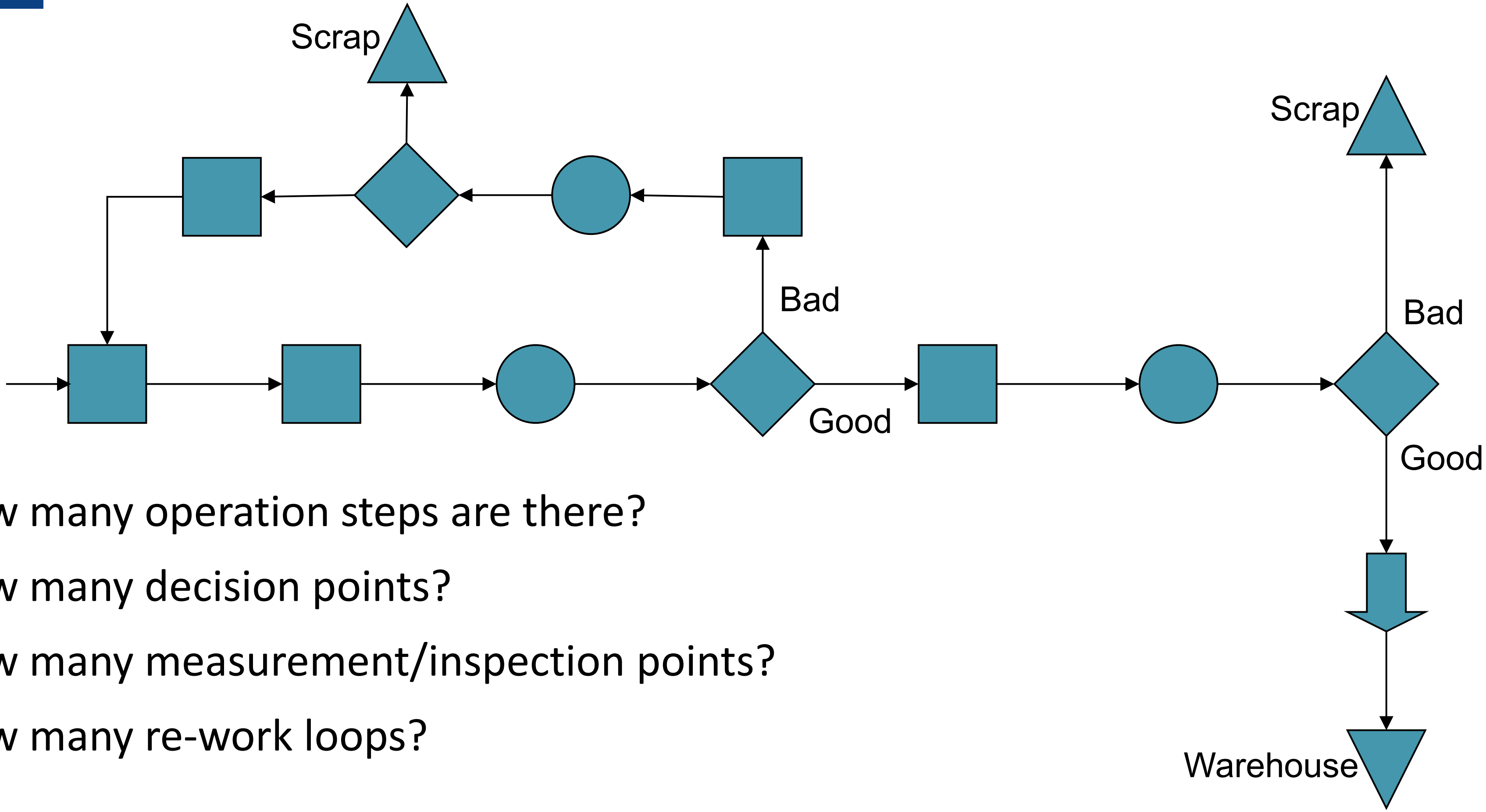








Define: Process Map



How many operation steps are there?

How many decision points?

How many measurement/inspection points?

How many re-work loops?

Define: Process FMEA

Failure Modes and Effect Analysis is a systematic approach to examine potential failures and prevent their occurrence

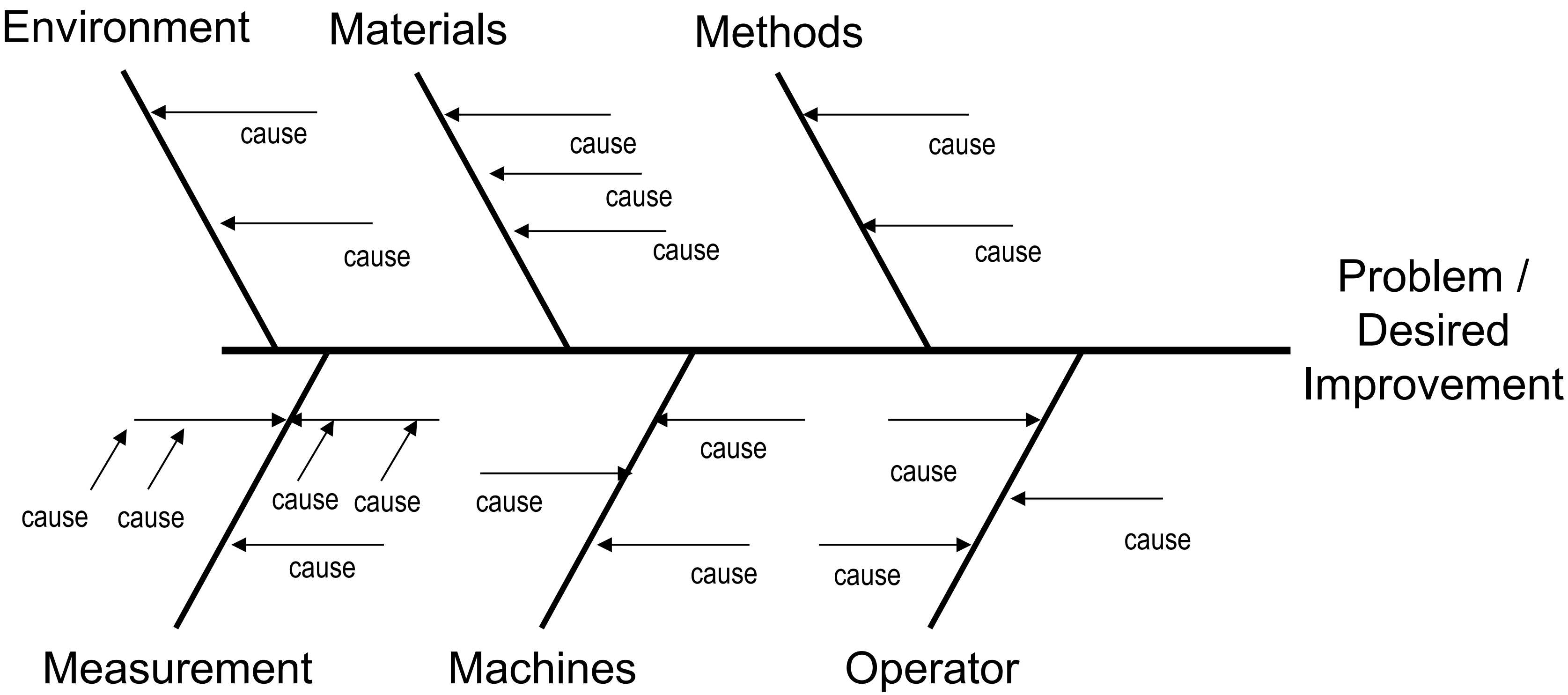
- Predict problems and prioritize them early, so the most likely failure modes can be addressed either by eliminating them or ensuring process controls.

FMEA is generally applied during the initial stages of the manufacturing process design. What can go wrong, eliminate it.

Company ____ Project ____	Plant ____ Product ____ Process ____	FMEA Number ____ Page ____ of ____ Prepared by ____ Date ____

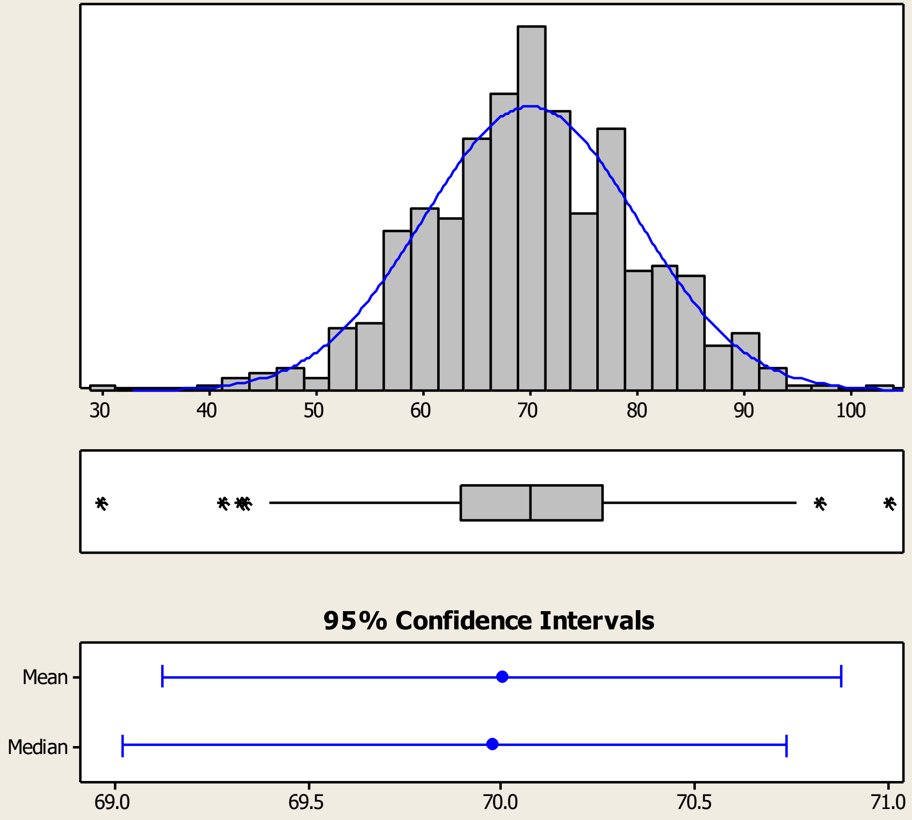
Item Function Step	Potential Failure	Potential Effect(s) of Failure	Sev	Class	Potential Cause(s) of Failure	Occur	Current Process Controls	Detect	RPN	Action Priority	Recommended Action(s)	Responsibility and Target Completion Date	Action Results: Actions Taken	Sev	Occur	Detect	RPN
									0								0

Measure: Case and Effect Diagram



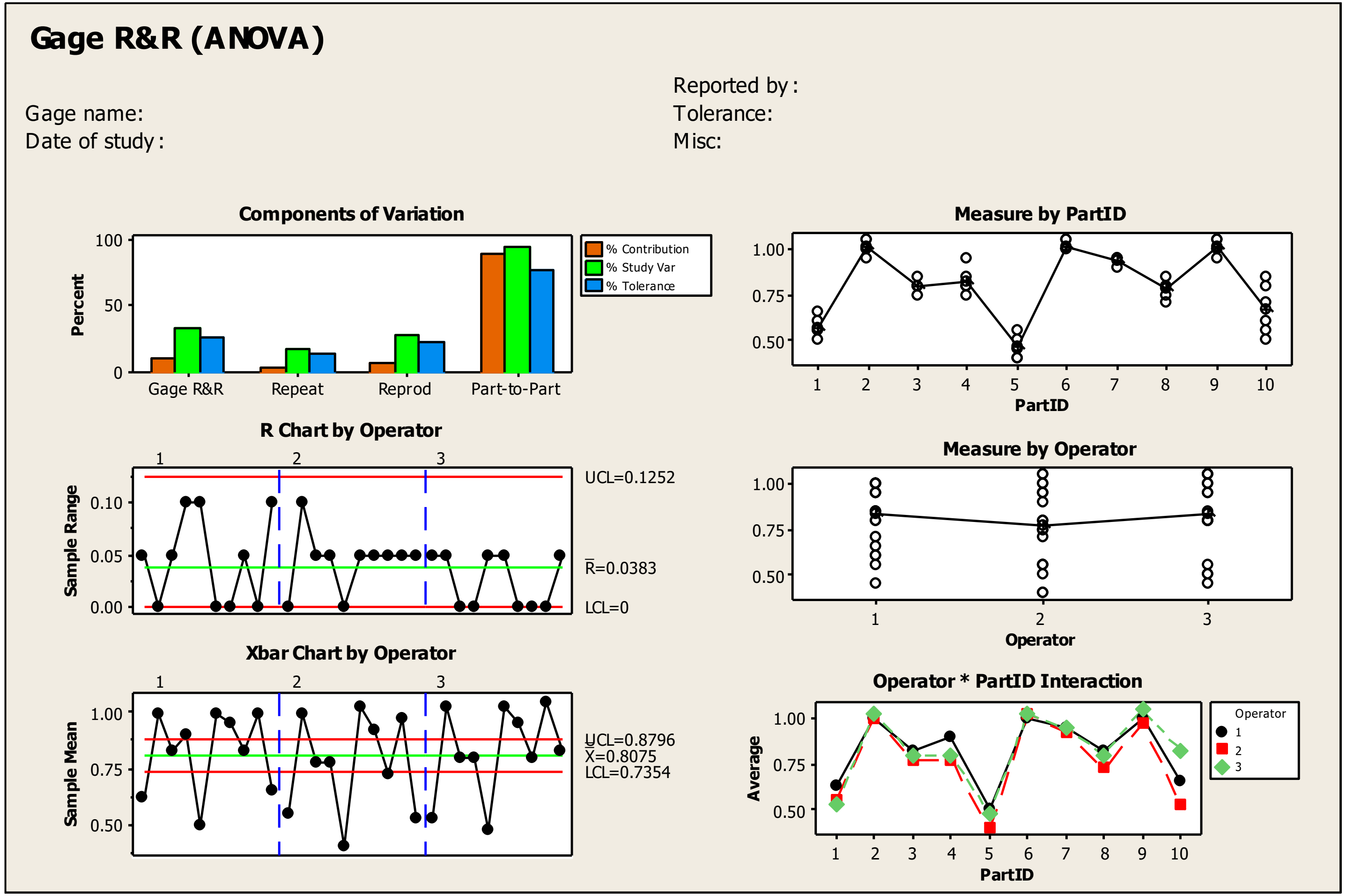
Measure: Statistics

Graphical Summary

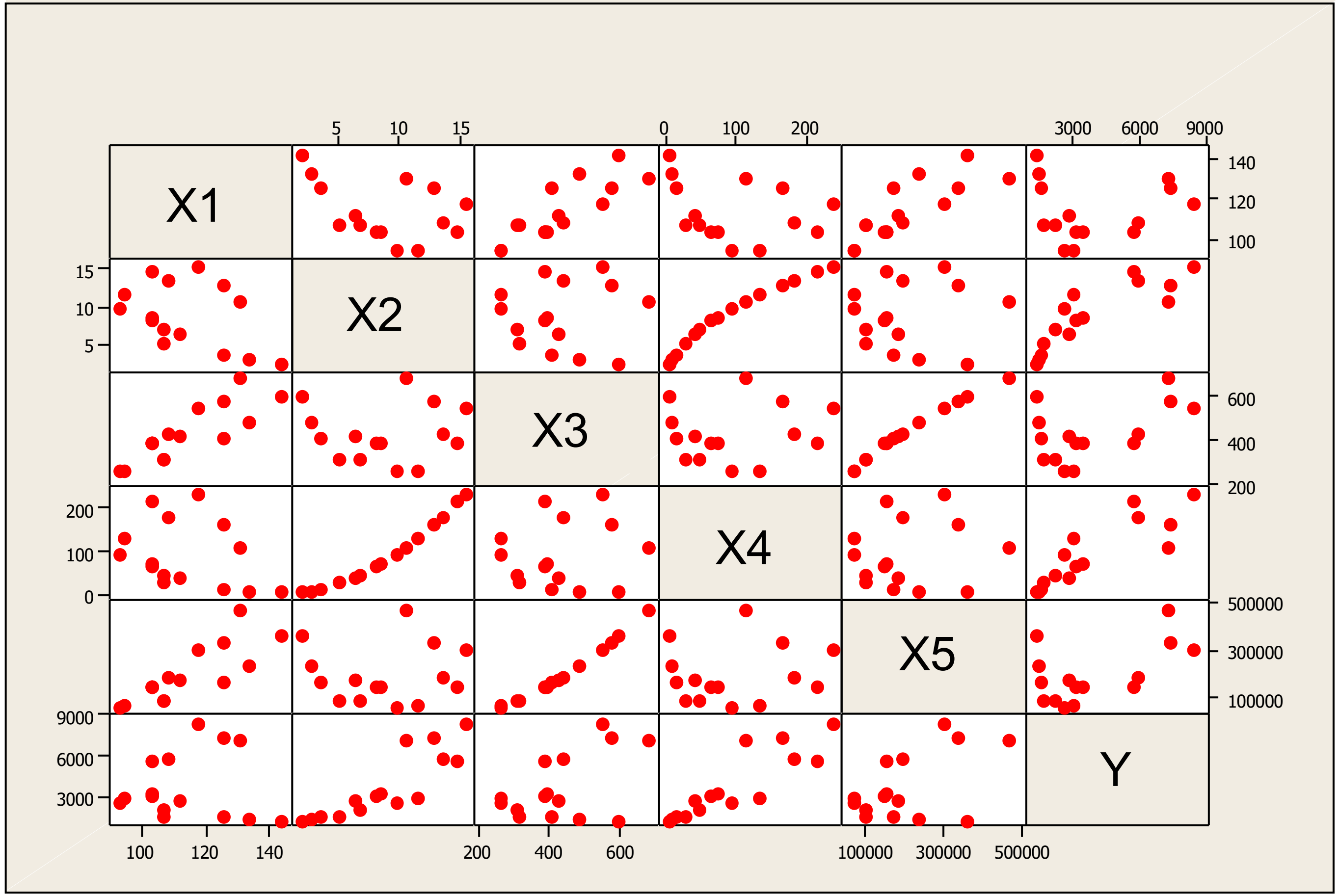


Anderson-Darling Normality Test	
A-Squared	0.42
P-Value	0.328
Mean	70.000
StDev	10.000
Variance	100.000
Skewness	-0.050008
Kurtosis	0.423256
N	500
Minimum	29.824
1st Quartile	63.412
Median	69.977
3rd Quartile	76.653
Maximum	103.301
95% Confidence Interval for Mean	
69.121	70.879
95% Confidence Interval for Median	
69.021	70.737
95% Confidence Interval for StDev	
9.416	10.662

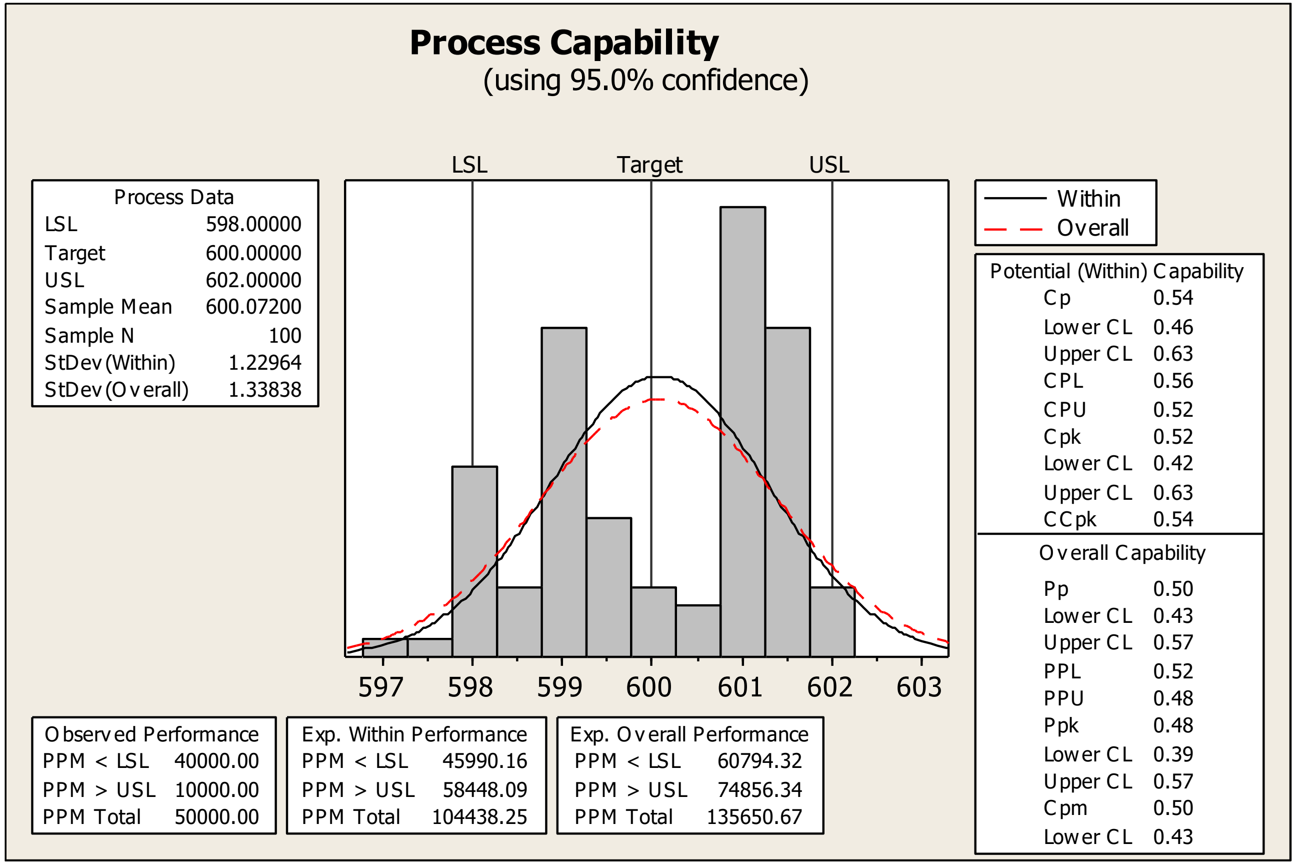
Measure: GRR Study



Analyze: Matrix Plots



Analyze: Process Capability Study



Improve: ANOVA Statistical Analysis

```
ANOVA: Output versus X1, X2

Factor     Type   Levels  Values
X1         fixed       4  1, 2, 3, 4
X2         fixed       5  1, 2, 3, 4, 5


Analysis of Variance for Response

Source     DF       SS      MS       F      P
X1          3   2377.8   792.6   49.62  0.000
X2          4  30648.7  7662.2  479.64  0.000
Error      12    191.7    16.0
Total      19  33218.2


S = 3.99687   R-Sq = 99.42%   R-Sq(adj) = 99.09%
```

Improve: Regression Statistical Analysis

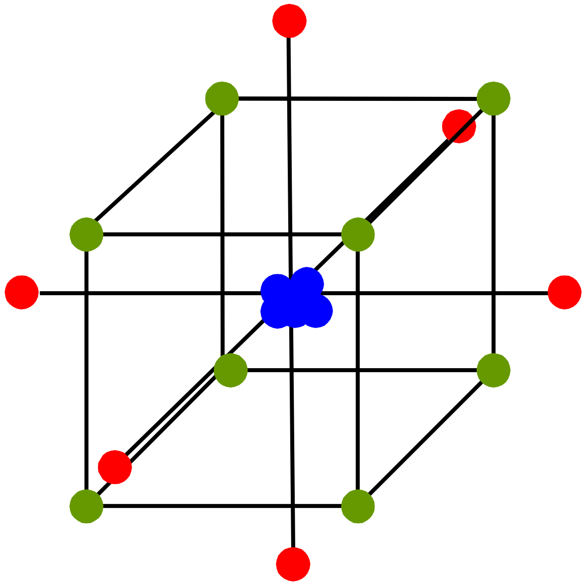
```
Regression Analysis: Output versus X1, X2, X3

The regression equation is
Output = - 505 + 1.13 X1 - 20.2 X2 + 144 X3

Predictor        Coef  SE Coef       T      P
Constant      -504.83    69.26   -7.29  0.002
X1            1.13458  0.08454   13.42  0.000
X2           -20.1671   0.8561  -23.57  0.000
X3            143.818    3.931   36.59  0.000

S = 24.7895   R-Sq = 98.2%   R-Sq(adj) = 95.4%
```

Improve: Design of Experiments



Std Order	A	B	C
1	-1	-1	-1
2	1	-1	-1
3	-1	1	-1
4	1	1	-1
5	-1	-1	1
6	1	-1	1
7	-1	1	1
8	1	1	1
9	-1.68	0	0
10	1.68	0	0
11	0	-1.68	0
12	0	1.68	0
13	0	0	-1.68
14	0	0	1.68
15	0	0	0
16	0	0	0
17	0	0	0

Project Outcomes

After a six sigma project, an increment of quality improvement will have become understood.

Following PDCA, this must be ACTed upon within the factory

Follow the learning and keep it learned by the organization.

To do this, ***Process Controls*** are used, documented with ***Process Control Plans***.

Control Plans

Control Plan Number:		Key Contact/Phone:	Date: Date (Rev):
Part Number:		Core Team:	Customer Engineering Approval/Date (If Req'd):
Part Name Description:		Supplier/Plant Approval/Date:	Customer Quality Approval/Date (If Req'd):
Supplier Plant:	Supplier Code:	Other Approval/Date (If Req):	Other Approval/Date (If Req'd):

			Characteristics				Methods					
									Sample			
Part / Process Number	Process Name / Operation Description	Machine, Device, Jig, Tools for Mfg.	No	Product	Process	Special Char. Class.	Spec Limits	Evaluation Measurement Technique	Size	Frequency	Control Method	Reaction Plan

Control Plans

A ***control plan*** is a list of KPOV / KPIVs at each process step, and what to when any has become out of control. Without revised improvements and use of the control plans, the organization has not learned. It is essential.

Control Plan Number:		Key Contact/Phone:	Date:	Date (Rev):
Part Number:		Core Team:	Customer Engineering Approval/Date (If Req'd):	
Part Name Description:		Supplier/Plant Approval/Date:	Customer Quality Approval/Date (If Req'd):	
Supplier Plant:	Supplier Code:	Other Approval/Date (If Req):	Other Approval/Date (If Req'd):	

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Control Plans

A ***control plan*** is a list of KPOV / KPIVs at each process step, and what to when any has become out of control.

“Inspect hole diameter. If 3 in a row oversized, replace drill bit”

“If vibration levels are exceeded, call Bob for maintenance”

“Inspect at the start of every shift. If out of calibration, adjust X until Y is on target”

Control Plan Number:		Key Contact/Phone:	Date:	Date (Rev):
Part Number:		Core Team:	Customer Engineering Approval/Date (If Req?d):	
Part Name Description:		Supplier/Plant Approval/Date:	Customer Quality Approval/Date (If Req?d):	
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Part / Process Number	Process Name / Operation Description	Machine, Device, Jig, Tools for Mfg.	Characteristics			Special Char. Class.	Methods					Reaction Plan
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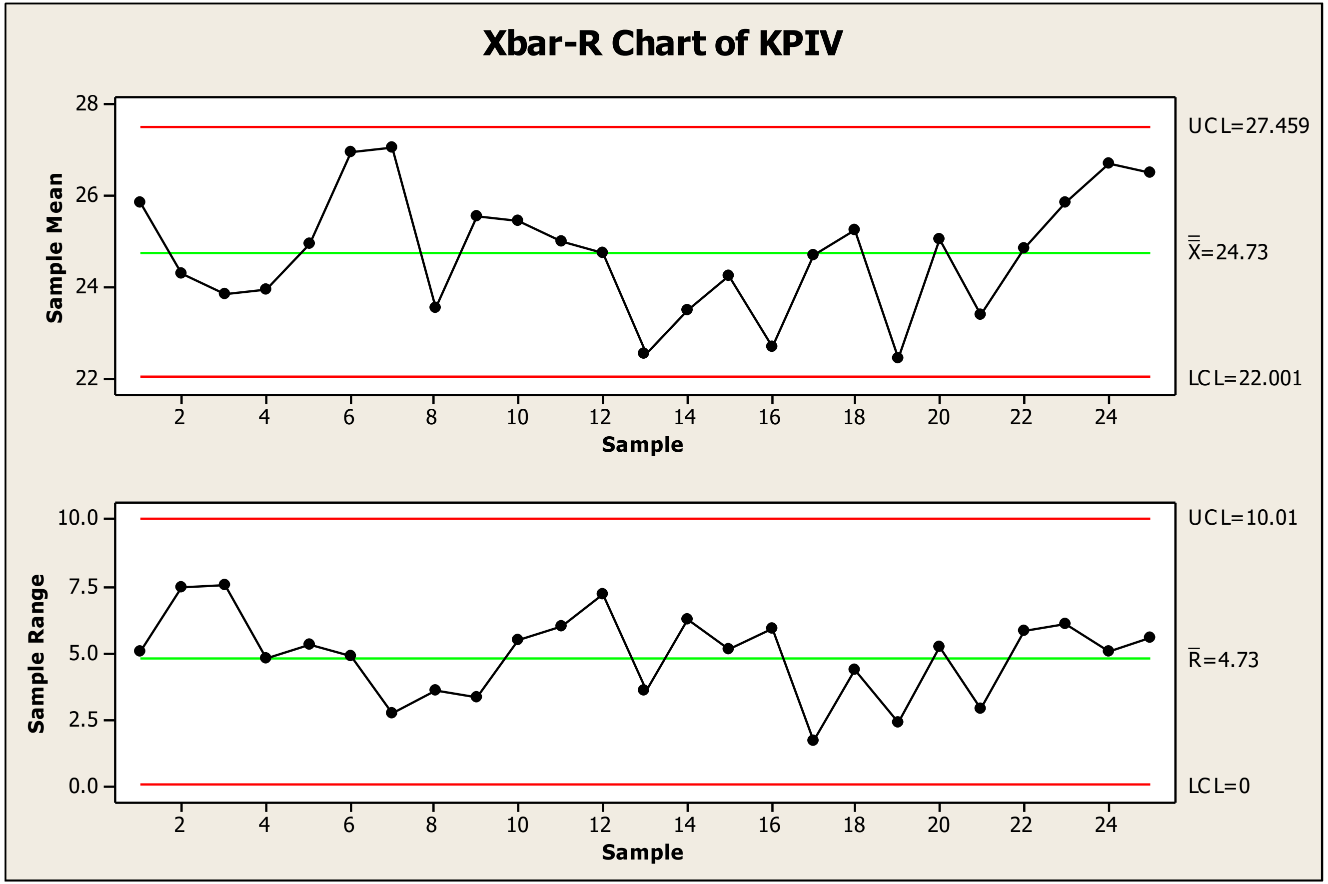
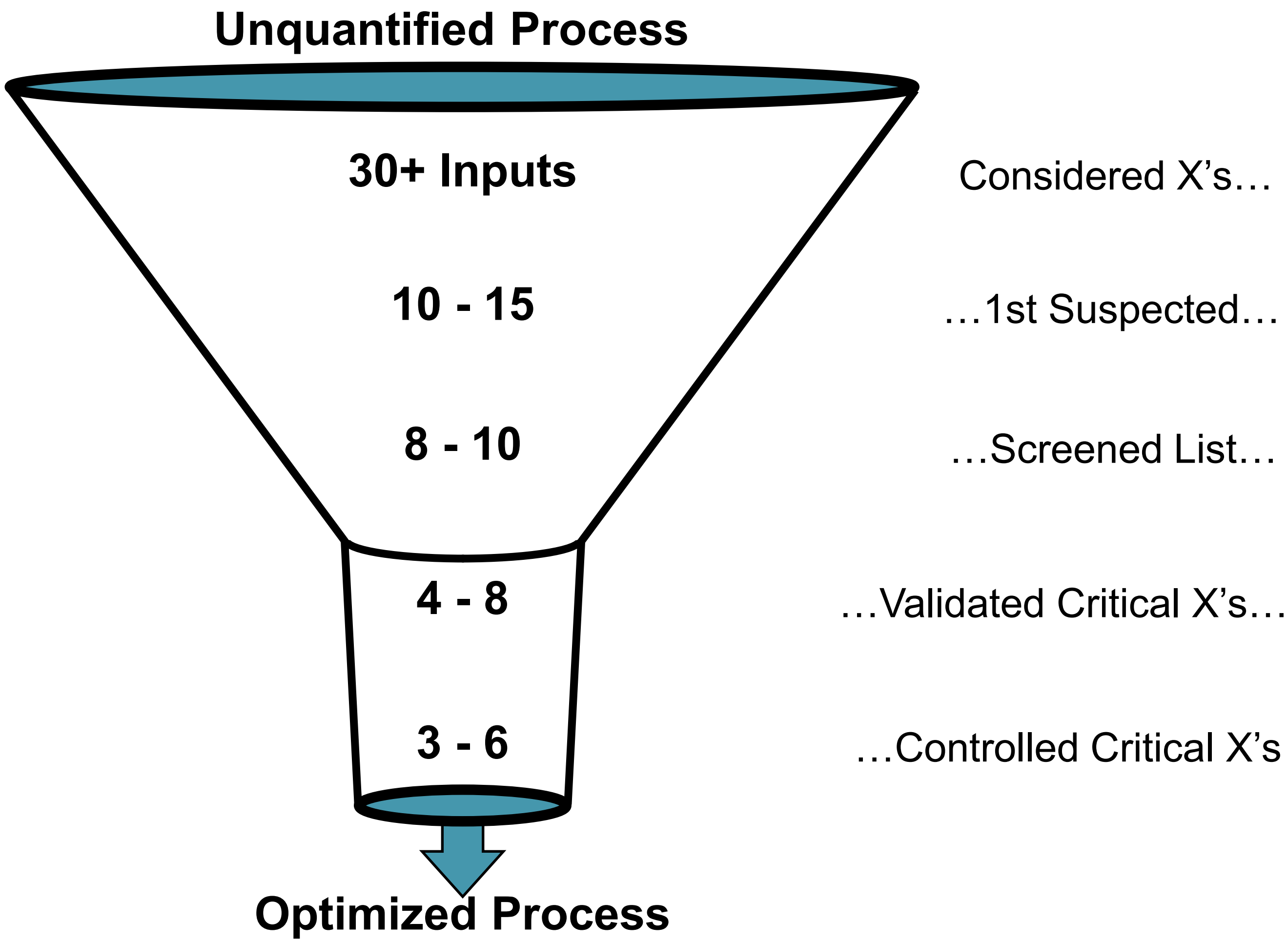
Statistical Process Control



In Search of the Vital Few Inputs

- Process Map
- CE Matrix, FMEA
- GRR, Capability
- Multi-Vari Studies Correlation
- Hypotheses
- Screening DOE
- DOE
- Capability
- Control Plans



The Improvement Strategy (DMAIC)

			Focus
PLAN	Define	Select product or process KPI; e.g. Y	Y
		Create Process Map	X
		Study or Create Process FMEA	Y
DO	Measure	Validate measurement system for Y	Y
		Establish process capability for Y	Y
	Analyze	Identify variation sources in Y	$X_1, \ldots, X_n$
		Screen potential causes for changes in Y & identify vital few X_i	$X_1, \ldots, X_n$
		Discover variable relationships between vital few X_i	Vital Few
CHECK	Improve	Establish operating tolerances on vital few X_i	Vital Few
		Validate measurement system for vital few X_i	Vital Few
ACT	Control	Determine ability to control vital few X_i	Vital Few
		Implement process control system on vital few X_i	Vital Few

Summary

DMAIC is PDCA for manufacturing <u>variability</u> problems

The emphasis is identifying critical input variables and replacing their pass/fail inspections with input variable measurements

Manage these critical inputs and the outputs will be fine

The six sigma process makes heavy use of statistics as is necessary to understand variability

The outcome of a six sigma effort should be much improved process controls, including measurements and control plans. Organizational Learning.

NEXT: Design of Experiments: ANOVA

EXERCISE

Suppose you start experiencing dry cookies.
Define a set of causes to explore changing to determine levels at which they cause dry cookies. Use this FMEA.

Step	Failure Mode`	Effect	Cause	Control
Melt Butter	Burnt butter	Burnt cookie taste	Stovetop too hot	Skill of cook
			Over-melt too long	Skill of cook
	Under-melted butter	Butter spots in cookie	Too short time on stove	Skill of cook
Refrigerate	Not cooled long enough	Dry cookies	Too short time	Skill of cook
	Cooled too cold	Dry cookies	Refrigerator set point	Skill of cook
Baking	Cookies not spaced right	Merged cookies	Cookie placement	Skill of cook
	Over baked	Burnt or dry cookie	Oven Temp set point	Skill of cook
			Time in oven	Skill of cook
	Underbaked	Partially raw cookie	Oven Temp set point	Skill of cook
			Time in oven	Skill of cook

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			Time in oven	Skill of cook
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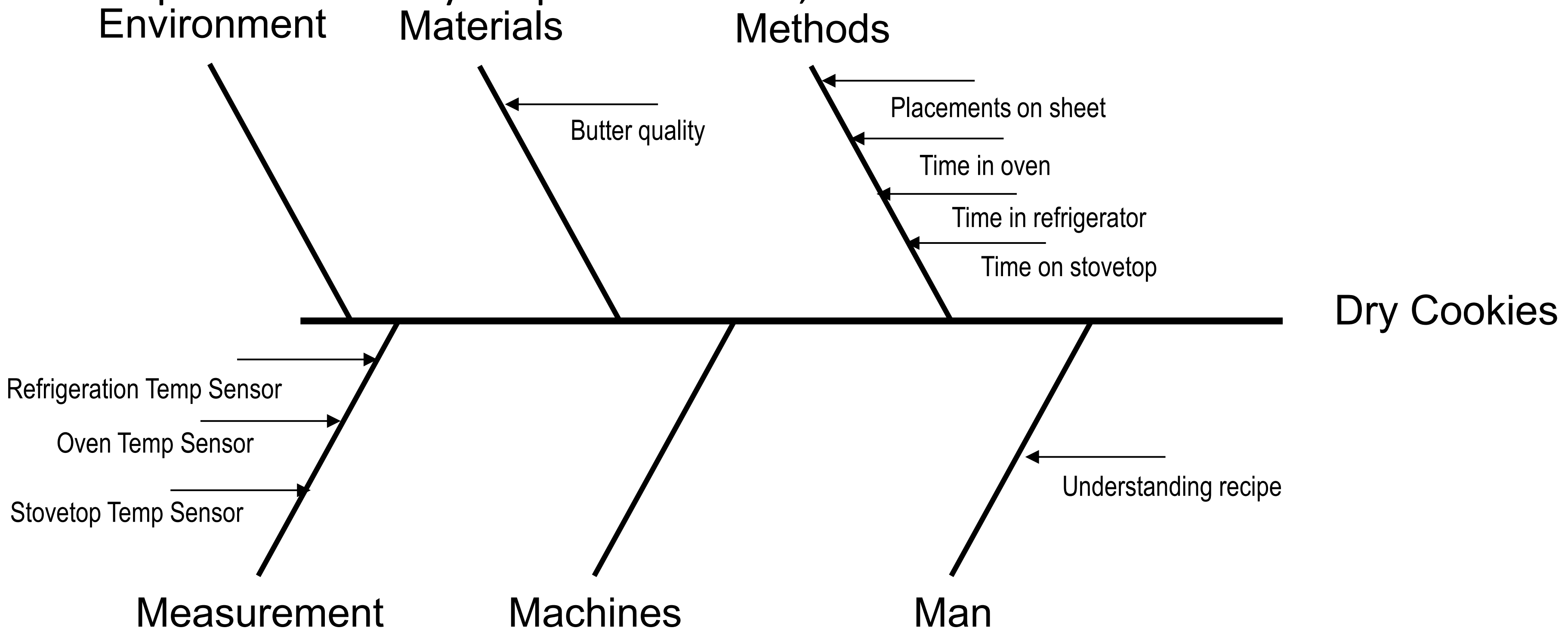
			Time in oven	Skill of cook

EXERCISE

The cause and effect diagram translates the FMEA causes into causal variables that can be studied as inputs. Based on your previous work, choose what to measure and record.



EXERCISE

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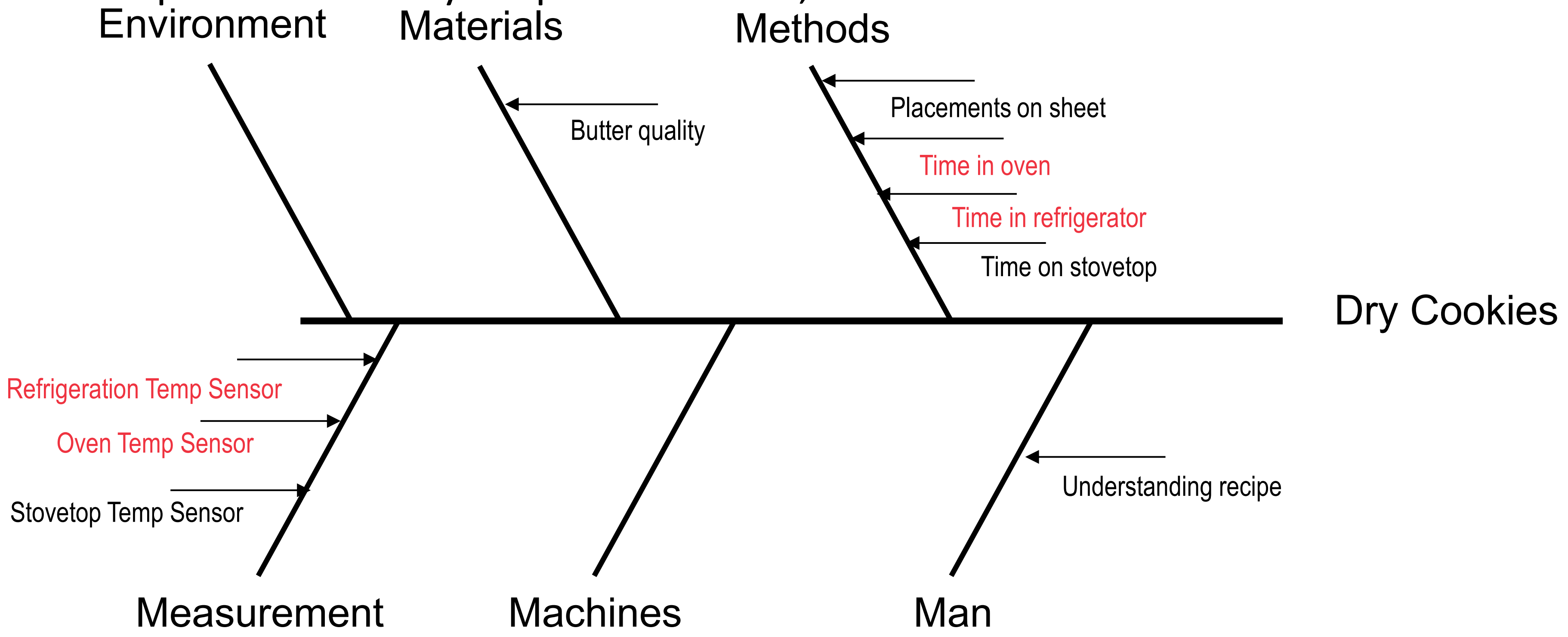
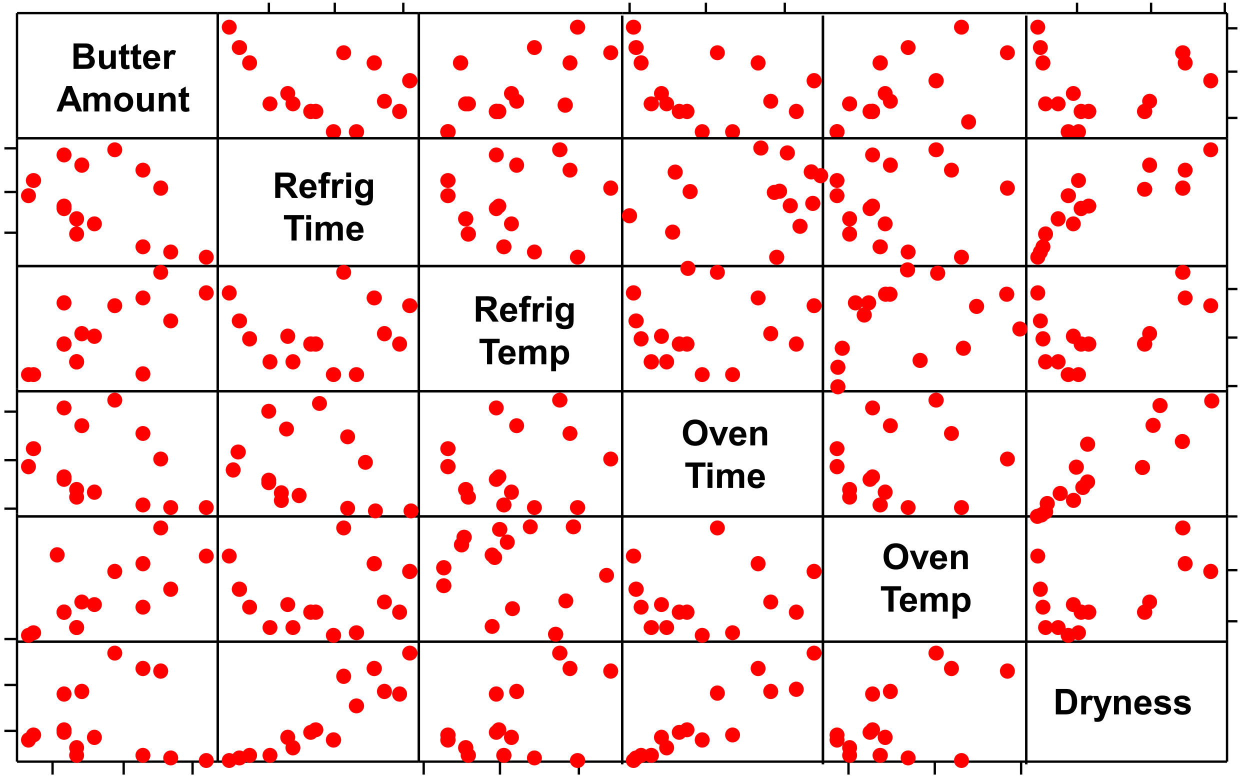


EXERCISE

Suppose you study the process over the last two weeks and record the ingredient levels and process setpoints, and you get the following. What inputs are important to dryness?



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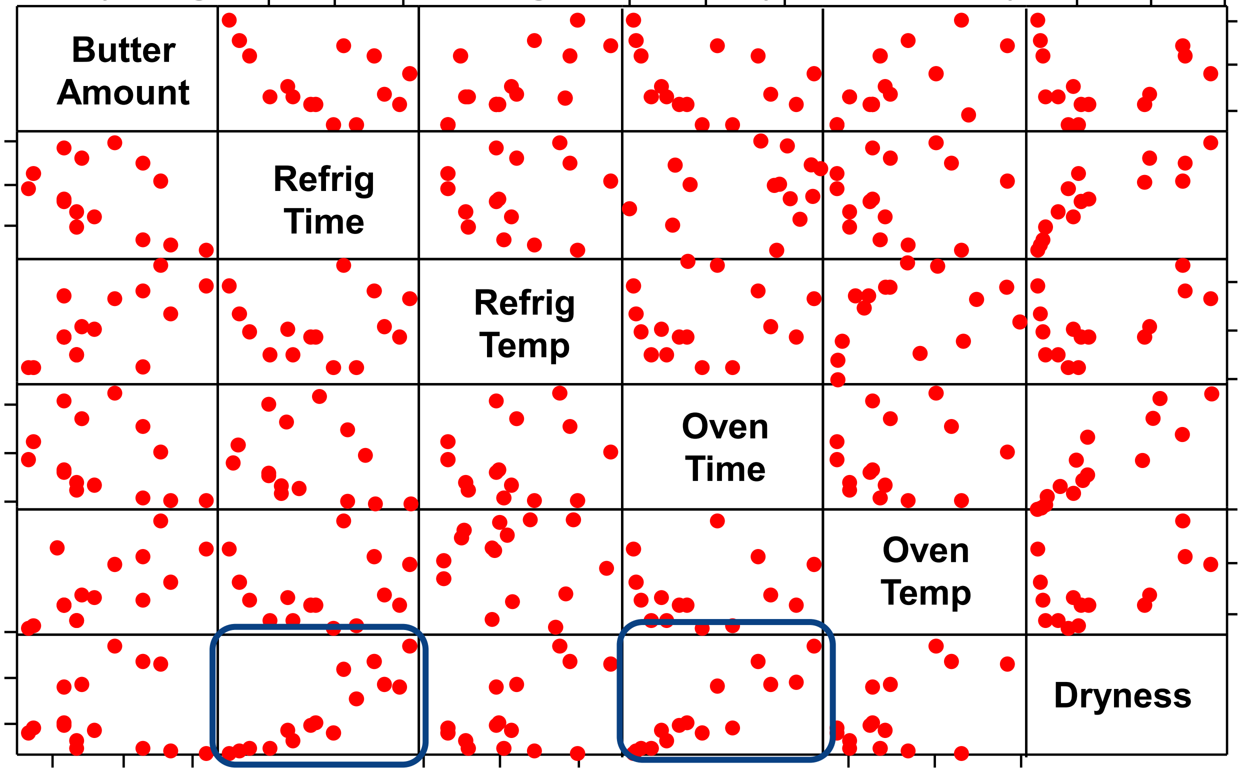

EXERCISE

Suppose the results of your cookie baking study show that the following 3 variables strongly affect the dry cookie defect.

- Insufficient melted liquid butter
- Refrigeration time under 2 hours
- Bake temperature over 175 C and time over 15 minutes

Add quality controls to the process map to ensure these process mistakes don’t happen.

EXERCISE

Suppose the results of your cookie baking study show that the following 3 variables strongly affect the dry cookie defect.

- Insufficient melted liquid butter
- Refrigeration time under 2 hours
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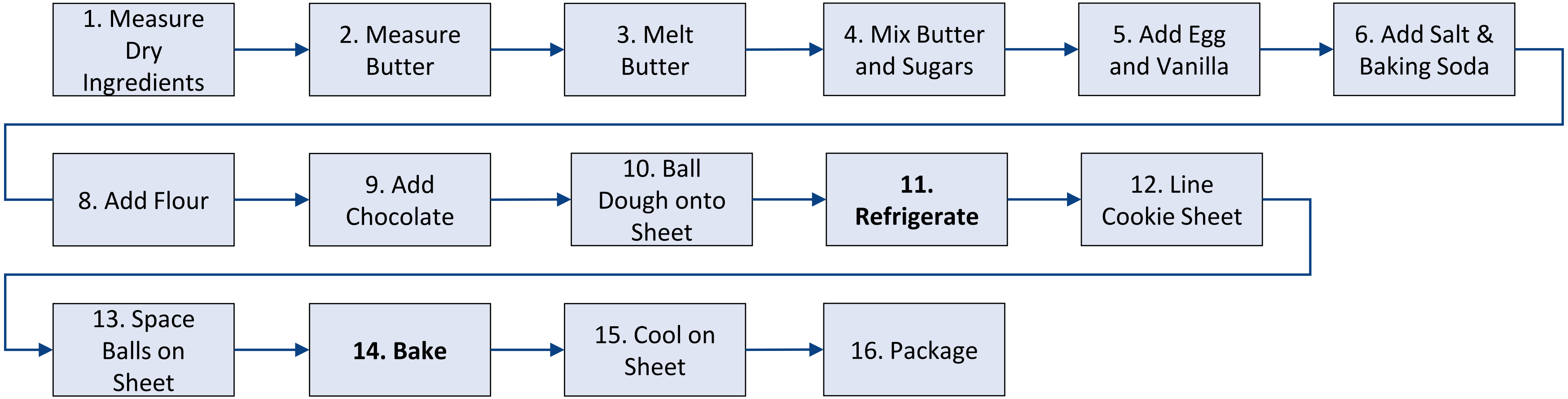
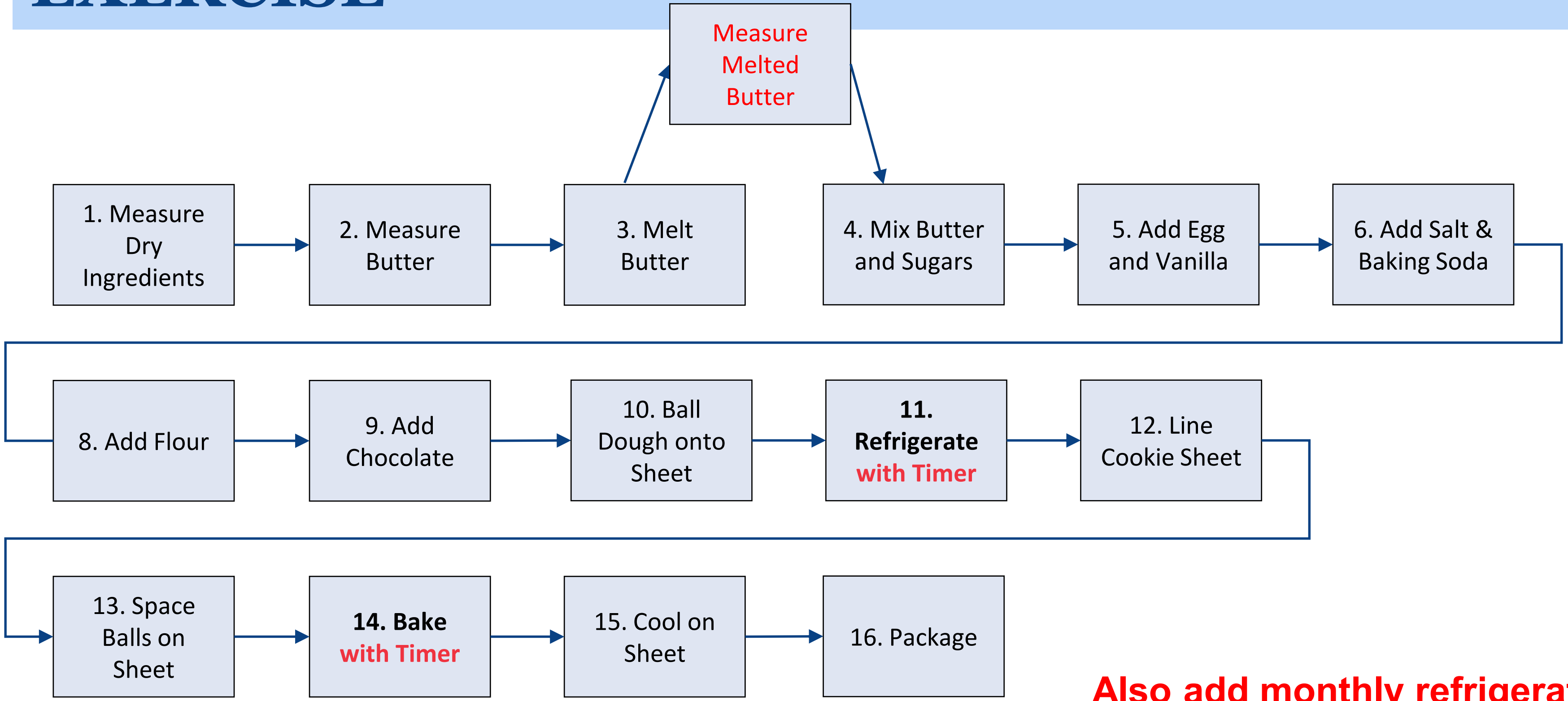
Add quality controls to this process map to ensure these process mistakes don’t happen.



EXERCISE



Also add clear set point labels to oven and refrigerator knobs.

Also add monthly refrigerator and stove temperature calibration procedures to maintenance schedules.